Modelling and Integrating Polarity, Subjectivity and Emotion of Sentiment Data in a Data Warehouse

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Abstract. The widespread adoption of digital platforms has produced an unprecedented volume of user-generated textual content, much of which carries opinions, attitudes and emotional responses expressed in natural language. Sentiment analysis has emerged as the primary computational approach for extracting this affective information, yet it has traditionally been confined to polarity classification, reducing the richness of human expression to a positive, negative or neutral label. Dimensions such as subjectivity and discrete emotional states, which carry significant analytical value, are rarely modelled in a systematic way.

At the same time, data warehousing provides mature and well-established mechanisms for integrating heterogeneous data and supporting multidimensional analytical processing, but affective information derived from text has seldom been treated as a first-class analytical element within these environments. This disconnect limits the ability to perform structured, historical and comparative analysis of sentiment alongside conventional business dimensions.

This dissertation proposes the modelling and integration of polarity, subjectivity and emotion within a dimensional data warehouse designed for the tourism and hospitality domain. A sentiment-aware analytical framework is developed following Kimball's dimensional modelling methodology, supported by ETL processes that transform unstructured review text into structured affective measures. The proposed model is evaluated through analytical queries executed using OLAP tools, assessing its ability to support multidimensional sentiment-aware analysis. The work aims to demonstrate that affective information, when modelled with the same rigour applied to conventional analytical data, can meaningfully extend the capabilities of business intelligence systems.

Keywords: Sentiment Analysis; Data Warehousing; Polarity; Subjectivity; Emotion; Business Intelligence.

1 Introduction

The widespread adoption of digital platforms over the past two decades has fundamentally changed how people express and share their experiences. Review platforms, social media and booking services now accumulate millions of user-generated texts carrying not just factual descriptions but opinions, emotional reactions and subjective judgments about products, services and places. For organisations that depend on understanding customer experience, such as hotels, restaurants or tourism boards, this textual data represents a significant and largely underexploited source of analytical insight.

The computational analysis of this kind of content falls under the field of sentiment analysis, which Liu (2012) defines as the study of opinions, sentiments and emotions expressed in text. Most practical applications of sentiment analysis reduce affective meaning to polarity: a positive, negative or neutral label assigned to a piece of text. This is computationally convenient but analytically limiting. Two reviews carrying the same negative polarity may communicate very different things: one expressing anger at rude staff, another expressing quiet disappointment at a deteriorating service. Treating them identically in an analytical system discards information that could meaningfully inform decisions.

Data warehousing addresses a different but related challenge: integrating data from heterogeneous sources into a consistent, historically maintained environment that supports complex analytical queries. Organised around fact and dimension tables, data warehouses have become the backbone of business intelligence systems in most large organisations (Kimball & Ross, 2013). They excel at handling structured, numerical and transactional data, but have been slow to incorporate affective information as an analytical element in its own right.

The result is a practical gap. Organisations collecting large volumes of customer reviews have limited means of querying that sentiment data with the same flexibility and historical depth they apply to their operational metrics. This dissertation addresses that gap by proposing the modelling and integration of polarity, subjectivity and emotion within a dimensional data warehouse environment. The following sections establish the context, motivation and objectives that underpin this work.

1.1 Context

Sentiment analysis has developed over the past two decades into one of the most active areas within natural language processing. Its central concern is the automatic extraction of affective information from text, determining not just what a piece of writing is about, but how the author feels about it. This involves tasks ranging from broad opinion classification at the document level to fine-grained aspect-level analysis, where sentiment is identified separately for each distinct characteristic of an entity being evaluated (Liu, 2012). The practical appeal is clear: platforms handling millions of reviews, social media posts or support tickets cannot rely on human reading, and automated sentiment extraction offers a scalable alternative (Wankhade, Rao, & Kulkarni, 2022).

Data warehousing occupies a different position in the analytical landscape. Where sentiment analysis is concerned with extracting meaning from unstructured text, data warehousing is concerned with storing, integrating and querying structured data at scale. Built around multidimensional schemas of facts and dimensions, data warehouses are designed to support the kind of historical, comparative and aggregative analysis that operational databases cannot efficiently handle (Kimball & Ross, 2013). They are the infrastructure through which business intelligence systems answer questions like: how did sales perform this quarter compared to last year, or which regions are underperforming relative to the national average?

These two fields have largely developed in parallel, with limited cross-pollination. Sentiment analysis has focused on extraction accuracy and linguistic coverage, producing increasingly sophisticated models but paying less attention to how results should be stored, versioned and queried over time. Data warehousing, for its part, has been built around structured and transactional data, and its modelling conventions have not naturally extended to the heterogeneous, linguistically-derived outputs of sentiment processing (Moalla, Nabli, & Hammami, 2022). The consequence is that sentiment data, even when carefully extracted, tends to be consumed immediately and discarded rather than integrated into the kind of analytical infrastructure that would allow it to be explored longitudinally and in combination with other business dimensions (Mohammad S. , 2021).

1.2 Motivation

The case for integrating sentiment into analytical systems is not purely academic. Organisations in experience-driven sectors such as tourism, hospitality and retail routinely collect thousands of reviews but rarely analyse them with the same rigour they apply to their financial or operational data. A hotel chain may track occupancy rates by region and season with precision yet have no structured way of asking whether guest sentiment toward cleanliness has deteriorated over the same period, or whether negative emotional responses are concentrated in a particular property or visitor segment. The data to answer these questions exists; the analytical infrastructure to support them typically does not.

Part of the problem is representational. When sentiment is extracted from text, it is most commonly reduced to a polarity score, a single number that collapses the full range of human affective expression into a positive-to-negative axis. This matters because polarity alone is a poor guide to action. A service receiving predominantly negative reviews driven by anger requires a different response than one receiving negative reviews driven by disappointment or resignation, yet a polarity-only system cannot distinguish between them (Mohammad S. , 2021). Subjectivity and emotion are not secondary attributes of sentiment; they are part of what sentiment means.

The motivation for this work therefore sits at the intersection of two needs. The first is the need for richer affective representations that go beyond polarity and treat sentiment as the multidimensional phenomenon it is. The second is the need for an analytical infrastructure that handles sentiment data with the same structural discipline applied to conventional business metrics, storing it consistently, maintaining its historical context and enabling it to be queried across multiple dimensions. Dimensional data warehousing, as established by Kimball and Ross (2013) provides precisely that kind of infrastructure. The challenge is adapting it to accommodate affective data in a way that preserves analytical expressiveness without sacrificing the modelling rigour on which warehouse design depends.

1.3 Objectives

General Objective

The general objective of this work is to design and validate a sentiment-aware data warehouse capable of integrating polarity, subjectivity and emotion extracted from unstructured textual data, with the aim of supporting richer and more expressive analytical processing within a multidimensional environment.

Specific Objectives

To achieve the general objective defined above, the following specific objectives are established:

- Review and analyze sentiment and emotion analysis techniques to identify approaches suitable for structured integration into analytical environments.
- Design a conceptual and dimensional data warehouse model that explicitly represents affective dimensions, including polarity, subjectivity and emotion.
- Develop extraction, transformation and loading (ETL) processes to integrate sentiment-enriched data from unstructured textual sources into the data warehouse.
- Implement a data warehouse prototype that supports sentiment-aware analytical processing.
- Validate the proposed solution through analytical queries and performance evaluation using online analytical processing tools.

1.4 Research Process

This dissertation follows a Design Science Research methodology (Hevner, March, Park, & Ram, 2004), which provides an appropriate framework for work that produces and evaluates an artefact, which in this case is a sentiment-aware data warehouse, as its primary contribution. Design Science Research is concerned not only with building systems but with demonstrating that they satisfy well-defined requirements and contribute knowledge beyond the specific instance constructed. The work is organised into successive phases that move from theoretical grounding through design and implementation to evaluation, with each phase informing the next.

The first phase consists of a state-of-the-art review covering sentiment analysis, affective computing and data warehousing. The review examines how sentiment-related information has been extracted and represented in the literature, and, more specifically, how it has been integrated into analytical environments. Rather than surveying these fields in isolation, the review is oriented toward identifying the limitations of existing integration approaches and establishing the gap that motivates the proposed work.

The second phase conducts a requirements analysis grounded in a concrete case study domain: tourism and hospitality services in Portugal. This domain was selected because experiential and affective evaluation play a central role in how services are perceived and reviewed, making it particularly well-suited to sentiment-aware analysis. The requirements analysis defines the analytical dimensions, data characteristics and query capabilities that the proposed warehouse must support, drawing on representative stakeholder perspectives and analytical use cases.

The third phase addresses the design of the data warehouse. Following Kimball's dimensional modelling methodology (Kimball & Ross, 2013), a multidimensional schema is developed in which polarity, subjectivity and emotion are modelled as first-class analytical measures rather than auxiliary attributes. Modelling decisions are justified analytically and in relation to the requirements established in the previous phase.

The fourth phase covers implementation and data integration. ETL processes are designed and implemented to transform synthetic review data into structured, sentiment-enriched records aligned with the dimensional model. Sentiment derivation relies on established lexicon-based tools, namely VADER for polarity, TextBlob for subjectivity and the NRC Emotion Lexicon for emotion, applied at the aspect level to preserve the fine-grained granularity of the warehouse.

The final phase evaluates the proposed system through analytical queries executed using OLAP tools. This evaluation assesses whether the warehouse supports the analytical requirements defined in phase two and examines the practical implications of the modelling decisions made in phase three.

2 State of the Art

This chapter reviews the foundational concepts and existing work relevant to the design of a sentiment-aware data warehouse. It is organised around five main areas. The first examines sentiment analysis as a field, covering its definitions, levels of granularity, output formats and limitations. The second explores the affective dimensions that lie beyond polarity, subjectivity and emotion, and the computational methods available for detecting them. The third considers sentiment from a data modelling perspective, examining what it means to treat affective information as structured analytical data. The fourth reviews the principles of data warehousing and multidimensional modelling that underpin the proposed design. The fifth surveys existing approaches to integrating sentiment into data warehouse environments, identifying recurring limitations and positioning the present work in relation to them.

2.1 Sentiment Analysis

Sentiment analysis provides the technical foundation for extracting affective information from text. This section examines the field from the perspective of this dissertation: the levels at which sentiment can be analysed, the forms in which outputs are produced and the limitations that motivate the richer affective representations developed in subsequent sections. The discussion is oriented toward analytical integration rather than extraction performance, the question then being not how well sentiment can be detected, but what kind of sentiment information is worth detecting and how it should be structured for downstream use.

2.1.1 Definition and Scope

Sentiment analysis, also referred to as opinion mining, is the subfield of natural language processing concerned with automatically identifying and interpreting affective information expressed in text. Rather than asking what a document is about, it asks how the author feels about it, detecting opinions, attitudes and emotional states embedded in written language. Liu (2012) formalised this scope through a structured characterisation of opinion: each opinion involves a holder, a target entity or aspect and an associated sentiment orientation reflecting the evaluative stance expressed. This decomposition is significant because it establishes opinion as a structured, multi-component phenomenon rather than a vague property of text, which has direct implications for how sentiment data can be represented, stored and reused analytically.

As a research field, sentiment analysis emerged in the early 2000s alongside the growth of user-generated content on the web. Pang and Lee (2008) document the foundational efforts of this period, in which researchers demonstrated that subjective information could be reliably classified using computational methods, establishing sentiment analysis as a legitimate area of inquiry within natural language processing. Since then, the field has expanded considerably in both technical sophistication and application scope, from customer feedback and brand monitoring to political discourse analysis and public health surveillance.

What has remained relatively constant across this expansion, however, is the dominance of polarity-based representations. The majority of sentiment analysis systems reduce affective meaning to a single evaluative dimension, classifying text as positive, negative or neutral. This choice reflects practical convenience, as polarity labels are easy to compute, aggregate and interpret, but it comes at a cost. How sentiment is defined, categorised and quantified directly determines what can be done with it analytically. A system that outputs only polarity labels cannot distinguish between different emotional states, cannot measure the degree of subjectivity in an expression and cannot support the kind of multidimensional analysis that structured analytical environments require. These representational limitations are the central motivation for the work presented in this dissertation.

2.1.2 The Different Levels of Sentiment Analysis

Sentiment analysis can be performed at different levels of granularity, each reflecting a different assumption about the unit of text that carries meaningful affective information. Liu (2012) identifies three primary levels: document, sentence and aspect. Some frameworks additionally distinguish a phrase level as an intermediate granularity between sentence and aspect, though it is less consistently adopted across the literature (Wankhade, Rao, & Kulkarni, 2022). These levels are introduced here from the perspective of sentiment extraction; their implications for analytical modelling and data grain are discussed in Section 2.3.3.

Document-Level Sentiment Analysis

Document-level analysis determines the overall sentiment expressed in an entire text, such as a review, article or social media post. It assumes that the document conveys a single dominant opinion toward a target entity, making it most appropriate for short, focused texts where sentiment is relatively uniform. Its simplicity and computational efficiency have made it the dominant approach in large-scale applications such as product review aggregation and brand perception monitoring (Pang & Lee, 2008). The limitation is precisely this assumption of homogeneity: when a document contains contrasting opinions toward different aspects of the same entity, document-level analysis collapses them into a single label, losing the distinctions that may be most analytically relevant.

Sentence-Level Sentiment Analysis

Sentence-level analysis assigns sentiment to individual sentences rather than to the document as a whole, allowing for finer-grained detection of opinion variation within a text. This is useful when a document contains multiple statements that express different or even opposing views (Birjali, Kasri, & Beni-Hssane, 2021). The challenge at this level is that sentences frequently rely on surrounding context for their meaning, may express sentiment implicitly or may convey purely factual information with no affective content at all, each of which complicates reliable classification.

Aspect-Level Sentiment Analysis

Aspect-level analysis, also referred to as feature-level analysis, is the most fine-grained of the three levels. Rather than characterising the sentiment of an entire text or sentence, it identifies the specific attributes or characteristics of an entity being discussed and determines the sentiment expressed toward each one independently. A hotel review, for example, may express positive sentiment toward location and negative sentiment toward cleanliness; only aspect-level analysis can capture both simultaneously. Liu (2012) notes that this level supports the most expressive and actionable sentiment representations, though it also introduces significant complexity in extraction, modelling and aggregation.

This level is of particular relevance to the present work. The data warehouse proposed in this dissertation operates at aspect-level granularity, meaning that each analytical record represents a sentiment observation associated with a specific service aspect within a review. The choice of this grain, and its analytical consequences, are discussed in detail in Sections 2.3.3 and 4.3.

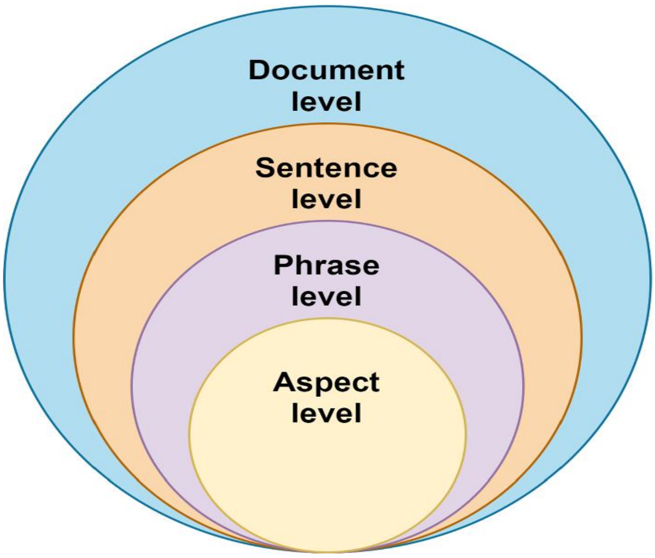

Fig. 1. Levels of Sentiment Analysis (Wankhade, Rao, & Kulkarni, 2022)

2.1.3 Analytical Implications of Granularity

The choice of sentiment analysis level is not merely a technical decision about extraction complexity; it is also a data modelling decision with direct consequences for how sentiment can be stored, queried and interpreted in analytical systems. Each level implies a different unit of analysis, a different record structure and a different set of constraints on what aggregations are valid.

At the document level, each sentiment observation corresponds to one review or post, producing a simple one-to-one relationship between source document and sentiment record. This structure is straightforward to store and aggregate but discards internal variation. When a reviewer expresses satisfaction with the location of a hotel but frustration with its staff, document-level analysis produces a single mixed or averaged sentiment value that obscures both signals. For decision-support purposes, this loss of detail may be acceptable in high-volume, low-depth scenarios such as brand tracking, but it is insufficient when the goal is to understand which specific aspects of a service drive positive or negative perception.

Sentence-level analysis introduces a one-to-many relationship between documents and sentiment records, since a single document may generate multiple sentence-level observations. This increases analytical expressiveness but also introduces new challenges. Sentences do not always map cleanly to distinct aspects of an entity, and the relationship between a sentence-level sentiment record and the broader context of the document it came from must be explicitly preserved to avoid misleading aggregations.

Aspect-level analysis takes this further, producing multiple sentiment observations per document, each associated with a specific attribute or characteristic of the entity being evaluated. This is the most analytically expressive level but also the most structurally demanding. A single review may generate five or six separate aspect-level records, each carrying its own affective values. In a data warehouse context, this fan-out effect must be accounted for carefully: naive aggregation across all records for a given service would weight reviews with more aspects more heavily than reviews with fewer, potentially distorting averages. Analysts working with aspect-level sentiment data must therefore apply appropriate filtering and grouping logic to ensure that aggregations remain semantically valid.

These considerations directly inform the grain declaration of the data warehouse proposed in this dissertation. Aspect-level granularity was selected because it preserves the maximum analytical detail and enables the most expressive queries, at the cost of requiring more careful query design. The implications of this decision are examined further in Section 4.3, where the grain is formally declared, and in Section 4.7, where the measures associated with each aspect-level observation are defined.

2.1.4 Polarity-based Sentiment Representation

Polarity is the most widely studied and practically adopted form of sentiment representation. At its core, it characterises the evaluative orientation of an expressed opinion, typically assigning one of three values: positive, negative or neutral. Liu (2012) describes polarity as reflecting the overall attitude of an opinion holder toward a target entity or aspect, which makes it a compact and interpretable summary of sentiment that scales well to large volumes of text.

Polarity can be expressed in two main forms. Categorical representations assign discrete labels to text segments, offering simplicity and ease of interpretation. Numerical representations express polarity as a continuous score within a defined range, typically from -1 to +1, capturing not just orientation but also the intensity of the expressed sentiment. In practice, both forms are used depending on the application context: categorical labels suit high-level dashboards and reporting, while numerical scores support more expressive analytical operations such as trend analysis, ranking and aggregation (Pang & Lee, 2008); (Wankhade, Rao, & Kulkarni, 2022).

The practical appeal of polarity-based analysis is clear. Polarity labels are easily understood by non-technical stakeholders, straightforward to compute at scale and readily aggregated into summaries, distributions and temporal comparisons. These properties have made polarity the default output of most commercial and research sentiment systems, from reputation monitoring platforms to customer satisfaction dashboards (Shayaa, et al., 2018).

To illustrate typical polarity outputs in the context of tourism and hospitality reviews, Table 1 presents examples of polarity classification applied to short opinionated texts representative of the domain considered in this work.

Table 1. Examples of polarity-based sentiment representation

Textual Opinion	Polarity Label
“The staff were attentive and the room immaculate”	Positive
“The location was convenient but nothing stood out”	Neutral
“"I would not recommend this restaurant to anyone"	Negative

The limitation of polarity-only representations becomes apparent when two reviews sharing the same label are compared more closely. A guest who writes "the room was dirty and the staff were dismissive" and a guest who writes "I expected more from a four-star hotel" both produce negative polarity, yet the first expresses frustration and the second expresses disappointment, states that may call for entirely different operational responses. Birjali, Kasri and Beni-Hssane (2021) note that polarity alone cannot distinguish between fundamentally different affective states sharing the same evaluative orientation, nor can it separate strongly subjective opinions from weakly opinionated or factual statements.

Despite these limitations, polarity remains a foundational component of sentiment analysis. It is the baseline upon which the richer affective representations discussed in the following sections are built, and it is retained as a core measure in the data warehouse model proposed in this dissertation.

2.1.5 Outputs of Sentiment Analysis

Sentiment analysis systems produce structured representations derived from unstructured textual input, encoding affective information in a form that can be interpreted, aggregated and reused in downstream analytical processes. Liu (2012) identifies three principal output formats: categorical labels, numerical scores and probability distributions, each reflecting different trade-offs between simplicity, expressiveness and analytical reusability.

Categorical labels assign each text unit a discrete class, most commonly positive, negative or neutral. Their simplicity makes them easy to interpret and straightforward to aggregate into summaries and distributions, which explains their prevalence in high-level reporting applications. Numerical scores express sentiment as a continuous value within a defined range, enabling finer distinctions between weak and strong opinions and supporting richer analytical operations such as ranking, trend analysis and weighted aggregation (Pang & Lee, 2008); (Shayaa, et al., 2018). Probability distributions go further still, expressing the confidence of a classification across all possible sentiment classes rather than committing to a single value. While distributions carry the most information, they introduce storage and aggregation complexity that makes them less practical in warehouse environments, where consistency and query efficiency are prioritised. For this reason, the data warehouse proposed in this dissertation uses numerical scores rather than probability distributions as its primary sentiment output format.

A property of sentiment outputs that is often overlooked in purely extraction-oriented work is their inherent contextual dependency. A sentiment score carries little analytical meaning in isolation; its value depends on knowing when the sentiment was expressed, by whom, about what entity and toward which specific aspect. This means that even the simplest polarity score is already multidimensional in nature, associated with a set of contextual attributes that must be preserved if the output is to support longitudinal and comparative analysis (Wankhade, Rao, & Kulkarni, 2022).

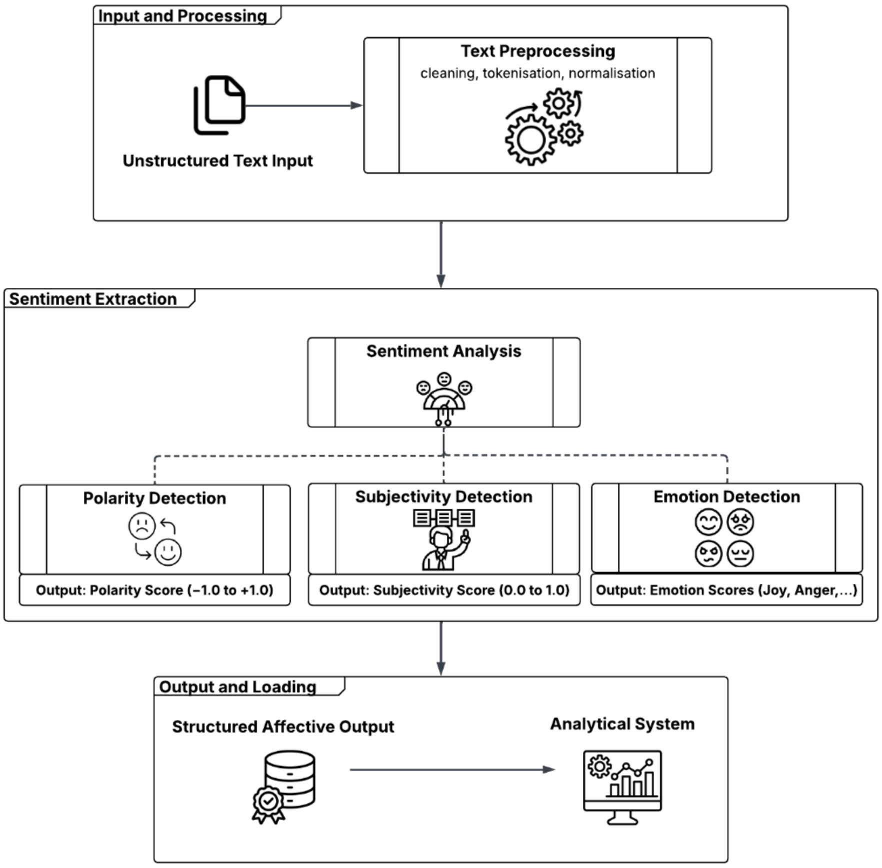

Fig. 2. General sentiment analysis pipeline illustrating the extraction of polarity, subjectivity and emotion from unstructured text and their transformation into structured affective outputs suitable for integration into analytical environments

In practice, however, the contextual richness of sentiment outputs is rarely preserved. Results are commonly aggregated immediately after extraction and presented as dashboard indicators or summary statistics, discarding the temporal, source and aspect information that would make them analytically reusable (Ciocodeică, et al., 2022). This pattern of immediate consumption reflects the broader disconnect between sentiment analysis as a processing task and the requirements of structured analytical environments, a disconnect that the present work directly addresses.

2.1.6 Limitations of Traditional Sentiment Analysis

Despite its maturity and wide adoption, sentiment analysis as conventionally practised exhibits several limitations that become particularly consequential when the goal is not just to extract sentiment but to store, query and reason over it analytically. These limitations are not primarily technical failures of classification accuracy; they concern the way affective information is conceptualised, represented and handled after extraction.

The most fundamental limitation is the reduction of sentiment to a single polarity dimension. As Mohammad S. (2021) argues, affective meaning is inherently multidimensional, encompassing not just evaluative orientation but also emotional type, emotional intensity and the degree of subjectivity with which an opinion is expressed. Collapsing all of this into a positive-to-negative scale means that reviews expressing anger, disappointment or resignation toward the same service aspect receive identical representations, despite carrying different implications for decision-making. This loss of affective detail is not a minor imprecision; it is a structural constraint that limits what questions can be answered from the data.

A related limitation concerns the inability to distinguish subjective opinions from objective or factual statements. Not all text that mentions a product or service expresses a genuine opinion about it, and treating factual descriptions as sentiment-bearing content introduces noise into any downstream analysis. Linguistic phenomena such as sarcasm, irony and domain-specific idioms compound this problem further, as surface-level analysis may systematically misclassify expressions whose meaning depends on context (Liu, 2012). These challenges are particularly relevant in tourism and hospitality reviews, where reviewers frequently mix factual description with evaluative comment within the same sentence.

Beyond representation, there is a structural problem with how sentiment outputs are typically handled after extraction. In most practical applications, sentiment values are computed on demand, immediately aggregated into summary statistics and presented on dashboards, without being systematically stored in a way that preserves their contextual attributes. This means that the temporal dimension of sentiment, how opinions evolve in response to service changes, seasonal patterns or external events, is effectively discarded. So too are the relationships between sentiment and other analytical variables such as reviewer demographics, geographic location or service category. Birjali, Kasri and Beni-Hsanne (2021) and Ciocodeică (2022) both identify this ad hoc treatment as a significant barrier to longitudinal and cross-dimensional sentiment analysis.

Finally, sentiment data is rarely designed for integration with the structured organisational data that surrounds it. Review sentiment exists alongside booking volumes, occupancy rates, pricing data and geographic information, yet most sentiment systems treat these as separate analytical worlds. The absence of a shared modelling framework means that sentiment cannot be systematically correlated with operational metrics, limiting its strategic value (Moalla, Nabli, & Hammami, 2022).

These limitations can be summarised with illustrative examples like so:

Polarity reduction

- **Description:** Sentiment is reduced to positive/negative labels, ignoring emotional diversity and intensity.
- **Illustrative Example:** Two hotel reviews both labelled negative: "The staff were rude and dismissive" (anger) and "I expected more from a four-star property" (disappointment) receive identical representations despite implying different operational responses.

Lack of Subjectivity Modelling

- **Description:** Difficulty distinguishing subjective opinions from objective or factual statements.
- **Illustrative Example:** "The hotel is located 200 metres from the beach" and "The hotel feels disappointingly far from the beach" are both about location, but only the second expresses a genuine opinion.

Context Sensitivity

- **Description:** Sarcasm, irony and domain-specific language reduce reliability of sentiment outputs.
- **Illustrative Example:** "Another unforgettable stay. The noise from the bar kept us awake until 3am" may be misclassified as positive due to the word "unforgettable".

Limited Analytical Reuse

- **Description:** Sentiment outputs are not systematically stored or modelled for longitudinal analysis.
- **Illustrative Example:** Daily average sentiment scores for a restaurant are displayed on a dashboard but discarded after the reporting period, making it impossible to detect whether sentiment toward food quality has declined since a chef change six months earlier.

Weak Integration with Structured Data

- **Description:** Sentiment data is maintained separately from core analytical systems.
- **Illustrative Example:** Negative sentiment toward cleanliness cannot be correlated with housekeeping staffing levels or occupancy rates because sentiment and operational data are stored in disconnected systems

These limitations collectively define the problem this dissertation sets out to address. Richer affective representation requires modelling polarity alongside subjectivity and emotion. Systematic analytical reuse requires a structured storage environment with a well-defined grain and consistent value representations. Integration with structured data requires a dimensional modelling framework that treats sentiment as a first-class analytical measure. Each of these requirements is addressed directly in the design presented in Chapters 3 and 4.

2.2 Beyond Polarity

The limitations of polarity-only sentiment representation identified in Section 2.1 point toward two affective dimensions that conventional sentiment analysis tends to overlook: subjectivity and emotion. This section examines each in turn, establishing what they contribute analytically, how they are modelled theoretically and how they can be detected computationally. It concludes by considering what it means to represent all three affective dimensions in a structured and consistent form suitable for integration into analytical systems.

2.2.1 Subjectivity in Sentiment Analysis

Subjectivity refers to the degree to which a textual expression reflects opinion, belief or evaluation rather than objective or verifiable fact. Where polarity captures the evaluative orientation of an opinion, subjectivity captures whether a genuine opinion is being expressed at all. Liu (2012) distinguishes between subjective text, which conveys private states such as assessments, feelings and beliefs, and objective text, which conveys factual information that can be independently verified. This distinction matters analytically because not all text that mentions a service or product is opinionated: a reviewer who writes "the hotel has 120 rooms and a rooftop terrace" is providing factual information, while one who writes "the hotel felt impersonal and corporate" is expressing a subjective judgment. Treating both identically in a sentiment system introduces noise that affects the reliability of any downstream analysis.

Subjectivity is not a binary property. Textual expressions fall along a continuum from strongly objective to strongly subjective, with many occupying an intermediate position. Mohammad (2021) observes that affective expressions range from weakly opinionated statements to highly charged personal evaluations, and that this gradation carries analytical significance. A cautiously positive review of a restaurant's food differs meaningfully from an enthusiastic one, even if both receive the same polarity label. Representing subjectivity as a continuous numerical dimension, rather than a binary flag, preserves this gradient and supports more expressive analytical operations such as filtering by opinion strength or tracking shifts in reviewer certainty over time.

Subjectivity detection is therefore often treated as a complementary task to polarity classification, enabling sentiment systems to focus analytical attention on genuinely opinion-bearing content and to weight that content appropriately (Birjali, Kasri, & Beni-Hssane, 2021). In the data warehouse proposed in this dissertation, subjectivity is retained as an independent numerical measure associated with each aspect-level observation, allowing analysts to distinguish strongly opinionated sentiment signals from weakly subjective ones when constructing queries or interpreting aggregated results.

Table 2. Examples of objective and subjective textual expressions

Textual Expression	Subjectivity
“The hotel is located 500 metres from Lisbon city centre.”	Objective
"The location could not have been more convenient for exploring the city"	Subjective
"The restaurant serves a three-course set menu for €25"	Objective
"The tasting menu was one of the most memorable dining experiences I have had"	Subjective

Incorporating subjectivity into sentiment analysis enables a clearer separation between factual descriptions and opinionated content, improving the reliability and interpretability of sentiment outputs. When considered alongside polarity and emotion, a subjectivity measure allows analytical systems to distinguish between reviews that express strong personal conviction and those that are more tentative or descriptive in tone, a distinction that is lost entirely when sentiment is reduced to polarity alone. This expressive quality makes subjectivity a meaningful component of the multidimensional affective representation developed in this work.

2.2.2 Emotion as a Multidimensional Affective Representation

Polarity and subjectivity together characterise the evaluative orientation and opinionated nature of a sentiment expression, but neither captures what kind of emotional state is being expressed. Two hotel guests may both leave negative, strongly subjective reviews, yet one is angry about a billing dispute while the other is saddened by a stay that failed to live up to a special occasion. These are not analytically equivalent signals, and a system that cannot distinguish between them is limited in the guidance it can offer to decision-makers. Emotion constitutes the third affective dimension required to produce a complete and analytically useful representation of expressed sentiment (Mohammad S. , 2021).

Unlike polarity, which positions sentiment along a single evaluative axis, emotion is inherently multidimensional. Emotional states differ in type, intensity and psychological function, which has led to the development of theoretical models designed to categorise and structure emotional experience systematically. Two families of models have been particularly influential: categorical models, which define a fixed set of discrete basic emotions, and dimensional models, which represent emotional states as positions along continuous psychological scales.

Ekman (1992) proposed the most widely cited categorical model, identifying six basic emotions (happiness, sadness, anger, fear, disgust and surprise) argued to be universally recognisable across cultures. The appeal of categorical models for computational and analytical work is clear: discrete emotion labels map naturally to structured data representations and are straightforward to store, query and aggregate. Their limitation is that they treat each emotion as a binary presence or absence, losing information about emotional intensity and the possibility that multiple emotions co-occur within the same expression.

Plutchik (1980) addressed this through a psychoevolutionary model in which eight primary emotions (joy, trust, fear, surprise, sadness, disgust, anger and anticipation) are organised according to opposing pairs and arranged at varying intensities. The model explicitly represents the fact that emotions can be stronger or weaker and that adjacent emotions combine to form more complex affective states. Unlike Ekman's categorical approach, Plutchik's framework supports numerical intensity scoring for each emotion rather than binary categorisation, enabling richer representation of emotional variation within a single observation. These properties make it a natural candidate for analytical contexts where emotional intensity and co-occurrence are relevant, as explored further in this work.

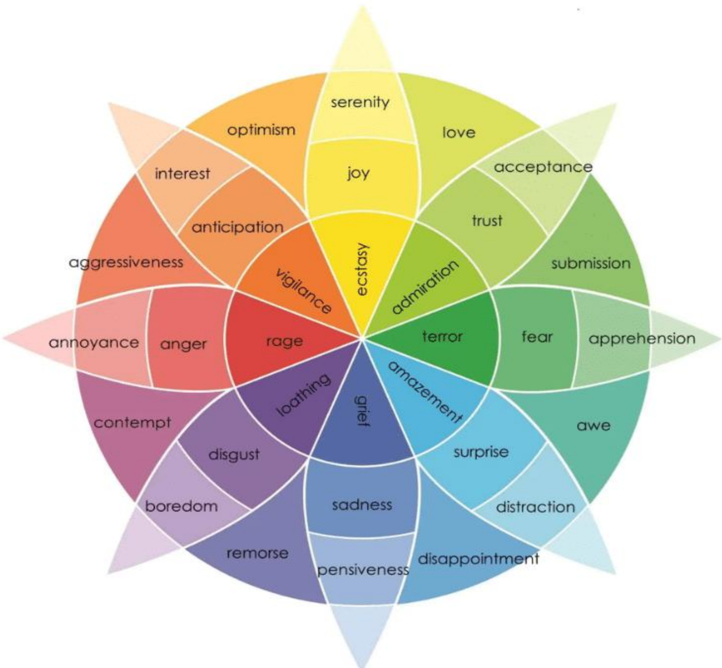

Fig. 3. Plutchik's wheel of emotions, illustrating basic emotions, their opposites and varying intensities. Adapted from (Plutchik, 1980).

The analytical value of emotion-aware representations is most visible precisely where polarity fails. Negative sentiment associated with anger toward a specific service aspect point toward a concrete operational problem requiring attention. Negative sentiment associated with sadness or disappointment may reflect unmet expectations rather than a service failure, suggesting a different response. Capturing these distinctions at the aspect level, as the proposed warehouse does, enables a qualitatively different kind of decision support than is possible with polarity alone.

Emotion modelling does introduce genuine challenges. Multiple emotions may co-occur within the same text, intensities must be normalised to a consistent scale and the reliability of automated emotion detection varies depending on the method used. These challenges are addressed further in this work, where the specific approach to emotion derivation and score normalisation is documented.

2.2.3 Computational Models for Emotion Detection

Theoretical emotion models provide the conceptual vocabulary for describing emotional experience but translating that vocabulary into computable representations requires dedicated detection methods. Computational approaches to emotion detection typically draw on some combination of linguistic resources, semantic knowledge and statistical learning to infer emotional states from text. The choice of method has direct consequences for how emotions are represented in the resulting outputs and how well those outputs serve downstream analytical purposes (Mohammad S. , 2021).

The most widely used approach in analytically oriented applications is lexicon-based emotion detection. In this paradigm, predefined resources map individual words or expressions to emotion categories or intensity scores, enabling emotions to be identified through lexical matching and aggregation. Resources such as the NRC Emotion Lexicon (Mohammad & Turney, 2013) provide mappings between terms and the eight Plutchik emotion categories, producing numerical scores that reflect the intensity of each emotion within a given text. Lexicon-based methods are attractive for analytical integration because their outputs are transparent, reproducible and structurally consistent, properties that are difficult to guarantee with more complex methods. Their principal limitation is sensitivity to context and domain variation: the emotional valence of a word may shift depending on usage, and lexicons built on general-purpose corpora may not fully capture the language patterns of a specific domain such as tourism and hospitality (Birjali, Kasri, & Beni-Hssane, 2021).

Knowledge-based and concept-level approaches attempt to address this limitation by associating emotions with concepts and semantic relationships rather than individual surface-level word forms. Frameworks such as Sentic Computing (Cambria & Hussain, 2015); (Cambria, Poria, Hazarika, & Kwok, 2018) combine common-sense reasoning with affective knowledge to detect emotion in a way that is more robust to linguistic variation and less dependent on exact term matching. The analytical appeal of concept-level approaches is that their outputs tend to be semantically grounded and naturally compatible with structured data models, though their implementation complexity is considerably higher than that of lexicon-based methods.

Machine learning approaches treat emotion detection as a multi-class or multi-label classification problem, learning associations between textual features and emotional categories from labelled training data. These methods frequently achieve strong performance in controlled settings but their outputs are opaque and highly sensitive to the characteristics of the training corpus, which complicates interpretability and reuse across different domains or time periods (Poria, Cambria, Bajpai, & Hussain, 2017). For applications where the primary goal is analytical reuse and longitudinal consistency rather than real-time classification accuracy, the transparency and reproducibility advantages of lexicon-based methods generally outweigh the performance gains of machine learning classifiers.

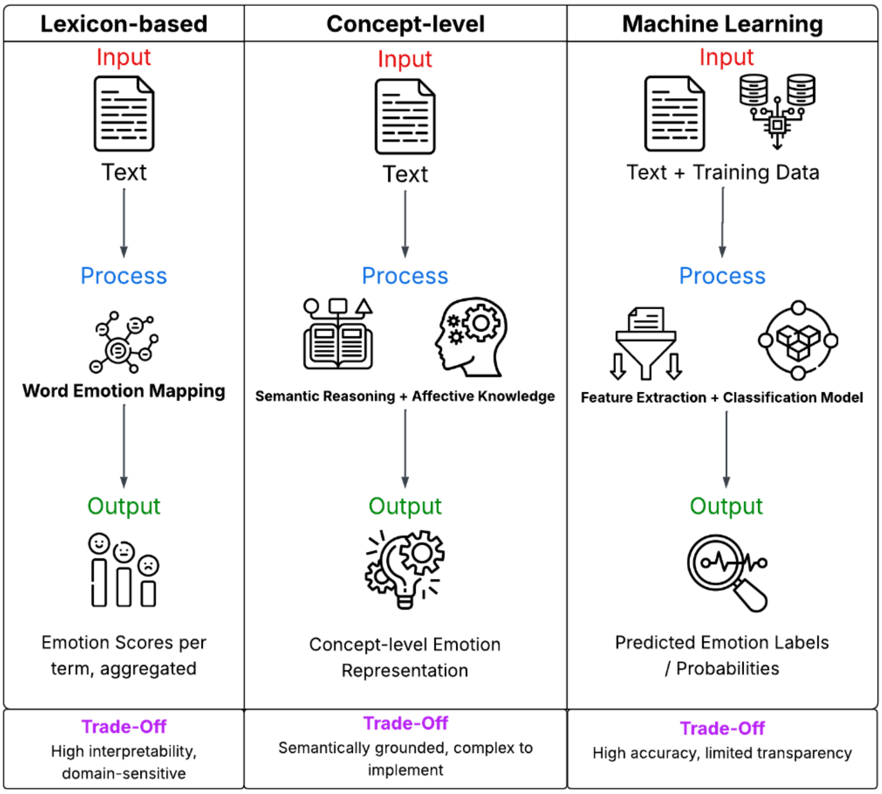

Fig. 4. Computational approaches to emotion detection, comparing lexicon-based, concept-level and machine learning methods across their processing characteristics and analytical trade-offs

Figure 4 summarises the three approaches comparatively, illustrating how they differ in their input requirements, processing logic and output characteristics. This comparison is useful for understanding the analytical trade-offs involved in selecting an emotion detection method for integration into a structured analytical environment.

The choice between these approaches is not purely technical as it also shapes the structure and reusability of the emotional data produced. Outputs may take the form of categorical labels, numerical intensity scores or probability distributions over emotion categories, each carrying different implications for storage, aggregation and query design. In analytically oriented systems, interpretability, output consistency and structural compatibility with the broader data model are often more decisive criteria than raw classification accuracy, a consideration that informs the approach adopted in this work.

2.2.4 Structured Representation of Affective Dimensions

The theoretical and computational foundations discussed in the preceding sections establish polarity, subjectivity and emotion as the three primary affective dimensions of sentiment. For these dimensions to support analytical processing rather than one-off extraction, they must be represented in a structured, consistent and well-defined manner. Ad hoc output formats, such as varying scales, inconsistent category definitions and tool-specific encodings, make it impossible to aggregate, compare or query sentiment data reliably across sources or time periods. Structuring affective dimensions explicitly is therefore not a secondary concern; it is a prerequisite for treating sentiment as analytical data.

Each affective dimension can be expressed in categorical or numerical form, and the choice between them carries analytical consequences. Polarity is commonly represented either as a discrete label (positive, negative or neutral) or as a continuous score within a defined range such as -1.0 to +1.0. Categorical labels are intuitive and easy to interpret, making them well suited to filtering and grouping operations. Numerical scores support more expressive operations including trend analysis, ranking and weighted aggregation, at the cost of requiring more careful interpretation (Liu, 2012); (Mohammad S. , 2021). Subjectivity follows a similar pattern: it can be modelled as a binary distinction between subjective and objective content, or as a continuous value capturing the degree of opinionated expression. Emotion presents the widest range of representational options, from discrete category labels corresponding to basic emotion types to numerical intensity scores associated with each emotion in a predefined set (Plutchik, 1980); (Ekman, 1992).

A critical property of affective dimensions in this context is that they are not mutually exclusive. A single review sentence may simultaneously carry a polarity orientation, a measurable degree of subjectivity and detectable intensities across multiple emotion categories. Representing these dimensions as separate but co-occurring structured attributes, rather than as isolated outputs of independent processing steps, enables them to be combined, filtered and aggregated within a unified analytical model (Cambria & Hussain, 2015); (Cambria, Poria, Hazarika, & Kwok, 2018). This is precisely the approach adopted in the data warehouse proposed in this dissertation, where polarity, subjectivity and each of the eight Plutchik emotion categories are stored as independent numerical measures associated with the same aspect-level observation.

Defining these representations explicitly at the modelling stage ensures that sentiment values carry consistent meaning regardless of which source document, time period or service they are associated with. This semantic consistency is what elevates affective information from a processing artefact to a reliable analytical asset, capable of supporting the kind of structured, historical and multidimensional analysis that the proposed warehouse is designed to enable.

2.2.5 Implications for Analytical and Decision-Support Systems

Treating polarity, subjectivity and emotion as co-occurring structured dimensions rather than independent outputs changes what is analytically possible. The shift is not merely one of expressiveness but a shift in the kinds of questions a decision-support system can answer.

Consider the difference between two queries over the same body of hospitality reviews. The first asks: what is the average sentiment toward cleanliness across all properties this quarter? The second asks: among reviews that express negative sentiment toward cleanliness, what proportion are driven by anger rather than mild dissatisfaction, and has that proportion changed since last quarter? The first query is possible with polarity alone. The second requires subjectivity scores to filter for strongly opinionated content and emotion scores to distinguish anger from disappointment. Neither dimension is redundant; each one opens a different analytical window onto the same underlying data.

This expressive gain has direct implications for how sentiment-aware systems should be designed. Sentiment dimensions must be stored as co-occurring attributes of the same observation rather than as separate outputs of independent processing pipelines. They must be normalised to consistent scales so that comparisons across sources and time periods are valid. And they must be integrated within a data model that allows them to be queried alongside contextual dimensions such as time, service category, geographic location and reviewer segment. As Ciocodeică et al. (2022) observe, sentiment outputs are frequently treated as auxiliary dashboard indicators rather than as core analytical elements, precisely because the infrastructure to support richer querying has not been put in place.

The practical consequence of this gap is that organisations collecting large volumes of affective data are unable to exploit it strategically. Moalla, Nabli and Hammami (2022) demonstrate that sentiment-aware analytical architectures, those that model affective dimensions consistently and integrate them with operational data, enable substantially more flexible and context-sensitive decision support than ad hoc sentiment dashboards. The present work is positioned as a contribution to closing this gap, through the design of a dimensional data warehouse in which polarity, subjectivity and emotion are modelled as first-class analytical measures from the ground up.

2.3 Sentiment as Analytical Data

The preceding sections examined what sentiment is, how it can be extracted and what richer affective representations look like. What they have not yet addressed is a more fundamental question: what does it mean to treat sentiment as analytical data? The answer has consequences for every modelling decision that follows, from how values are normalised and at what granularity observations are recorded, to what contextual dimensions they must carry and what an analytical system needs to do to exploit them effectively.

2.3.1 Sentiment as Structured Analytical Information

Section 2.2.5 established that treating polarity, subjectivity and emotion as co-occurring dimensions changes what analytical questions can be answered. This section examines the structural implications of that shift more precisely: if sentiment is to function as analytical data, what properties must it have, and what does that mean for how it is modelled and stored?

The starting point is a distinction that is often overlooked in practice. Sentiment analysis is frequently used as a processing step rather than as a data-producing activity. Results are computed on demand, surfaced in a dashboard and discarded once the reporting cycle ends. No persistent record is kept of which aspect was evaluated, at what time, by which reviewer, on which platform or with what degree of subjectivity. This matters because without that contextual record, sentiment values cannot be compared across time periods, cannot be filtered by reviewer segment and cannot be correlated with operational data. The information has been extracted but not preserved in a form that supports analysis (Liu, 2012); (Birjali, Kasri, & Beni-Hssane, 2021).

Treating sentiment as analytical data requires a different approach. Each sentiment observation should be modelled as a structured record characterised by a set of well-defined attributes: the affective values themselves (polarity, subjectivity and emotion scores), the temporal context in which the sentiment was expressed, the source from which it was extracted, the entity or service it refers to and the specific aspect to which it pertains. This is not a novel idea in data management, it is how transactional and operational data has been modelled for decades. The novelty lies in applying the same discipline to affective information, which has traditionally resisted structured treatment due to its linguistic and subjective nature (Shayaa, et al., 2018).

When modelled in this way, sentiment data becomes persistent, queryable and combinable with other analytical dimensions. It can support longitudinal trend analysis, comparative evaluation across services or regions and historical reconstruction of how public perception evolved in response to specific events. These capabilities are precisely what the data warehouse proposed in this dissertation is designed to provide, and the modelling decisions described in subsequent sections (normalisation, granularity selection and dimensional schema design) all follow from this foundational commitment to treating sentiment as a first-class analytical asset.

2.3.2 Normalisation and Standardisation of Sentiment Values

Sentiment analysis systems vary considerably in the formats and scales they use to express their outputs. Polarity may be expressed as a categorical label, a score in the range -1.0 to +1.0, a score in the range 0.0 to 1.0 or a probability distribution over sentiment classes. Subjectivity may be modelled as a binary flag or as a continuous value. Emotion may be represented as discrete category labels, as intensity scores tied to a fixed emotion set or as multi-label annotations. When sentiment data from different sources or tools is combined without addressing these differences, comparisons become meaningless and aggregated results unreliable (Liu, 2012); (Birjali, Kasri, & Beni-Hssane, 2021).

Normalisation addresses the scale and range problem by mapping sentiment values to well-defined, bounded domains. For polarity, this means defining a consistent numerical range with clearly specified semantics at its boundaries. For subjectivity, it means mapping to a bounded interval where 0.0 represents fully objective content and 1.0 represents maximally subjective expression. For emotion, it means expressing each emotion's intensity as a score within a consistent range, regardless of which detection tool produced it. The goal is not uniformity for its own sake but analytical operability: values that occupy consistent ranges can be aggregated, ranked and compared across sources, time periods and services without introducing scale-dependent bias (Mohammad S. , 2021).

Standardisation addresses a related but distinct problem: semantic consistency. Even when values occupy the same numerical range, they may not mean the same thing across different extraction methods or application domains. A polarity score of 0.7 produced by one tool may reflect a different threshold of positive sentiment than the same score produced by another. Establishing explicit coding conventions and value semantics, documenting precisely what each value means and how it was derived, is therefore a prerequisite for reliable multi-source sentiment integration (Shayaa, et al., 2018).

Table 3 illustrates the normalised value conventions adopted for each affective dimension in the data warehouse proposed in this dissertation.

Table 3. Normalised value conventions for affective dimensions in the proposed warehouse

Affective Dimension	**Normalised Range**	**Boundary Semantics**	**Representation in Warehouse**
Polarity	-1.0 to +1.0	• -1.0 = maximally negative; • 0.0 = neutral; • +1.0 = maximally positive	Decimal numerical score (polarity_score)
Subjectivity	0.0 to 1.0	• 0.0 = fully objective; • 1.0 = maximally subjective	Decimal numerical score (subjectivity_score)
Emotion (each of 8)	0.0 to 1.0	• 0.0 = emotion not expressed; • 1.0 = maximum intensity	Eight separate decimal scores (joy_score, trust_score, etc.)
Confidence	0.0 to 1.0	• 0.0 = no confidence; • 1.0 = maximum confidence	Decimal numerical score (confidence_score)

Together, normalisation and standardisation transform sentiment from loosely defined processing outputs into analytically operable variables. They make it possible to treat polarity, subjectivity and emotion scores as proper numerical measures within a dimensional data model, supporting the full range of analytical operations (aggregation, filtering, ranking and trend detection) that structured analytical environments are designed to enable.

2.3.3 Granularity and Data Grain of Sentiment Information

Granularity was introduced in Section 2.1.3 as a property of sentiment extraction, distinguishing document-level, sentence-level and aspect-level analysis. Here it is examined from a strictly different angle: as a data modelling decision that determines the structure of every record in the analytical system and constrains every aggregation performed over it.

The concept of data grain defines what a single record in a dataset represents. In a sentiment-aware analytical system, declaring the grain means answering a precise question: does one record represent the overall sentiment of a review, the sentiment of a sentence within that review, or the sentiment expressed toward a specific aspect within a sentence? Each answer implies a fundamentally different schema, a different relationship between source documents and stored observations and a different set of valid analytical operations.

Document-level grain is the simplest choice. Each review produces exactly one sentiment record, making aggregation straightforward and storage compact. The cost is expressiveness: a review that praises the location of a hotel while criticising its service quality collapses into a single mixed sentiment value, losing the distinction between two analytically meaningful signals. Aspect-level grain preserves that distinction by producing multiple records per document, one for each identified aspect, each carrying its own affective measures. This is the richest and most analytically expressive option, but it introduces a one-to-many relationship between source documents and fact records that must be explicitly managed (Birjali, Kasri, & Beni-Hssane, 2021).

One consequence of aspect-level grain that deserves attention is the risk of unintended weighting in aggregations. A review containing five aspect-level observations contributes five times as much to a service-level average as a review containing only one. Analysts working with aspect-level data must therefore apply appropriate grouping logic to ensure that aggregations over service sentiment are not inadvertently dominated by reviews with high aspect density. This is not a reason to avoid aspect-level grain, as its analytical benefits outweigh this complexity, but it is a modelling constraint that must be acknowledged and addressed in query design.

Table 4 summarises the three granularity levels and their principal analytical implications.

Table 4. Granularity levels of sentiment data and analytical implications

Granularity Level	Data Grain	Analytical Implications
Document-level	One sentiment record per document	Straightforward aggregation and compact storage, but internal variation across aspects or sentences is lost
Sentence-level	One sentiment record per sentence	Captures within-document variation, but sentence boundaries do not always align with meaningful analytical units

Aspect-level	One sentiment record per aspect per document	Maximum analytical expressiveness and aspect-specific querying, but introduces one-to-many relationships and requires careful aggregation logic

The grain declaration is one of the most consequential decisions in the design of a sentiment-aware data warehouse. Once established, it shapes the schema, the ETL process and the valid interpretation of every query result. The choice made in this work is motivated by the analytical requirements established in Section 2.3.5 and formalised in the dimensional model developed later in this work.

2.3.4 Multidimensional Nature of Sentiment Data

Sentiment observations do not exist in a vacuum. A polarity score of -0.6 carries different analytical meaning depending on when it was recorded, which service it refers to, which aspect of that service it concerns and on which platform the review appeared. These contextual attributes are the dimensions along which sentiment data becomes comparable, filterable and analytically useful. A system that stores sentiment scores without preserving their contextual associations produces data that cannot be meaningfully queried beyond the most basic aggregations.

The contextual dimensions typically associated with sentiment observations include temporal information, the source or platform from which the text was extracted, the entity being evaluated and the specific aspect to which the sentiment pertains. Each of these dimensions enables a different class of analytical query. The temporal dimension supports trend analysis and longitudinal comparison. The entity and aspect dimensions support drill-down from service-level to feature-level sentiment. The source dimension supports cross-platform comparison and bias detection. Together they provide the contextual scaffolding within which affective values acquire analytical meaning (Liu, 2012).

The affective dimensions themselves add a second layer of analytical structure. These dimensions characterise the nature of the expressed sentiment rather than its context, and they operate at the same level of granularity as the observation they describe. Each one contributes information the others cannot supply: polarity establishes whether the sentiment is positive or negative, subjectivity indicates how strongly opinionated the expression is and emotion identifies which specific affective states are present. A query that uses all three simultaneously can answer questions that none of them could address alone (Mohammad S. , 2021).

Table 5 organises these dimensions by type, illustrating their analytical roles within a structured sentiment data model.

Table 5. Contextual and affective dimensions associated with sentiment data

Dimension Type	Example Dimensions	Analytical Role
Contextual	Time, Entity, Aspect	Provide analytical context and enable comparison across periods, services and features

Affective	Polarity, Subjectivity, Emotion	Characterise the expressed sentiment across its three analytical dimensions
Source	Platform, Document Type, Reviewer Segment	Enable cross-source comparison, bias detection and demographic filtering

Structuring sentiment data around these dimensions aligns it with the analytical paradigms used in multidimensional data warehousing, where facts are always interpreted in relation to the dimensional context that surrounds them. The dimensional model developed later in this work operationalises this structure directly, mapping each dimension type identified here to a concrete dimension table within the warehouse schema.

2.3.5 Analytical Requirements for Structured Sentiment

The preceding subsections have examined sentiment from four complementary angles: as structured information that must be persistently modelled, as data requiring consistent normalisation, as observations with a well-defined grain and as inherently multidimensional content. This section draws those threads together by identifying the analytical requirements that a structured sentiment representation must satisfy if it is to support decision-making rather than just information retrieval.

Five requirements emerge from the preceding discussion.

Comparability

For sentiment values to support reliable analysis, they must carry consistent meaning regardless of when they were recorded, which source they came from or which extraction method produced them. A system that aggregates polarity scores derived from different tools without normalising their scales may produce results that reflect measurement variation rather than genuine differences in how services are perceived. Standardised encodings and a clearly declared grain are the minimum conditions under which aggregated sentiment indicators can be trusted (Liu, 2012); (Birjali, Kasri, & Beni-Hssane, 2021).

Aggregability

Storing sentiment values is not sufficient if those values cannot be validly summarised. Aggregation operations on sentiment data are only meaningful when the grain is explicitly declared, and normalisation conventions are consistently applied. Without these constraints, averaging across records of different types or scales produces numbers that are arithmetically plausible but semantically misleading.

Context Preservation

A sentiment score carries no analytical meaning in isolation. Its value depends entirely on knowing when the opinion was expressed, which service it concerned and which specific aspect it addressed. Without those associations, the score cannot be filtered by service category, compared across time periods or placed in relation to operational data. Context is not metadata appended to sentiment; it is the condition under which sentiment becomes analytically usable (Shayaa, et al., 2018).

Longitudinal Analysis

Much of the strategic value of sentiment data lies not in any single observation but in how opinion evolves over time. Detecting whether guest satisfaction with a particular aspect has declined following a management change, or whether emotional responses to a service category shift with the seasons, requires that observations be persistently stored with explicit temporal context. Sentiment treated as a disposable reporting signal cannot support this kind of analysis.

Integrability

Sentiment data does not exist in isolation from the operational environment that surrounds it. Review sentiment is collected alongside booking volumes, occupancy rates, pricing data and geographic information, and its strategic value depends on being queryable in combination with these dimensions. This requires that affective measures conform to the same modelling principles applied to conventional analytical data, supporting consistent joins and aggregations across heterogeneous sources. The absence of this integration is one of the most consistently identified limitations in existing sentiment-aware analytical systems (Moalla, Nabli, & Hammami, 2022); (Ciocodeică, et al., 2022).

These requirements are not independent. Aggregability depends on comparability. Longitudinal analysis depends on context preservation. Integrability depends on all four of the preceding requirements having been satisfied. They form a layered set of constraints that together define what it means for sentiment data to be analytically operable rather than merely available.

Data warehousing, examined in the following section, provides precisely the kind of infrastructure these requirements call for: a structured, historically maintained and multidimensionally organised environment in which sentiment data can be stored, versioned and queried with the same rigour applied to any other analytical asset.

2.4 Data Warehousing

Structured sentiment data, as Section 2.3 established, imposes a set of requirements that go well beyond extraction accuracy. It needs to be persistently stored, consistently normalised, declared at an appropriate grain and embedded within a contextual framework that allows it to be queried and compared. Data warehousing has been the dominant paradigm for exactly this kind of analytical environment for three decades, and its core design principles align closely with what structured sentiment requires.

2.4.1 Data Warehousing: Purpose and Characteristics

Data warehousing is the discipline concerned with consolidating data from multiple heterogeneous sources into a consistent, historically maintained environment designed specifically for analytical querying rather than transactional processing. Where operational databases are optimised for inserting, updating and retrieving individual records in real time, data warehouses are built to support complex aggregations, longitudinal comparisons and multidimensional exploration across large volumes of data. Their primary purpose is to give decision-makers a stable, integrated and historically grounded view of organisational information (Inmon, 2002)

Inmon (2002) characterises a data warehouse through four defining properties. Subject orientation means that data is organised around analytical subjects such as customers, services or events rather than around the application processes that generated it. Integration means that data arriving from different source systems is reconciled into consistent representations, with unified naming conventions, formats and semantics. Time variance means that the warehouse retains historical snapshots rather than overwriting current values, enabling analysts to examine how measures have changed over time. Non-volatility means that once data is loaded it is not modified or deleted in the normal course of operation, ensuring the stability and reproducibility of analytical results.

These four properties collectively distinguish data warehouses from the operational systems that feed them and from the ad hoc analytical environments (spreadsheets, reporting databases, disconnected dashboards) that they are designed to replace. For the purposes of this work, the most consequential of the four is time variance. Sentiment data is inherently dynamic: public perception of a service changes in response to operational decisions, seasonal patterns and external events. A warehouse that preserves the historical record of sentiment observations, rather than overwriting them with the most recent values, is a prerequisite for the kind of longitudinal sentiment analysis this dissertation aims to support.

Two major methodological traditions have shaped how data warehouses are designed. Inmon's enterprise warehouse approach advocates for a centralised, normalised repository from which subject-specific data marts are derived. Kimball's dimensional modelling approach, by contrast, advocates for building subject-oriented dimensional schemas directly, organising data around fact tables and surrounding dimension tables in a structure optimised for analytical querying (Kimball & Ross, 2013). The relative merits of these approaches are examined in the following subsection, where the choice of Kimball's methodology for the present work is justified.

2.4.2 Multidimensional Modelling Concepts

Multidimensional modelling is the design methodology through which data warehouse schemas are constructed. Its central organising principle is the separation of analytical content into two distinct types of tables:

- **fact tables:** which store quantitative observations of analytical interest;
- **dimension tables:** which provide the contextual axes along which those observations are interpreted;

A fact table recording sales transactions, for example, would be surrounded by dimensions for time, product, customer and location, allowing analysts to answer questions such as: how did sales of a particular product category perform in a specific region during the previous quarter? This separation of measures from context is what makes multidimensional schemas both analytically expressive and computationally efficient (Kimball & Ross, 2013).

The most widely used schema pattern in multidimensional modelling is the star schema, in which a central fact table is directly joined to a set of denormalised dimension tables. The resulting structure is simple, readable and optimised for the kinds of aggregative queries that analytical systems must support. Snowflake schemas normalise the dimension tables further, reducing storage redundancy at the cost of increased join complexity. For analytical workloads, where query performance and interpretability are the primary concerns, star schemas are generally preferred (Kimball & Ross, 2013).

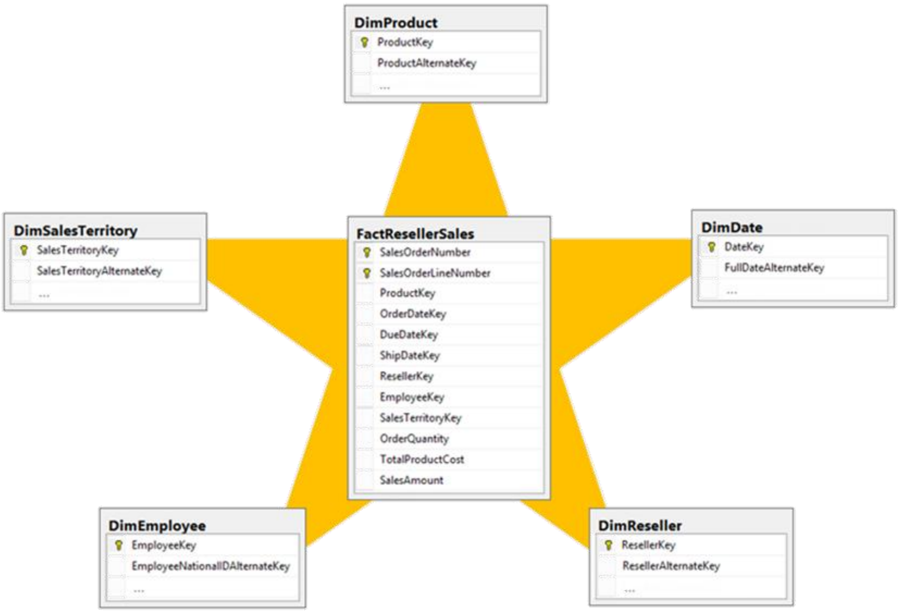

Fig. 5. Conceptual representation of a star schema, illustrating the relationship between a central fact table and surrounding dimension tables (Fernando, 2020)

The schema shown in Figure 5 uses a sales domain for illustration, but the star schema pattern is domain-agnostic. The same structure can be applied to any analytical subject where quantitative observations need to be examined across multiple contextual dimensions. In the context of sentiment-aware analytics, the fact table would store affective measures associated with individual observations, while the surrounding dimensions would capture the temporal, entity-level, source and aspect context required to make those measures analytically meaningful. The multidimensional modelling approach therefore provides a natural and well-validated framework for the kind of structured sentiment analysis this work aims to support.

Multidimensional schemas also support a well-established set of OLAP operations: slicing to filter observations by a specific dimension value, dicing to apply multiple simultaneous filters, drilling down to finer levels of granularity and rolling up to higher levels of aggregation. These operations are precisely what is needed to explore sentiment data analytically. For instance, they allow comparing emotional responses across service types, tracking polarity trends over time or identifying which aspects of a property generate the most negative subjectivity in a given season.

The choice between the two principal approaches to data warehouse design (Inmon's enterprise warehouse model and Kimball's dimensional modelling approach) has direct consequences for how a sentiment-aware system would be structured. Inmon advocates for a centralised, fully normalised enterprise data warehouse from which subject-specific data marts are derived, prioritising data consistency and enterprise-wide integration. Kimball advocates for building dimensional schemas directly, prioritising analytical usability, query performance and incremental deliverability. For the present work, Kimball's approach is adopted. The analytical requirements established in Section 2.3.5 call for a system that is directly queryable, dimensionally expressive and practically implementable within the scope of a single research project. Kimball's star schema pattern satisfies all three criteria, while Inmon's enterprise-first architecture would introduce infrastructure complexity that is disproportionate to the scope and goals of this work.

2.4.3 Temporal Analysis and Historical Data Management

Time occupies a privileged position in data warehouse design. While every dimension provides analytical context, the time dimension is distinctive in that it is present in virtually every analytical query: aggregating by period, comparing across intervals, detecting trends and identifying seasonality all require temporal structure. Kimball and Ross (2013) observe that the date dimension is typically the most heavily used dimension in any data warehouse, a pattern that reflects the centrality of temporal reasoning to analytical decision-making.

In practice, temporal information in a data warehouse is modelled through a dedicated date dimension that exposes multiple levels of granularity within a single structure. A well-designed date dimension allows analysts to aggregate observations at the day, week, month, quarter or year level without requiring separate queries or schema changes, simply by selecting the appropriate attribute. As illustrated in Figure 6, each granularity level enables a qualitatively different class of analytical query: annual aggregation supports year-over-year trend comparison, seasonal grouping enables the detection of recurring patterns, period comparison supports month-level performance tracking, rolling analysis enables moving average calculations and event-level granularity allows individual observations to be examined in relation to specific dates or occurrences. For sentiment data specifically, this hierarchical structure enables questions that a flat timestamp field could never support efficiently.

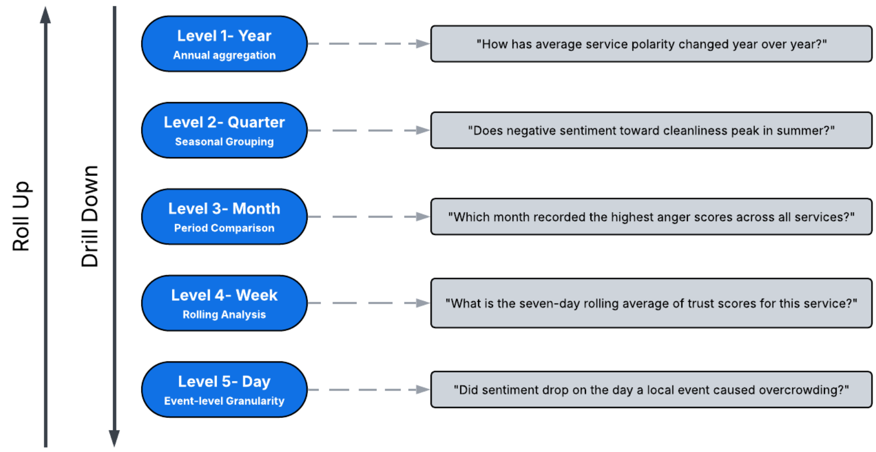

Fig. 6. Time dimension hierarchy illustrating how temporal granularity levels enable progressively detailed analytical queries in a data warehouse environment

A more complex temporal challenge arises when the entities being analysed are themselves subject to change over time. A hotel that changes ownership, a restaurant that relocates or a service that rebrands presents a modelling problem: historical sentiment observations associated with the old entity must remain analytically valid while new observations reflect the updated attributes. The standard solution in dimensional modelling is the Slowly Changing Dimension, which tracks attribute changes by versioning dimension records rather than overwriting them. Type 2 Slowly Changing Dimensions, the most commonly used variant, maintain a full history of attribute changes by creating new dimension records for each change and flagging the current version explicitly (Kimball & Ross, 2013). This mechanism is directly relevant to the proposed warehouse, where service-level attributes such as ownership, category and location may change over the analytical period covered.

Together, the date dimension and slowly changing dimension patterns provide the temporal infrastructure through which a data warehouse moves beyond static snapshots toward a genuinely historical analytical environment, one capable of reconstructing past states, tracking evolution and supporting the kind of longitudinal sentiment analysis that ad hoc approaches cannot deliver.

2.4.4 Conceptual Design versus Implementation Perspectives

Designing a data warehouse involves moving through three successive levels of abstraction, each addressing different concerns and serving different purposes. Golfarelli and Rizzi (2009) articulate this progression through the conceptual, logical and physical design perspectives, a distinction that is particularly important when the system being designed must accommodate enriched or non-traditional data such as sentiment measures alongside conventional analytical content.

The conceptual design perspective is concerned with capturing analytical requirements and domain semantics independently of any technological constraints. At this level the focus is on identifying what entities, measures and relationships are analytically relevant, not on how they will be stored or queried. Conceptual models serve as a shared reference between stakeholders and designers, ensuring that the analytical intent of the system is clearly articulated before any structural or technical decisions are made. For a sentiment-aware warehouse, the conceptual model must represent not only conventional entities such as services, locations and time periods but also the affective dimensions and their relationship to the aspects and reviews from which they are derived.

The logical design perspective translates the conceptual model into a specific data modelling paradigm. In the case of dimensional modelling, this means mapping analytical entities to fact and dimension tables, defining their relationships and specifying the grain of the fact table. Logical models remain independent of physical storage considerations, but they introduce structural constraints that shape how the system will behave analytically. The choice of star schema over snowflake, for instance, is a logical design decision with implications for query performance and usability that persist through to implementation.

The physical design perspective addresses the implementation-specific concerns that determine how the logical model is realised in a particular database environment: storage formats, indexing strategies, partitioning schemes and query optimisation mechanisms. These decisions are typically deferred until the logical model is stable, since premature physical optimisation risks introducing constraints that limit analytical flexibility later.

Figure 7 illustrates this progression through successive refinements of the same analytical domain, showing how an abstract conceptual representation is translated into a logical schema and ultimately into a physical implementation.

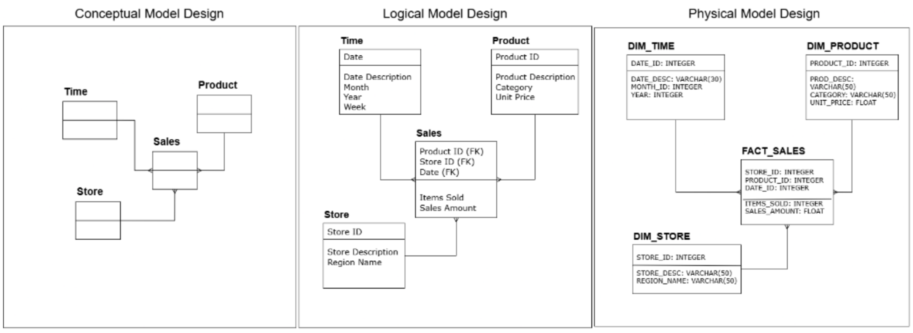

Fig. 7. Illustration of conceptual, logical and physical design perspectives

Maintaining this separation of concerns throughout the design process is not merely a methodological convention, as it has practical consequences for the quality and adaptability of the resulting system. A warehouse designed by working carefully through all three levels is more likely to remain aligned with its analytical requirements as those requirements evolve, and more capable of accommodating new data sources or affective dimensions without requiring structural redesign. This is the approach followed in this dissertation, where conceptual modelling in Chapter 3 precedes and informs the logical and physical design decisions developed subsequently.

2.4.5 Relevance of Data Warehousing for Sentiment-Aware Analytics

The four subsections of Section 2.4 have established the core principles of data warehousing: subject-oriented integration, historical consistency through time variance and non-volatility, multidimensional schema design and the disciplined separation of conceptual, logical and physical concerns. Taken individually, each of these properties addresses a limitation identified in Section 2.3.5 for structured sentiment data. Taken together, they describe an analytical infrastructure that is well matched to the requirements of sentiment-aware analytics in ways that ad hoc processing environments are not.

The integration capability of data warehouses addresses the integrability requirement directly. Sentiment data derived from unstructured text sources does not exist in isolation, as it acquires analytical value when examined alongside conventional business dimensions such as service category, geographic location, reviewer segment and time period. Data warehousing principles of semantic consistency and conformed dimensions provide precisely the mechanisms needed to join affective measures with these contextual dimensions in a reliable and repeatable way (Kimball & Ross, 2013).

The time-variant nature of data warehouses addresses the longitudinal analysis requirement. Tracking how guest sentiment toward a specific service aspect evolves over successive quarters, or how emotional responses shift following a change in management, requires that historical observations be preserved rather than overwritten. Ad hoc sentiment dashboards, which typically display only current or recent aggregates, cannot support this kind of analysis. A properly designed data warehouse can (Inmon, 2002).

The multidimensional modelling paradigm addresses the aggregability and context preservation requirements simultaneously. By separating affective measures from the dimensional context that gives them meaning, a star schema allows analysts to slice sentiment by time, filter by service category, drill down from regional to property level and compare emotional profiles across reviewer segments, all within a single coherent analytical model.

The challenge, and the motivation for the present work, is that these properties do not extend naturally to affective data under conventional warehouse designs. Traditional data warehousing practices have been built around structured, numerical and transactional content, and the modelling conventions they have established do not readily accommodate the multi-dimensional, linguistically derived outputs of sentiment analysis. Integrating polarity, subjectivity and emotion into a dimensional warehouse requires deliberate modelling decisions that go beyond simply adding a sentiment column to an existing fact table. It is precisely these decisions that the design work presented in this dissertation addresses.

2.5 Existing Approaches to Sentiment-Aware Analytical Systems

The previous sections built the theoretical case for what a sentiment-aware analytical system should look like: affective dimensions modelled at aspect-level grain, normalised to consistent scales, embedded within a dimensional infrastructure that connects them to service, time and reviewer context. What remains is to ask how far existing work has moved in this direction. The survey that follows is not concerned with sentiment analysis as a classification problem, but with how sentiment outputs have been stored, modelled and made available for analytical querying.

2.5.1 Sentiment Integration in Business Intelligence and Reporting Environments

The most widespread approach to connecting sentiment analysis with analytical decision-making involves embedding sentiment outputs into existing business intelligence pipelines. In this model, sentiment classification is treated as a preprocessing step: text is collected, polarity is extracted, scores are aggregated and the resulting summaries are surfaced in dashboards or reporting panels alongside conventional operational metrics. The practical appeal of this architecture is considerable. It demands no structural changes to existing analytical infrastructure, introduces no new modelling concerns and produces outputs that are immediately legible to business users. Sentiment values are treated as additional columns in reporting tables rather than as elements of a purposefully designed analytical model, which means the integration can be achieved with minimal friction (Borukhson, Glaum, Weinhardt, & Fegert, 2025).

The analytical cost of this convenience is significant. Dashboard-oriented architectures are designed for consumption rather than exploration. Aggregated summaries are computed and displayed, but the granular observations from which those summaries were derived are not preserved in any structured form. Once the reporting cycle closes, it becomes impossible to re-aggregate at a different level of granularity, apply a filter that was not anticipated at design time, or reconstruct the historical distribution of sentiment values for a specific aspect and time period. The observation that drives the dashboard is discarded the moment it has been presented. This treatment of sentiment as a transient reporting signal rather than a persistently stored analytical measure is a structural choice with lasting analytical consequences, and one that several studies on sentiment adoption in organisational contexts have identified as a key barrier to deriving strategic value from affective data (Reddy, Agarwal, Parashar, & Arora, 2025).

A further limitation concerns representational scope. In virtually all dashboard-oriented implementations, polarity is the only affective dimension captured. Subjectivity and emotion are absent not because they lack analytical value, as Section 2.2 made clear, but because the reporting model has no structural place for them. A system that displays average sentiment toward a service aspect cannot distinguish whether negative scores reflect mild dissatisfaction or sustained anger, cannot filter for strongly opinionated content or identify factual statements that should not carry sentiment weight, and cannot track how the emotional character of negative feedback shifts in response to operational changes. These are precisely the questions that a richer affective representation, stored at the right granularity within a properly designed dimensional model, would support. The reduction of sentiment to polarity in reporting environments is not a deliberate design choice so much as a structural default, one that forecloses analytical possibilities that the organisations collecting this data may not even be aware they are losing (Hong, et al., 2025); (Chai, Xie, & Qin, 2025).

2.5.2 Sentiment-Augmented Data Warehouse Architectures

A smaller but more structurally ambitious body of work has moved beyond reporting pipelines to propose data warehouse designs that incorporate sentiment or opinion measures directly into dimensional schemas. These proposals share a recognition that the limitations of dashboard-oriented approaches are architectural rather than superficial, and that addressing them requires applying dimensional modelling discipline to affective data from the ground up. The work of Moalla, Nabli and Hammami (2022) represents one of the more developed examples of this direction. Their proposed architecture constructs independent data mart schemas for three social media platforms (Facebook, Twitter and YouTube) using Kimball's bottom-up methodology, then integrates these schemas into a global warehouse design through a defined set of semantic mapping rules. The central fact table in the resulting schema, Fact-opinion, is surrounded by dimensions representing the social media platform, user, date and content, and captures opinion measures in the form of aggregate counts of positive, negative and neutral comments per post. The architecture is implemented in a MongoDB document-oriented NoSQL store and evaluated through a set of analytical queries examining comment distribution across time periods, platforms and content types (Moalla, Nabli, & Hammami, 2022).

This is a meaningful structural advance over dashboard-only approaches. By modelling opinion data within a dimensional schema rather than treating it as a transient reporting signal, the architecture preserves observations in a structured form and enables queries that cross the dimensions of time, platform and content. It demonstrates that dimensional modelling principles can accommodate opinion measures and that the resulting schema supports a level of analytical flexibility that flat aggregation cannot.

The limitations, however, are significant and the paper itself makes several of them explicit. The architecture does not support OLAP operations. The authors identify the development of OLAP operators for document-oriented warehouses as future work, and the analytical queries demonstrated in the paper are basic MongoDB aggregations rather than multidimensional drill-down or roll-up operations. The opinion measures are aggregate counts at post level, not granular sentiment scores at the level of individual aspects or even individual comments. A post that draws three hundred responses receives a single set of count measures; the variation in opinion within those responses, the aspects they address, the intensity of the views expressed and the emotional character of the feedback are all structurally invisible. The grain declaration effectively forecloses the class of analytical questions that require aspect-level or comment-level resolution (Rana, et al., 2021) (Schouten & Frasincar, 2016).

Representational scope is equally constrained. The opinion measures capture only polarity, classifying comments as positive, negative or neutral. Subjectivity and emotion are absent from the schema entirely, and the paper makes no reference to them as analytical dimensions of interest. The architecture therefore inherits the same affective incompleteness that characterises dashboard-oriented approaches, even as it improves upon their structural organisation.

Other proposals in this space reflect comparable patterns. Architectures that integrate temporal dimensions and entity-level context into dimensional schemas demonstrate that multidimensional opinion analysis enables richer queries than flat aggregation (Kraiem, Feki, & Ravat, 2025). Where emotion or subjectivity are acknowledged in this literature, they are typically treated as secondary concerns outside the scope of the proposed system rather than as co-equal analytical dimensions requiring their own modelling decisions (Rana, et al., 2021). The cumulative result is a body of work that has established the structural feasibility of incorporating opinion measures into dimensional environments while consistently leaving the full affective space underrepresented.

2.5.3 Recurring Limitations of Existing Approaches

The approaches surveyed in Sections 2.5.1 and 2.5.2 represent different levels of architectural ambition, but they share a set of structural limitations that persist across both categories. These limitations are not incidental shortcomings of individual implementations. They reflect recurring design assumptions that the existing literature has not yet challenged systematically. Three stand out as directly consequential for the design goals of the present work.

Affective Incompleteness

Across both dashboard-oriented integrations and the more structurally ambitious warehouse proposals, polarity is treated as the entirety of the affective signal worth capturing. Subjectivity is occasionally referenced as a preprocessing filter, a mechanism for discarding objective statements before polarity classification is applied, but it is not retained as an analytical dimension in its own right. Emotion is largely absent from both the schema designs and the analytical ambitions of existing work. This is not because the literature is unaware of richer affective representations. Section 2.2 established that subjectivity and emotion carry distinct and complementary analytical value, and research on affective representation in adaptive systems reinforces that collapsing this richness into a single polarity dimension produces outputs that are insufficiently attuned to the complexity of human emotional expression ((Zhang, Bae, Chung, & Dey, 2025). Mohammad (2021) argues explicitly that reducing affective meaning to a polarity dimension discards information that is consequential for decision-making. The gap is not one of awareness but of implementation: the modelling conventions that would allow subjectivity and emotion to be stored, normalised and queried as first-class analytical measures have not been developed in the existing warehouse-oriented literature.

Coarse Analytical Grain

Where dimensional models for sentiment or opinion data have been proposed, they consistently declare their grain at a level that forecloses aspect-level analysis. Dashboard approaches operate on aggregated review-level summaries, and the more structured warehouse proposals reviewed in Section 2.5.2 operate on post-level aggregate counts. Neither preserves the granular observation at the level of an individual aspect within an individual review, which would allow analysts to attribute sentiment to specific service components, compare aspect-level profiles across categories or track how opinion toward a particular aspect evolves over time independently of overall scores. As Section 2.1.3 established, grain declaration is not a refinement decision that can be corrected downstream through re-aggregation. A warehouse designed at review or post level cannot answer aspect-level questions, because the data those questions require was never recorded. The systematic absence of aspect-level grain from existing proposals is one of the clearest gaps between what the sentiment analysis literature has demonstrated to be analytically valuable and what the warehousing literature has actually implemented (Mansilla, Meng, Milligan, & Smith, 2025).

Weak Dimensional Conformance

Existing sentiment-aware warehouse designs tend to treat affective measures as additions appended to an existing schema rather than as elements of a coherent, fully conformable dimensional model. Opinion counts are associated with posts and platforms, but the dimensional infrastructure that would allow sentiment to be reliably joined to broader operational context, including service category, geographic location, reviewer segment and historical service changes, is absent or underdeveloped. The integrability requirement identified in Section 2.3.5 is not simply a matter of coexistence within the same database. It demands that sentiment dimensions be modelled with the same rigour applied to conventional warehouse dimensions: consistent grain declarations, surrogate key management, conformed dimension design and slowly changing dimension handling where entities evolve over time. Without this discipline, sentiment measures can be aggregated in isolation but cannot be meaningfully correlated with the operational and contextual data that would give them strategic value (Kimball & Ross, 2013); (Yaakub, Li, Algarni, & Peng, 2012).

Together these three limitations define a consistent gap in the existing literature. Individually, each one constrains what analytical questions can be answered. Collectively, they describe a body of work that has not yet produced a warehouse architecture in which polarity, subjectivity and emotion are co-modelled as conformable measures at aspect-level grain, integrated within a dimensional framework that connects affective data to the full range of contextual dimensions that decision-making requires.

2.5.4 Positioning of the Present Work

The approaches surveyed in Sections 2.5.1 and 2.5.2 share a common ceiling: they improve on ad hoc sentiment reporting without fully committing to the modelling discipline that structured affective analytics requires. The present work departs from these conventions at three points, each corresponding to one of the limitations identified in Section 2.5.3.

The first departure concerns representational scope. Rather than treating polarity as the primary affective signal and relegating subjectivity and emotion to secondary status, the proposed warehouse models all three dimensions as concurrent measures within the same fact table. Each observation carries a polarity score, a subjectivity score and a set of emotion scores aligned with Plutchik's eight primary emotions, normalised to the conventions established in Section 2.3.2. The practical consequence of this choice is that queries become possible that a polarity-only model cannot support: distinguishing sustained anger from mild dissatisfaction toward the same service aspect, filtering for strongly opinionated content within a body of mixed reviews, or tracking how the emotional character of negative feedback shifts over time.

Grain is the second departure. The warehouse operates at aspect level, meaning that each fact table record represents a single aspect identified within a single review. A hotel review addressing cleanliness, staff behaviour and breakfast quality generates three distinct analytical observations. This is not a refinement of review-level approaches but a structural choice that opens a qualitatively different class of questions: how does sentiment toward a specific service component evolve across quarters, which aspects drive the strongest emotional responses within a given service category, and where do polarity and emotion diverge in ways that aggregate scores would obscure. None of these questions is answerable from a post-level or review-level model, regardless of how richly the sentiment values within it are represented, because the required granularity was never preserved.

The third departure is one of conformance discipline. Sentiment measures in the proposed warehouse are not appended to an existing schema as supplementary columns. They are embedded within a star schema designed from the ground up using Kimball's dimensional modelling methodology, connected to a full set of conformable dimensions covering service entity, geographic location, review platform, temporal context, reviewer segment and aspect category. Surrogate key management, slowly changing dimension handling for service attributes and a versioned dimension design for review metadata are applied consistently throughout. The result is a schema in which affective measures participate fully in the analytical model rather than sitting alongside it.

Taken together, these decisions do not resolve every open problem in sentiment-aware analytics. Real-time ingestion, cross-lingual consistency and the automated extraction of implicit aspects remain outside the scope of this work. What they do establish is a dimensional foundation in which polarity, subjectivity and emotion are treated with the same modelling rigour applied to any other analytical asset, at the granularity level at which affective data is most analytically meaningful.

2.6 Summary

The five sections of this chapter have moved from the technical foundations of sentiment analysis through the representational case for richer affective dimensions, the structural requirements that sentiment data imposes when treated as an analytical asset, the warehousing infrastructure best suited to meeting those requirements and finally the gap that existing integration attempts have left open. Taken together, they establish the theoretical and conceptual basis on which the design work presented in the following chapters rests. The problem is not that sentiment analysis and data warehousing are incompatible disciplines. It is that their integration has not yet been approached with the modelling rigour that both fields individually demand. The chapters that follow address that directly.

3 Case Study and System Specification

The theoretical and conceptual groundwork laid in Chapter 2 points toward a system with specific characteristics: a dimensional warehouse operating at aspect-level grain, storing polarity, subjectivity and emotion as co-occurring analytical measures, integrated within a conformable dimensional framework. What that groundwork does not yet provide is a concrete analytical context in which those design decisions can be grounded, evaluated and justified. That is the work of this chapter. It introduces the case study domain, establishes the scenario within which the proposed system is situated, identifies the stakeholders whose analytical needs the system must serve and derives from those needs the functional, analytical and data requirements that will govern every subsequent design decision. The requirements established here are not a formality; they are the analytical contract against which the warehouse design presented in Chapter 4 is measured.

3.1 Case Study Domain

Selecting a domain for this case study was not a neutral decision. The proposed warehouse is designed to demonstrate that polarity, subjectivity and emotion can be modelled as first-class analytical measures within a dimensional environment, and that claim is only as convincing as the domain in which it is tested. The domain needed to satisfy three conditions: it had to generate rich volumes of user-generated textual content carrying genuine affective variation, it had to present clearly identifiable analytical entities and dimensions that map naturally to a warehouse schema, and affective information had to be analytically consequential within it rather than incidental.

Several candidates were considered.

- **E-commerce Product Reviews** offer abundant textual data and have been widely studied in the sentiment analysis literature, but the analytical environment around them is already dominated by numerical ratings, pricing data and transactional metrics. The affective dimensions this work proposes to model would be supplementary rather than central in that context.
- **Social Media Discourse** around brands presents similar issues: the data is voluminous but the analytical entities are diffuse, the dimensions are hard to standardise and the relationship between a post and a specific service characteristic is rarely explicit.
- **Entertainment Review Platforms**, covering films, video games or music, were also considered but present a similar limitation: evaluation in those domains tends to be holistic rather than aspect-specific, which works against the aspect-level grain that is foundational to this work.

Tourism and Hospitality presents a different picture. Service quality in this domain is not determined primarily by objective or measurable characteristics. It is shaped by subjective perception, emotional response and the gap between expectation and experience (Pine & Gilmore, 2011). Online platforms play a decisive role in how tourism services are perceived and how consumers make decisions about them, with user-generated reviews functioning as primary sources of experiential information rather than supplementary feedback (Xiang & Gretzel, 2010). Reviews in this domain are inherently aspect-rich: a single hotel review routinely addresses room quality, staff behaviour, cleanliness, location and value, each carrying its own affective signal. This multiplicity is not a complication to be managed but a structural advantage, since it means that the aspect-level grain of the proposed warehouse maps directly and naturally onto the way users actually write about their experiences.

The domain also offers clear analytical entities, a stable geographic hierarchy and well-defined service categories, all of which translate cleanly into dimensional structures. Temporal patterns are analytically meaningful here in a way that is domain-specific: seasonality, peak travel periods and the effects of service changes over time are genuine decision-support concerns for the operators and analysts who would use a system like the one proposed. Together these properties make tourism and hospitality not just a plausible testbed for the proposed design but the most analytically appropriate one among the candidates considered.

Due to legal and ethical constraints associated with the use of real user-generated content, all data used in this case study is synthetically generated. The synthetic dataset is constructed to replicate realistic tourism and hospitality review patterns, including variation in sentiment distribution, emotional expression and aspect coverage, ensuring analytical validity while avoiding licensing and privacy concerns.

3.2 Case Study Scenario

The case study is framed around a fictional sentiment-aware analytical platform for the tourism and hospitality sector in Portugal. The platform is conceived as a national-level decision-support system that aggregates user-generated reviews associated with tourism and hospitality services, enriches them with affective information and makes the resulting data available for structured multidimensional analysis. Its purpose is not operational reporting but analytical exploration: understanding how services are experienced, how that experience varies across locations and time periods and what emotional responses different service characteristics tend to generate.

The platform covers three categories of service. Accommodation establishments include hotels, guesthouses and similar lodging providers. Food and beverage venues encompass restaurants, bars and cafés. Tourist sites include museums, historical monuments, parks and other points of interest typically visited by travellers. These categories were selected because they are the primary subjects of user-generated reviews in the tourism domain and because they present meaningfully different experiential profiles, making cross-category comparison analytically interesting rather than redundant.

Each service category is associated with a set of aspects that reviewers commonly address. Accommodation reviews tend to cover room quality, cleanliness, staff behaviour, location and value. Food and beverage reviews typically address food and drink quality, service efficiency, atmosphere and pricing. Reviews of tourist sites most frequently reference accessibility, preservation state, informational quality, crowding and overall enjoyment. These aspect sets are not exhaustive, but they represent the recurring evaluative dimensions that define user experience within each category and that the warehouse is designed to capture and analyse at the individual aspect level.

The primary data source within this scenario consists of textual reviews associated with the services described above. Each review represents an individual evaluation expressed by a user at a specific point in time and linked to a specific service. Beyond the raw textual content, reviews carry structured contextual attributes including service type, geographic location and temporal information. From the textual content, affective attributes are derived: sentiment polarity, degree of subjectivity and emotional scores across the eight Plutchik categories. The presence of multiple aspects within a single review means that one review may generate several analytical observations, each carrying its own affective profile.

Although the platform is conceptually national in scope, the case study focuses on a representative subset of Portuguese regions and cities for demonstration and validation purposes. This allows the modelling and analytical capabilities of the proposed warehouse to be illustrated in a realistic and controlled manner without requiring exhaustive geographic coverage.

Figure 8 presents a conceptual overview of the scenario, illustrating the flow from tourism services and user-generated reviews through sentiment extraction to integration within the warehouse.

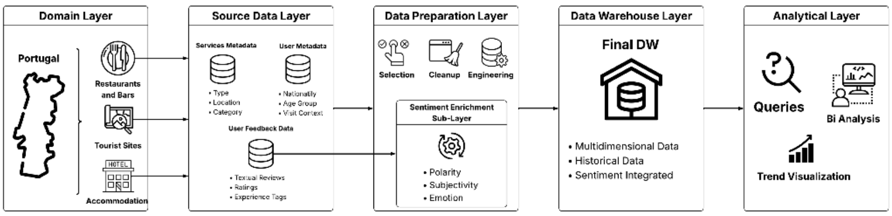

Fig. 8. Layered architecture of the proposed case study scenario, illustrating the flow from the Portuguese tourism and hospitality domain through source data collection, sentiment-enriched data preparation and dimensional warehouse integration to multidimensional analytical processing.

3.3 Stakeholders

The analytical platform proposed in this case study is designed to serve several distinct groups, each approaching tourism and hospitality data from a different organisational perspective. Understanding their needs is not a formality: the requirements derived in Section 3.5 follow directly from what these groups need to be able to ask of the system.

Tourism Analysts and Strategic Planning Teams operate at a regional or national level and are concerned with the broad picture of how tourism services are perceived across the country. Their work involves identifying how sentiment patterns shift across seasons, comparing emotional responses between regions and detecting whether particular service categories are consistently underperforming relative to others. For these users the warehouse needs to support aggregation across large volumes of observations and comparison across geographic and temporal dimensions simultaneously.

Service Managers and Operators, including hotel managers, restaurant owners and administrators of tourist sites, work at a much finer level of detail. They are less interested in national trends and more concerned with understanding which specific aspects of their own service drive satisfaction or dissatisfaction. A hotel manager asking whether complaints about cleanliness have increased since a staffing change, or a restaurant owner trying to understand why emotional responses to the dining atmosphere have shifted over the past quarter, needs aspect-level sentiment data that can be filtered, compared over time and placed in relation to operational context. This group benefits most directly from the aspect-level grain of the proposed warehouse.

Experience Quality and Marketing Teams occupy a middle position. Their interest is in how experiences are perceived and communicated, which means they draw on both the broad patterns that strategic analysts use and the aspect-level detail that operators rely on. They are particularly interested in the emotional dimensions of sentiment, since understanding what drives joy, anticipation or disappointment in a tourism context shapes how experiences are designed and how they are promoted. For this group, the ability to query polarity, subjectivity and emotion together within the same analytical context is what distinguishes the proposed system from a conventional sentiment dashboard.

3.4 Analytical Use Cases

The use cases presented here are not exhaustive but representative. They are organised around the three stakeholder groups identified in Section 3.3, reflecting the analytical questions each group is most likely to bring to the system. Together they define the analytical scope that the warehouse design must support and provide the basis from which the requirements in Section 3.5 are derived.

Strategic and Comparative Analysis

These use cases reflect the needs of tourism analysts and planning teams operating across regions and service categories:

- How does overall user sentiment differ between accommodation services, restaurants and bars and tourist sites?
- How do emotional responses expressed in user reviews vary across regions and cities?
- Which regions or cities consistently exhibit stronger positive or negative emotional responses across multiple service categories?
- How does user sentiment evolve over time for specific service categories or locations, particularly across different tourism seasons?

Aspect-level and Operational Analysis

These use cases reflect the needs of service managers and operators working at the level of individual services and their characteristics:

- Which service aspects are most frequently associated with negative sentiment or negative emotions in accommodation services?
- Are certain emotions more prevalent in specific types of services or tourist sites?
- How does sentiment change following service improvements or operational decisions?

Affective and Segmentation Analysis

These use cases reflect the needs of experience quality and marketing teams interested in emotional drivers and visitor profiles:

- How does the degree of subjectivity influence the interpretation of aggregated sentiment trends?
- How do polarity, subjectivity and emotion interact when analysing sentiment patterns across service aspects and visitor segments?
- How does sentiment differ between solo travellers, families and business travellers for the same service category?

The use cases above represent the core analytical scope of this work, but the warehouse is designed to support additional analyses that emerge naturally from its dimensional structure. The Travel Segment dimension, for instance, enables exploration of which visitor profiles are associated with particular service types, how demographic groups differ in their emotional responses to the same service category or whether business travellers and families exhibit systematically different subjectivity patterns in their reviews. These analyses are not the primary focus of this dissertation, but they are supported by the schema without requiring any structural changes.

From a data warehousing perspective, these use cases map directly onto standard OLAP operations. Comparative questions across service categories and regions correspond to slice and dice operations. Aspect-level exploration requires drill-down from service-level aggregates to individual characteristics. Temporal analyses of sentiment evolution rely on roll-up and drill-down along the time dimension. This alignment confirms that the proposed warehouse, designed around Kimball's dimensional modelling methodology, provides the right analytical infrastructure for the questions these stakeholders need to answer.

3.5 Requirements

The requirements presented in this section were derived from the case study scenario, the stakeholder groups identified in Section 3.3 and the analytical use cases established in Section 3.4. They are organised into three categories that reflect different dimensions of the system specification: functional requirements define what the system must be capable of doing, analytical requirements define how the stored data must be exploitable for multidimensional analysis and decision support, and data requirements define what information must be available and how it must be structured. Together they constitute the analytical contract against which the warehouse design presented in Chapter 4 is evaluated.

3.5.1 Functional Requirements

Functional requirements specify the core capabilities the system must provide independently of any particular implementation or modelling approach. They define the operational boundaries of the warehouse: what it must store, what it must support and how it must handle the specific characteristics of sentiment-enriched review data.

Table 6 presents the functional requirements of the proposed system.

Table 6. Functional Requirements

ID	Requirement Description	Scope	Rationale
FR-01	The system must store information about tourism and hospitality services, including accommodation, restaurants and bars and tourist sites.	Data management	Required to represent the core entities of the analytical domain.
FR-02	The system must support the storage of user-generated reviews and their structured analytical metadata, including star ratings, associated with tourism and hospitality services.	Data management	Reviews and their structured attributes constitute the primary source of experiential and sentiment-related information.
FR-03	The system must support the structured storage of sentiment-related measures, including polarity, subjectivity and emotion, derived from review content.	Sentiment integration	Enables sentiment-aware analysis beyond simple numerical ratings.
FR-04	The system must store a confidence indicator associated with each derived sentiment observation, reflecting the reliability of the sentiment derivation process.	Sentiment integration	Supports quality-aware analytical interpretation of sentiment measures.
FR-05	The system must support aggregation of sentiment-related data across multiple analytical dimensions including time, service, location and aspect.	Analytical processing	Aggregation across dimensions is the fundamental operation of multidimensional analysis.
FR-06	The system must support comparison of sentiment indicators across different services, locations and time periods.	Analytical processing	Comparative analysis is required to identify patterns and differences in user experience.
FR-07	The system must support multiple sentiment observations per review when distinct service aspects are evaluated independently.	Granularity management	Enables both high-level evaluation and detailed analysis of individual service characteristics.
FR-08	The system must support temporal analysis of sentiment data to identify trends and seasonal patterns.	Temporal analysis	Tourism experiences are strongly influenced by temporal factors including seasonality and peak travel periods.
FR-09	The system must allow the execution of analytical queries suitable for decision-support and business intelligence purposes.	Decision support	Ensures alignment with data warehousing and OLAP objectives.
FR-10	The system must support the identification of edited reviews and maintain logical grouping between original and edited versions for analytical traceability.	Analytical traceability	Allows edited reviews to be correctly handled and their analytical impact assessed.

3.5.2 Analytical Requirements

Analytical requirements describe how the data stored in the warehouse must be exploitable for multidimensional analysis. Where functional requirements are concerned with system capabilities, analytical requirements are concerned with the quality and expressiveness of the analytical operations those capabilities must support. They reflect directly the use cases established in Section 3.4 and the informational needs of the stakeholder groups identified in Section 3.3.

Table 7 presents the analytical requirements of the proposed system.

Table 7. Analytical Requirements

ID	Requirement Description	Analytical Focus	Rationale
AR-01	The system must support comparative analysis of sentiment indicators across different service categories.	Cross-service analysis	Enables evaluation of experiential differences between accommodation, restaurants and bars and tourist sites.
AR-02	The system must support aspect-level analysis of sentiment, enabling independent aggregation and comparison of sentiment across individual service characteristics.	Aspect-level analysis	Allows identification of the specific drivers of satisfaction and dissatisfaction within services.
AR-03	The system must support spatial analysis of sentiment indicators across geographic areas at region, district and municipality level.	Spatial analysis	Enables comparison of user experience across different areas of the country.
AR-04	The system must support temporal analysis of sentiment evolution over time, including seasonal and longitudinal comparisons.	Temporal analysis	Required to identify trends, seasonal patterns and the effects of service changes on user perception.
AR-05	The system must support multidimensional aggregation and exploration of sentiment data through roll-up, drill-down, slice and dice operations.	Multidimensional analysis	Enables flexible OLAP-style exploration of sentiment across all analytical dimensions.
AR-06	The system must support identification of recurring emotional patterns associated with services, locations or aspects.	Emotion analysis	Supports detection of persistent experiential issues or strengths at the aspect and service level.
AR-07	The system must support segmentation-based analysis of sentiment according to visitor characteristics such as travel party type, trip purpose and nationality.	Segmented analysis	Enables exploration of how different visitor profiles experience and evaluate tourism services.
AR-08	The system must support the combined analysis of polarity, subjectivity and emotion within the same analytical query context.	Affective analysis	Supports simultaneous multidimensional affective analysis using all three sentiment dimensions.
AR-09	The system must support comparative analysis of structured star ratings alongside derived sentiment measures to enable rating-sentiment correlation analysis.	Rating-sentiment analysis	Enables investigation of alignment and divergence between explicit numerical ratings and derived affective indicators.

3.5.3 Data Requirements

Data requirements specify the information that must be available and how it must be structured to satisfy the functional and analytical requirements established above. They are concerned with the content and organisation of the data rather than with system capabilities or analytical operations, and they are expressed independently of any specific warehouse modelling or implementation decision.

Table 8 presents the data requirements of the proposed system.

Table 8. Data Requirements

ID	Requirement Description	Data Category	Rationale
DR-01	The system must store information about tourism and hospitality services, including accommodation, restaurants and bars and tourist sites.	Core entities	Required to represent the primary objects of analysis within the domain.
DR-02	The system must store textual user reviews associated with tourism and hospitality services.	Review data	Reviews constitute the main source of experiential and sentiment-related information.
DR-03	The system must store star ratings associated with each review as a structured evaluative attribute.	Review data	Enables comparison of explicit numerical ratings with derived sentiment measures.
DR-04	The system must store temporal information associated with each review, including the date of publication.	Temporal data	Required to support trend analysis and seasonal comparisons.
DR-05	The system must store geographic information associated with services, including region, district and municipality.	Spatial data	Enables spatial and hierarchical analysis of sentiment across locations.
DR-06	The system must store service categorisation information, allowing services to be grouped by type.	Classification data	Required to support comparative analysis across service categories.
DR-07	The system must store sentiment polarity information derived from review content as a normalised numerical score	Sentiment data	Enables evaluation of positive, negative or neutral sentiment patterns.
DR-08	The system must store subjectivity information associated with each aspect-level observation as a normalised numerical score.	Sentiment data	Supports interpretation of sentiment by distinguishing strongly opinionated content from weakly subjective or factual expressions.
DR-09	The system must store emotion-related information derived from review content as a set of normalised numerical scores.	Sentiment data	Required to capture the full emotional dimension of user experience at the aspect level.
DR-10	The system must store a confidence score associated with each sentiment observation reflecting the reliability of the derivation process.	Sentiment data	Supports quality-aware filtering and interpretation of sentiment measures in analytical queries.
DR-11	The system must support the association of a single review with multiple service aspects when applicable.	Aspect data	Enables fine-grained analysis of sentiment related to individual service characteristics.
DR-12	The system must store visitor profile information associated with reviews, including travel party type, trip purpose and nationality where available.	Visitor data	Required to support segmentation-based analysis across different types of travellers.
DR-13	The system must support the storage of synthetic data generated to simulate realistic tourism and hospitality review scenarios.	Data constraints	Ensures compliance with legal and ethical constraints while maintaining analytical relevance.

3.6 System Constraints and Assumptions

The system described in this dissertation was designed and evaluated within a set of clearly defined boundaries. These boundaries are not limitations to be apologised for but deliberate scope decisions that allow the work to focus on its core contribution: the dimensional modelling and analytical integration of affective data.

All data used in this case study is synthetically generated. No real user-generated content or personal data is included at any stage. The synthetic dataset is constructed to approximate realistic tourism and hospitality review patterns, including variation in sentiment distribution, emotional expression and aspect coverage, ensuring that the analytical scenarios demonstrated are meaningful without introducing legal or ethical complications associated with the use of real user data.

The analytical scope is limited to three service categories within the tourism and hospitality domain: accommodation, restaurants and bars and tourist sites. Geographically the case study operates at a national level for Portugal, demonstrated using a representative subset of regions and cities. Adjacent domains such as transportation, pricing and booking logistics are outside the scope of this work and are not modelled or evaluated.

A further assumption concerns sentiment derivation. Polarity, subjectivity and emotion are treated as available outputs of a prior extraction process rather than as something this dissertation designs or evaluates. The tools used for derivation, VADER for polarity, TextBlob for subjectivity and the NRC Emotion Lexicon for emotion, are applied as established instruments whose outputs feed the warehouse. The design and comparative evaluation of sentiment extraction methods is a separate research concern (Cambria, Poria, Hazarika, & Kwok, 2018), and one that falls outside the scope of a dissertation focused on analytical integration rather than extraction performance.

Finally, the system is intended exclusively for analytical and decision-support purposes. Real-time ingestion, operational reporting and transactional workloads are not considered. Evaluation is limited to the validation of analytical capabilities and modelling adequacy rather than performance benchmarking under large-scale or production conditions. These boundaries are consistent with the Design Science Research methodology adopted in Section 1.4, which evaluates artefacts against the requirements they were designed to satisfy rather than against generalised performance criteria.

3.7 Summary

Chapter 3 established the analytical and operational foundation against which the warehouse design is evaluated. The case study domain and scenario grounded the work in a concrete real-world context, defining the boundaries of the system in terms of service categories, geographic scope and data characteristics. The stakeholder groups and analytical use cases translated that context into specific informational needs, which were then formalised into a structured set of functional, analytical and data requirements. The chapter closed by making explicit the constraints and assumptions that delimit the scope of the work, clarifying what the proposed system is and is not intended to address. Together these elements constitute the analytical contract that the dimensional design presented in the following chapter is built to satisfy.

4 Data Warehouse Design Methodology

Chapter 3 established the analytical contract against which the proposed warehouse must be measured: a system operating at aspect-level grain, capable of supporting comparative, temporal and affective analysis across the tourism and hospitality domain, with polarity, subjectivity and emotion treated as co-occurring first-class measures. Chapter 2 identified the structural shortcomings of existing approaches, namely their affective incompleteness, their coarse grain and their weak dimensional conformance, and argued that addressing them requires applying dimensional modelling discipline to affective data from the ground up. This chapter is where that argument is translated into a concrete design.

The design follows Kimball's four-step dimensional modelling methodology, which structures the chapter's progression. The business process is selected and its analytical character established. The grain is formally declared, anchoring every subsequent modelling decision. The dimensions that provide contextual scaffolding for sentiment observations are identified and characterised in detail, including the outrigger structures that handle geographic and reviewer-segment information. The facts are then identified and their representation justified, with particular attention to the decisions involved in modelling polarity, subjectivity and emotion as quantitative measures within the same analytical record. The chapter concludes with a set of schema representations that consolidate the design into visual form and prepare the transition to data integration.

4.1 Dimensional Modelling Approach

The design of the proposed sentiment-aware data warehouse follows the dimensional modelling methodology described by Kimball and Ross (2013). This methodology structures warehouse development around four sequential steps that collectively ensure the resulting schema is analytically coherent, requirement-driven and capable of supporting the multidimensional querying it is designed to serve. Its particular relevance to this work lies in the discipline it imposes on design decisions that would otherwise be difficult to justify systematically, especially when the measures involved are affective rather than transactional.
The four steps are applied as follows:

1. **Select the Business Process:** identifies the analytical event that the fact table will represent. In this work, the relevant process is not a transaction but an experiential evaluation: the assessment of a specific characteristic of a tourism or hospitality service, expressed within a user-generated review and enriched with derived affective attributes. Defining this process precisely is what allows the subsequent steps to remain focused on a coherent analytical subject.

2. **Declare the Grain:** specifies exactly what one record in the fact table represents. This is the most consequential decision in the design, since every dimension, every measure and every valid aggregation follows from it. In a sentiment-aware context, the grain declaration must resolve the tension between analytical expressiveness and modelling complexity, particularly given that a single review may carry sentiment toward several distinct service characteristics simultaneously.

3. **Identify the Dimensions:** establishes the contextual perspectives along which sentiment observations can be explored, grouped and compared. In this work, dimensions are derived directly from the functional, analytical and data requirements established in Chapter 3, and include temporal, geographic, service-level, platform and reviewer-segment contexts, as well as the aspect structure that defines the analytical unit of each observation.

4. **Identify the Facts:** specifies the measurable elements stored in the fact table. Here, the central challenge is representing polarity, subjectivity and emotion as concurrent quantitative measures within a single record, in a way that supports aggregation, comparison and multidimensional analysis without losing the affective richness that distinguishes this model from conventional sentiment reporting.

4.2 Selection of the Business Process

Kimball and Ross (2013) define the business process as the analytical event that a fact table represents, and its selection shapes every design decision that follows, from grain declaration through to measure identification. In most conventional warehouse designs, the business process corresponds to a well-defined operational event such as a sale, a booking or a delivery, each producing a structured record with a clear timestamp, a measurable outcome and a finite set of participants. The process modelled in this work has a different character.

The selected business process is the evaluation of a specific characteristic of a tourism or hospitality service, expressed within a user-generated review and enriched with affective attributes derived from its textual content. Each review represents a user's experiential response to a service rather than a discrete operational transaction. Its primary output is unstructured text carrying implicit affective information, which must be extracted and structured before it can function as the basis for a fact table record.

A further consequence of this choice is that one instance of the process does not necessarily produce a single analytical observation. A single review in the tourism and hospitality domain typically addresses several distinct service characteristics within the same text, each carrying its own affective signal. This multiplicity is a defining property of the process being modelled and the principal reason why the grain declaration in the following section cannot default to one record per review. Recognising it at this stage grounds the subsequent design in the structural reality of the data rather than in a simplified abstraction of it.

4.3 Declaring the Grain

The grain of the fact table defines precisely what one record represents, and Kimball and Ross (2013) are explicit that this declaration must precede dimension and measure identification. Once the grain is fixed, it constrains every aggregation, every valid join and every analytical question the warehouse can answer. Changing it later would require restructuring the schema from the ground up.

In the proposed warehouse, the grain is declared as follows:

“One sentiment observation associated with a specific aspect extracted from a single user-generated review of a tourism or hospitality service.”

This places the warehouse at aspect-level granularity, meaning that a review addressing three distinct service characteristics generates three separate fact records, each carrying its own affective measures. The choice was reached by evaluating two alternatives. A document-level grain, where one record represents one review, is the simpler option and would make aggregation straightforward. Its cost is that internal variation within a review is lost: a guest who praises the location while criticising the cleanliness produces a single mixed sentiment value that obscures both signals. A sentence-level grain offers finer resolution but introduces a different problem, since sentences do not map cleanly onto analytically meaningful units and their relationship to specific service characteristics is often ambiguous. Aspect-level grain preserves the distinctions that matter for the analytical use cases established in Chapter 3, particularly those requiring independent aggregation of sentiment across individual service characteristics, without descending to a granularity level that the data cannot reliably support.

One consequence of this choice warrants attention. Because a single review may generate multiple fact records, a naive aggregation of sentiment scores across all records for a given service would weight reviews with more identified aspects more heavily than reviews with fewer. This is not a reason to avoid aspect-level grain, since the analytical requirements of this work justify the added complexity, but it is a constraint that must be observed in query design. Aggregations intended to reflect overall service sentiment should group by review before averaging across services, rather than averaging directly across all fact records.

4.4 Identifying the Dimensions

With the grain declared, the dimensions can be identified. Their selection is not independent of the grain, because each fact record represents a sentiment observation associated with a specific aspect within a single review, the dimensions must collectively capture everything needed to interpret that observation. The analytical requirements established in Chapter 3 provide the basis for this identification, translating the stakeholder questions defined there into the contextual perspectives the warehouse must support.

Seven dimensions are identified: Date, Service, Aspect, Review and Platform, together with Location and Travel Segment as outrigger dimensions.

The **Date** dimension provides temporal context for each sentiment observation. In the tourism and hospitality domain, where seasonal patterns and peak travel periods are central to how services are perceived and evaluated, temporal granularity is particularly important. The dimension supports analysis across days, months, quarters, years and tourism seasons, enabling the longitudinal and seasonal comparisons required by AR-04.

The **Service** dimension represents the tourism or hospitality entity being evaluated. It holds the descriptive and classificatory attributes needed to compare sentiment across service categories and subcategories, addressing AR-01 and AR-02 directly. Geographic information is deliberately separated from this dimension into its own structure, discussed below.

The **Aspect** dimension captures the specific service characteristic referenced within a review. Given that the grain is defined at the aspect level, this dimension is central to the model: without it, the fine-grained sentiment analysis that distinguishes this warehouse from review-level approaches is not possible. It supports the aspect-level aggregation and comparison required by AR-02.

The **Review** dimension preserves the link between fact records and their source content. Since a single review may generate multiple fact records, this dimension provides the shared context that ties those records together and supports the traceability required by FR-10. It also carries the reviewer's star rating, allowing the rating-sentiment correlation analysis required by AR-09.

The **Platform** dimension identifies the digital source from which each review was obtained. Different platforms exhibit different user demographics and review behaviours, making platform a relevant analytical axis for cross-source comparison.

Two dimensions are connected to the model as outriggers rather than as direct dimensions of the fact table. An outrigger is a dimension associated with another dimension rather than directly with the fact table, used when the attributes in question describe a property of the parent dimension rather than a property of the analytical observation itself.

Location is connected through Service because geographic attributes describe where a service operates: they are a property of the service entity, not of the individual sentiment observation. Attaching Location directly to the fact table would require repeating the same geographic foreign key across every fact record produced by the same service, introducing redundancy without analytical benefit. Connecting it through Service preserves clean access to geographic context while keeping the fact table lean.

Travel Segment follows the same logic in relation to the Review dimension: segmentation attributes such as nationality, age group and travel party type describe the reviewer behind a review, not the individual aspect observation extracted from it. Connecting Travel Segment through Review avoids replicating those attributes across every fact record that the same review generates.

Two candidate dimensions were evaluated and excluded. A time-of-day dimension was considered but rejected because the available data does not support meaningful analysis at that level of temporal granularity. Emotion was also considered as a potential dimension but rejected because emotional values are derived outcomes of the sentiment analysis process rather than independent contextual attributes. Representing emotion as a dimension would require either selecting a single dominant emotion per observation, losing the multidimensional character of emotional expression, or constructing a degenerate or junk dimension with no analytical hierarchy. Storing emotion as a set of numerical measures in the fact table, which is the approach adopted here, better reflects its nature and better supports the aggregation and comparison operations the warehouse is designed to enable.

4.5 Characterising the Dimensions

The following subsections document each dimension using a standardised specification template. The template covers four elements: general information describing the dimension's purpose, expected size and growth behaviour; an attribute table listing each field with its type, domain and examples; an index table defining the indexing strategy; and a hierarchy table recording any analytical hierarchies the dimension supports. This consistent format makes the transition from conceptual design to logical and physical implementation more tractable and ensures that modelling decisions are documented at the level of detail required for implementation.

Table 9. Specification Table Template

Dimension						
ID						
Description						
Type						
Size						
Growth						
Attributes						
Nr	**ID**	**Description**	**Key [Type]**	**Domain [Size] [Range]**	**Variation**	**Examples**
1						
2						
3						
4						
5						
6						
7						
8						
Indexes						
Nr	**ID**	**Index**	**Type**			
1						
2						
3						
4						
Hierarchies						
Nr	**ID**	**Scheme**				
1						
2						
3						
4						

4.5.1 Date Dimension

The Date dimension provides temporal context for every sentiment observation in the warehouse. Each record represents a single calendar day, identified by a surrogate key in YYYYMMDD integer format. This format was chosen because it supports efficient range queries and joins while remaining human-readable without conversion.

The dimension is pre-populated to cover a five-year window from 2022 to 2026, producing approximately 1,825 records. This range extends beyond the temporal coverage of the synthetic dataset to allow the warehouse to accommodate future review activity without requiring calendar extension in the near term. No growth is expected within that window after initial loading; extension beyond 2026 would require appending additional calendar records.

Beyond standard calendar attributes (day, month, quarter and year), the dimension includes a season attribute that classifies each date into one of four tourism seasons. This attribute is specific to the analytical domain: seasonal patterns in the tourism and hospitality sector influence both the volume and character of reviews and supporting season-level aggregation without requiring analysts to derive it at query time is a practical analytical convenience.

Table 10. Specification of the Date dimension

Date Dimension	
ID	D_DATE
Description	Calendar dimension used to analyze reviews across time. It supports grouping and filtering by year, quarter, month and season, enabling trend and seasonality analysis without relying on raw timestamps.
Type	Calendar Dimension
Size	1.8k rows (5-year window: 2022–2026)
Growth	No regular growth is expected after initialization. Additional rows may be added if the temporal coverage of the warehouse is extended.

Attributes

Nr	**ID**	**Description**	**Key**	**Domain**	**Variation**	**Examples**
1	date_key	Surrogate key in YYYYMMDD format	PK	INT (8 digits)	…	20240215
2	full_date	Full calendar date	A	DATE	…	2/15/2024
3	day_of_month	Day number within the month	A	TINYINT (1–31)	…	15
4	day_name	Name of the weekday	A	VARCHAR(10)	…	Thursday
5	month	Month number	A	TINYINT (1–12)	…	2
6	month_name	Name of the month	A	VARCHAR(10)	…	February
7	quarter	Quarter number of the year	A	TINYINT (1–4)	…	1
8	year	Calendar year	A	SMALLINT (e.g., 2022–2026)	…	2024
9	season	Tourism season classification	A	VARCHAR(10) {Winter, Spring, Summer, Autumn}	…	Winter

Indexes

Nr	**ID**	**Index**	**Type**
1	PK_D_DATE	(date_key)	Clustered Primary Key
2	IDX_D_DATE_YEAR	(year)	Non-clustered
3	IDX_D_DATE_YEAR_MONTH	(year, month)	Non-clustered

Hierarchies

Nr	**ID**	**Scheme**
1	H_DATE_Y_Q_M_D	Year → Quarter → Month → Day
2	H_DATE_Y_S_M_D	Year → Season → Month → Day

4.5.2 Location Dimension

The Location dimension represents the geographic context of the service being evaluated. As established in Section 4.4, it is implemented as an outrigger connected through the Service dimension rather than as a direct dimension of the fact table, reflecting the fact that geographic attributes describe where a service operates rather than properties of the individual sentiment observation.

The dimension models a three-level Portuguese geographic hierarchy: region, district and municipality. This hierarchy supports progressive geographic aggregation, allowing analysts to examine sentiment patterns at the municipality level and roll up through district to region without requiring additional transformations at query time. The hierarchy is aligned with the administrative structure of Portugal, which ensures consistency with external geographic reference data and simplifies the population of the dimension during the ETL process.

Geographic boundaries within Portugal are treated as stable for the purposes of this design. Administrative divisions do change in practice, but at a frequency that does not justify the overhead of a slowly changing dimension implementation. The dimension is therefore modelled as SCD Type 1: updates to geographic records overwrite existing values, and no historical versioning of location attributes is maintained.

Table 11. Specification of the Location Dimension

Location Dimension						
ID	D_LOCATION					
Description	Geographic dimension representing the location of a tourism or hospitality service in Portugal. It enables spatial analysis of sentiment observations across region, district, municipality and tourism region levels.					
Type	Static Outrigger Dimension					
Size	308 rows corresponding to all municipalities in Portugal					
Growth	No growth expected after initialization					
Attributes						
Nr	**Attribute**	**Description**	**Key**	**Domain**	**Variation**	**Example**
1	location_key	Surrogate key identifying the location record	PK	INT	...	27
2	municipality_code	Natural key corresponding to the official administrative municipality code	NK	VARCHAR(10)	...	303
3	region	NUTS II statistical region where the service is located	A	VARCHAR (40) {Norte, Centro, Lisboa, Alentejo, Algarve, Região Autónoma dos Açores, Região Autónoma da Madeira}	...	Norte
4	district	Portuguese administrative district where the municipality is located	A	VARCHAR(40)	...	Braga
5	municipality	Municipality where the service operates	A	VARCHAR(80)	...	Guimarães

Indexes			
Nr	**Name**	**Index**	**Type**
1	PK_D_LOCATION	(location_key)	Clustered Primary Key
2	UQ_D_LOCATION_NK	(municipality_code)	Unique
3	IDX_D_LOCATION_REGION	(region)	Non-clustered
4	IDX_D_LOCATION_DISTRICT	(district)	Non-clustered

Hierarchies		
Nr	**ID**	**Scheme**
1	H_LOC_REGION_DISTRICT_MUNICIPALITY	Region → District → Municipality

4.5.3 Service Dimension

The Service dimension represents the tourism or hospitality entity being evaluated in each review. It covers hotels, restaurants, bars and tourist attractions operating within Portugal, and carries the descriptive and classificatory attributes needed to group and compare sentiment observations across service categories, subcategories and individual establishments.

The dimension is modelled as a Slowly Changing Dimension Type 2. Service attributes such as category, subcategory, price range and operational status are subject to change over time: a restaurant may be reclassified, a hotel may change ownership or be rebranded, and a venue's price range may shift. If these changes were handled by overwriting existing records, sentiment observations made before the change would be retroactively associated with attributes that did not describe the service at the time the review was written, undermining the historical accuracy of any longitudinal analysis. SCD Type 2 addresses this by inserting a new dimension record when a tracked attribute changes, preserving the previous record with its original values and marking it as inactive through a validity period. Fact records written before the change retain their foreign key pointing to the original dimension record, ensuring that historical queries return results consistent with the service's state at the time of each observation.

Each service record carries a surrogate key as its primary key, alongside a platform_service_id that stores the business identifier assigned by the source platform. The surrogate key decouples the warehouse from platform-specific identifiers and accommodates the SCD Type 2 versioning pattern, since a single service may have multiple warehouse records over its history. A composite unique index on the combination of platform_service_id and valid_to ensures that no two active records for the same service can coexist: because valid_to is set to a future sentinel date for the current version and to a specific end date for expired versions, this constraint guarantees uniqueness across both active and historical records without requiring a separate active flag as the sole uniqueness criterion.

Table 12. Specification of the Service Dimension

Service Dimension						
ID	D-SERVICE					
Description	Dimension representing the analysed service entity within the tourism and hospitality domain, including accommodation, restaurants, bars and tourist sites. It provides descriptive and categorical information to support segmentation, comparison and performance evaluation.					
Type	Slowly Changing Dimension Type 2					
Size	5,000–10,000 rows (depending on number of services included across selected Portuguese regions)					
Growth	Estimated 5–10% per year (new services entering the market and existing services closing or changing category)					
Attributes						
Nr	**ID**	**Description**	**Key**	**Domain**	**Variation**	**Examples**
1	service_key	Surrogate key generated for DW	PK	INT	…	10245
2	platform_service_id	Business identifier from source platform	NK	VARCHAR(50)	…	TRIP_458921
3	location_key	Reference to the Location dimension identifying where the service operates	FK	INT	…	145
3	service_name	Official name of the service	A	VARCHAR(150)	Rare. May change due to service rebranding or name correction.	Hotel Atlântico
4	service_category	High-level classification of service	A	VARCHAR(30) {Accommodation, Restaurant/Bar, Tourist Site}	Very rare. May change if the service is reclassified.	Accommodation
5	service_subcategory	Detailed classification within category	A	VARCHAR(50)	Rare. May change if the service positioning evolves.	Boutique Hotel
6	star_rating	Official star classification where applicable	A	TINYINT (1–5, nullable)	Occasionally. May change when official classification is updated.	4
7	price_tier	Relative price classification	A	VARCHAR(10) {Low, Medium, High, Luxury}	Occasionally. May change if pricing strategy evolves.	High
8	is_current	Flag Indicating active version of record	A	BOOLEAN	System-managed. Updated when a new SCD2 record is created.	TRUE
9	valid_from	Start date of record validity (SCD2)	A	DATE	System-managed. Set when a new record version is created.	1/1/2023

10	valid_to	End date of record validity (SCD2)	A	DATE	System-managed. Updated when the record is superseded.	12/31/9999
Indexes						
Nr	**ID**	**Index**	**Type**			
1	PK_D_SERVICE	(service_key)	Clustered Primary Key			
2	UQ_D_SERVICE_NK	(platform_service_id, valid_to)	Unique / Non-clustered			
3	IDX_D_SERVICE_CATEGORY	(service_category)	Non-clustered			
Hierarchies						
Nr	**ID**	**Scheme**				
1	H_SERVICE_CAT_SUB	Category → Subcategory → Service				
2	H_SERVICE_CAT_SERVICE	Category → Service				

4.5.4 Aspect Dimension

The Aspect dimension captures the specific service characteristic referenced within each review. Since the grain of the fact table is declared at the aspect level, this dimension is structurally central to the model: it is what distinguishes one sentiment observation from another when multiple observations originate from the same review. Its attributes include a standardised aspect name, a broader category grouping and a textual description. Examples of aspects include cleanliness, staff quality, location, price and food quality.

The values in this dimension constitute a controlled analytical taxonomy rather than a record of real-world events. Aspect names and categories are defined by the design of the warehouse and refined as the extraction process matures; they do not reflect external changes in the world that an analyst would want to track historically. A change to an aspect's name or category represents a correction or refinement of the taxonomy itself, not a meaningful analytical event that should be preserved as a separate version. For this reason, the dimension is modelled as SCD Type 1: updates overwrite existing records, and no historical versioning is maintained.

This contrasts with the treatment of the Service dimension, where attribute changes carry genuine analytical significance because they reflect real-world changes to the service being described. The distinction between dimensions that record external reality and dimensions that define internal analytical structure is what drives the different SCD choices between the two.

The dimension is expected to remain small and relatively stable once the initial taxonomy is established. Growth occurs only when new aspect categories are introduced as the system is extended to cover additional service types or analytical domains.

Table 13. Specification of the Aspect Dimension

Aspect Dimension	
ID	D_ASPECT
Description	Dimension representing the specific aspect evaluated within a review. An aspect corresponds to a particular characteristic of a tourism and hospitality service, such as cleanliness, staff behaviour or accessibility. This dimension enables fine-grained analytical exploration by supporting aggregation and comparison at the aspect and aspect-category levels.
Type	Slowly Changing Dimension Type 1
Size	30-60 Rows (depends on defined Taxonomy for Aspects)
Growth	Very low. New aspects may be added occasionally as the taxonomy evolves.

Attributes

Nr	**ID**	**Description**	**Key**	**Domain**	**Variation**	**Examples**
1	aspect_key	Surrogate key generated in the data warehouse	PK	INT	…	15
2	aspect_code	Stable identifier representing the aspect in the taxonomy and ETL mappings	NK	VARCHAR(30)	…	CLEANLINESS
3	aspect_name	Human-readable name of the aspect	A	VARCHAR(100)	Rare. May be updated to improve clarity or standardisation.	Cleanliness
4	aspect_category	High-level grouping used to support roll-up analysis by category	A	VARCHAR(50) {Location, Physical Environment, Staff and Service, Core Offering, Value and Pricing, Experience and Atmosphere}	Rare. May be updated to improve classification clarity.	Physical Environment
5	aspect_description	Short description clarifying the aspect's meaning and analytical scope	A	VARCHAR(255)	Rare. May be refined to better describe the aspect.	"Perceived cleanliness of rooms, facilities or common areas."

Indexes

Nr	**ID**	**Index**	**Type**
1	PK_D_ASPECT	(aspect_key)	Clustered Primary Key
2	UQ_D_ASPECT_NK	(aspect_code)	Unique / Non-clustered
3	IDX_D_ASPECT_CATEGORY	(aspect_category)	Non-clustered

Hierarchies

Nr	**ID**	**Scheme**
1	H_ASPECT_CATEGORY	Aspect Category → Aspect

4.5.5 Travel Segment Dimension

The Travel Segment dimension captures reviewer demographic and trip context attributes associated with each review. As established in Section 4.4, it is implemented as an outrigger connected through the Review dimension, reflecting the fact that segmentation attributes describe the reviewer behind a review rather than properties of the individual sentiment observation extracted from it.

The dimension's attributes cover nationality, age group, gender, travel party type and trip purpose. These attributes support the segmentation analyses required by AR-07 and AR-08, allowing sentiment patterns to be examined across different reviewer profiles. A solo business traveller and a family on a leisure trip may evaluate the same service very differently and isolating those differences requires reviewer context to be available as an analytical dimension rather than buried in review metadata.

Not all source reviews include complete demographic information. Where attributes are unavailable, the dimension accommodates null values rather than forcing imputation, preserving the accuracy of segmentation queries by avoiding the introduction of artificial demographic signals.

The dimension carries no analytical hierarchy. Its attributes are independent categorical descriptors of the reviewer and their travel context rather than levels in a structured classification, and no meaningful aggregation path exists across them.

Table 14. Specification of the Travel Segment Dimension

Travel Segment Dimension						
ID	D_TRAVEL_SEGMENT					
Description	Dimension representing segmentation attributes associated with the reviewer and the context of the trip. This dimension supports analytical grouping of sentiment observations according to demographic and travel-related characteristics. It is connected to D-REVIEW as an outrigger dimension.					
Type	Static Outrigger Dimension					
Size	Depends on the number of distinct combinations of segmentation attributes present in the dataset					
Growth	Low. New records are created only when new combinations of segmentation attributes appear in the dataset.					
Attributes						
Nr	**ID**	**Description**	**Key**	**Domain**	**Variation**	**Examples**
1	travel_segment_key	Surrogate key generated in the data warehouse	PK	INT	…	42
2	travel_segment_code	Stable identifier representing the combination of segmentation attributes	NK	VARCHAR(80)	…	PT_25_34_FAMILY_LEISURE
3	nationality	Nationality associated with the reviewer	A	VARCHAR(60)	…	Portuguese
4	age_group	Age group associated with the reviewer	A	VARCHAR(20) {<18, 18–24, 25–34, 35–44, 45–54, 55–64 and 65+}	…	25–34
5	gender	Gender associated with the reviewer	A	VARCHAR(20) {Female, Male and Other/Unknown}	…	Female
6	travel_party_type	Type of travel party associated with the review	A	VARCHAR(30) {Solo, Couple, Family and Friends}	…	Family
7	trip_purpose	Purpose of the trip associated with the review	A	VARCHAR(30) {Leisure and Business}	…	Leisure
Indexes						
Nr	**ID**	**Index**	**Type**			
1	PK_D_TRAVEL_SEGMENT	(travel_segment_key)	Clustered Primary Key			
2	UQ_D_TRAVEL_SEGMENT_NK	(travel_segment_code)	Unique / Non-clustered			

4.5.6 Review Dimension

The Review dimension preserves the contextual metadata associated with each source review. Since the grain of the fact table is declared at the aspect level and a single review may generate multiple fact records, this dimension provides the shared context that ties those records together. Its primary role is traceability: allowing any sentiment observation in the fact table to be linked back to the review from which it was extracted, along with the reviewer's star rating, the review date and the original text.

The star rating is stored in this dimension rather than as a fact measure. Although a numeric value, the rating reflects a discrete evaluative judgement made by the reviewer at the time of writing rather than a derived analytical measure. Storing it in the Review dimension keeps it associated with its source context and supports the rating-sentiment correlation analysis required by AR-09 without introducing a measure that does not conform to the aspect-level grain of the fact table. This attribute should not be confused with the star_rating attribute carried by the Service dimension, described in Section 4.5.3. D_Review’s star_rating records the score a guest assigns to their own stay or visit. D_Service’s star_rating records the official classification awarded to the establishment, such as a hotel's four-star rating from the national tourism authority. The two are independent: the property's official classification does not change because of how individual guests rate their experience, and a guest's personal rating is not constrained by the property's formal class.

Some review platforms permit users to edit previously submitted reviews. Where edited versions are available, the warehouse preserves both the original and the revised content rather than overwriting the original. This behaviour is implemented through two attributes: review_group_id, which assigns a shared identifier to all versions of the same review, and is_edit, which distinguishes the original submission from subsequent revisions. This design is referred to here as a versioned dimension. It resembles SCD Type 2 in that multiple records may represent the same real-world entity over time, but it differs in its analytical intent. SCD Type 2 is concerned with tracking changes to descriptive attributes of an entity so that historical fact records retain accurate dimensional context. The versioned dimension pattern here is concerned with preserving the integrity of the source content itself: an edited review is not a correction of the Review dimension's attributes but a distinct analytical object that may carry different sentiment signals from its predecessor.

The dimension does not support analytical hierarchies. Reviews are not a level in any aggregation path above aspects: a query grouping sentiment observations by service cannot roll up through reviews to reach service-level results, because reviews are a lateral source context rather than a structural layer above the aspect. Hierarchical aggregation from aspect to service is handled through the Service dimension directly.

Table 15. Specification of the Review Dimension

Review Dimension	
ID	D_REVIEW
Description	Dimension representing the review from which one or more aspect-level sentiment observations are extracted. Each row corresponds to a single review instance. If a review is edited, the edited version is stored as a new review record and is linked to the original through a review group identifier.
Type	Versioned Dimension
Size	10k-20k Rows
Growth	High. The number of records grows continuously as new reviews and edited versions are added.

Attributes

Nr	**ID**	**Description**	**Key**	**Domain**	**Variation**	**Examples**
1	review_key	Surrogate key generated in the data warehouse	PK	INT	…	10452
2	platform_review_id	Source/platform identifier of the review instance	NK	VARCHAR(50)	…	TA_93810271
3	travel_segment_key	Reference to the Travel Segment dimension associ-ated with the review	FK	INT	…	7
4	review_group_id	Identifier used to group multiple versions of the same review (original and edits)	A	VARCHAR(50)	Stable. Assigned when the review chain is created.	RG_000183
5	is_edit	Indicates whether this review instance is an edited version of a previous review in the same group	A	BOOLEAN	…	TRUE
6	review_text	Full textual content of the review	A	TEXT / VARCHAR(MAX)	New record created when content chang-es.	"The staff were amaz-ing…"
7	star_rating	Review rating provided by the platform	A	DECIMAL (2,1)	New record created when rating changes.	4.5
8	review_language	Language detected	A	VARCHAR(10) (ISO code)	Rare. May differ if the review text is edited.	pt, en, esp

Indexes

Nr	**ID**	**Index**	**Type**
1	PK_D_REVIEW	(review_key)	Clustered Primary Key

2	UQ_D_REVIEW_NK	(platform_review_id)	Unique / Non-clustered
3	IDX_D_REVIEW_GROUP	(review_group_id)	Non-clustered

4.5.7 Platform Dimension

The Platform dimension identifies the digital source from which each review was obtained. Different review platforms serve different user demographics, apply different moderation policies and structure their review content in different ways. Capturing platform as a dimension allows analysts to account for these differences when interpreting sentiment patterns and supports cross-platform comparisons of the kind required by AR-05.

Each platform record carries attributes describing the platform's name, type, base URL and the language or languages in which it primarily operates. The dimension is expected to remain small, reflecting the limited number of platforms included in the scope of this work. Growth would occur only if additional platforms were integrated as data sources in a future extension of the system.

The dimension is modelled as SCD Type 1. Platform attributes such as name and URL are treated as descriptive reference data rather than analytically significant historical facts. If a platform were rebranded or its URL changed, overwriting the existing record is sufficient, since the platform itself remains the same analytical entity and preserving the previous attribute values would not support any meaningful historical query.

The Platform dimension carries no analytical hierarchy. Platforms are treated as a flat enumeration of sources rather than as a structured classification.

Table 16. Specification of the Platform Dimension

Platform Dimension						
ID	D_PLATFORM					
Description	Dimension representing the digital platform from which a review was obtained. This dimension supports comparative analysis across platforms and provides contextual information regarding the origin of sentiment observations.					
Type	Slowly Changing Dimension Type 1					
Size	15-20 Rows					
Growth	Low. New platforms may be added depending on the scope of analysis and data sources considered.					
Attributes						
Nr	**ID**	**Description**	**Key**	**Domain**	**Variation**	**Examples**
1	platform_key	Surrogate key generated in the data warehouse	PK	INT	…	3
2	platform_code	Stable platform identifier used in ETL mappings	NK	VARCHAR(30)	…	TRIPADVISOR
3	platform_name	Human-readable name of the platform	A	VARCHAR(100)	Rare. May change due to rebrand-ing.	TripAdvisor
4	platform_type	Category of the platform	A	VARCHAR(50) {Review Website, Social Media, Booking Platform, Other}	Rare. May change if classification is updated.	Review Website
5	platform_website	Official platform website domain	A	VARCHAR(150)	Rare. May change if domain or ownership changes.	tripadvisor.com
Indexes						
Nr	**ID**	**Index**	**Type**			
1	PK_D_PLATFORM	(platform_key)	Clustered Primary Key			
2	UQ_D_PLATFORM_NK	(platform_code)	Unique / Non-clustered			

4.6 Integration of Dimensions

Taken individually, each dimension provides a single contextual perspective on a sentiment observation. What the dimensional model enables, however, is the simultaneous application of multiple perspectives to the same analytical question. A query examining how sentiment toward cleanliness varies across service categories, geographic regions and traveller profiles is not executing three separate dimensional analyses but one, drawing on Date, Service, Aspect, Location and Travel Segment in combination. The analytical value of the model lies in this cross-dimensional expressiveness, and understanding how the dimensions relate to one another is necessary for understanding what kinds of questions the warehouse can reliably answer.

The five dimensions directly connected to the fact table (Date, Service, Aspect, Review and Platform) each contribute a foreign key to every fact record. This means that every sentiment observation is simultaneously addressable by time, by the service being evaluated, by the aspect under analysis, by its source review and by the platform from which it was collected. Any combination of these axes is a valid grouping for aggregation, and the star schema pattern ensures that such combinations can be resolved through simple joins without intermediate aggregation steps.

The two outrigger dimensions introduce a structural consideration worth making explicit. Because Location is accessed through Service and Travel Segment is accessed through Review, queries that filter or group by geographic or reviewer-segment attributes require a join through the parent dimension before reaching the outrigger. This is a deliberate design choice that keeps the fact table lean and avoids replicating outrigger foreign keys across every fact record, but it means that Location and Travel Segment are not directly addressable from the fact table in the way that the five core dimensions are. In practice this has no impact on analytical expressiveness, since the geographic context of a service does not change between the multiple fact records that the same service generates, and the reviewer profile associated with a review does not change between the multiple fact records that the same review generates. The outrigger pattern preserves that stability without introducing redundancy.

One conformance consideration is also relevant here. The Service and Date dimensions are the primary axes along which sentiment observations are expected to be aggregated in the use cases established in Chapter 3. Both are fully conformed: Date uses a standard calendar structure applicable across any business process, and Service carries a stable surrogate key that persists across SCD Type 2 versions, allowing historical and current observations to be grouped by the same logical service entity when required. This conformance ensures that aggregations across time and service remain analytically consistent even as service attributes evolve.

4.7 Identifying the Facts

The final step of the dimensional modelling process consists of identifying the facts, that is, the measurable elements that characterise the analytical event being represented. In the context of this dissertation, this step is particularly significant, since the fact table must not only support traditional analytical operations, but also structurally represent the affective information extracted from user-generated reviews. Based on the previously defined grain, the measures associated with each fact record are designed to capture polarity, subjectivity and emotion, while also including complementary indicators that support richer sentiment-aware analysis.

4.7.1 Definition of the Fact Table

The fact table is the central analytical structure of the proposed data warehouse, integrating the contextual dimensions defined in the preceding sections with the quantitative measures that characterise each sentiment observation. In accordance with Kimball and Ross (2013), its definition follows directly from the declared grain: each record in F_Aspect_Sentiment_Observation represents one sentiment observation associated with a specific aspect extracted from a single user-generated review of a tourism or hospitality service.

Each record carries five foreign keys referencing the Date, Service, Review, Aspect and Platform dimensions, with geographic and reviewer-segment context accessible through the outrigger relationships established in Section 4.4. These keys define the analytical position of each observation within the multidimensional space of the warehouse. The measures stored alongside them fall into two categories. The affective measures, comprising polarity, subjectivity and the eight emotion scores, constitute the analytical core of the model and represent the structured affective content derived from the review text. The supporting measures complement these by providing indicators of sentiment intensity and extraction reliability, enhancing the interpretability of the affective data without introducing additional affective dimensions. Both categories are quantitative, ensuring compatibility with the aggregation and comparison operations that multidimensional analysis requires. The specific modelling decisions and justifications for each measure are addressed in the subsections that follow.

4.7.2 Measuring Polarity

As established in Section 2.1.4, polarity captures the evaluative orientation expressed in a piece of text, representing whether the sentiment conveyed is positive, negative or neutral. The theoretical grounding for polarity and the representational alternatives commonly adopted in sentiment analysis are discussed there in detail. The concern here is narrower: how polarity should be represented as a measure within the proposed fact table, and why.

The two principal alternatives are categorical and numerical representation. A categorical approach assigns a discrete label to each observation, which is interpretable and simple to communicate but loses the intensity distinctions that are analytically relevant in this context. Two reviews expressing weakly positive and strongly positive sentiment toward the same aspect would receive identical labels, making it impossible to distinguish between them in aggregation. Given that the analytical use cases established in Chapter 3 include longitudinal trend analysis, seasonal comparison and cross-service sentiment ranking, a representation that collapses intensity differences at the point of storage would limit the warehouse before any query is written.

A continuous numerical score preserves those distinctions and is directly compatible with the aggregation operations the warehouse is designed to support. Averaging polarity scores across a dimension such as service category or time period produces a meaningful result that reflects both the direction and the magnitude of sentiment across observations. VADER, the tool selected for polarity derivation, produces a compound score normalised within the range of -1.0 to 1.0, where negative values indicate negative sentiment and positive values indicate positive sentiment, with magnitude reflecting intensity. This output is stored directly as the polarity_score measure, requiring no further transformation of the representation at load time.

In the proposed data warehouse, polarity is therefore modelled as a single continuous numerical measure per fact record, normalised within a bounded range to ensure comparability across observations regardless of their source.

4.7.3 Measuring Subjectivity

Subjectivity and its role in sentiment analysis are discussed in Section 2.2.1, including the distinction between subjective and objective expression and the representational alternatives available for capturing it. The question addressed here is how subjectivity should be represented as a measure within the fact table.

The choice is between a binary classification, which assigns each observation to either subjective or objective, and a continuous score reflecting the degree of opinionated expression. A binary representation is interpretable but discards meaningful variation: a review written in measured, restrained language and one written in strongly personal terms would be treated as analytically equivalent if both cross the same subjectivity threshold. For a warehouse designed to support nuanced cross-dimensional comparison, that loss of resolution is difficult to justify.

A continuous score captures the range of variation between objective and highly opinionated expression and remains directly compatible with aggregation operations. Averaging subjectivity scores across service categories, platforms or time periods produces results that reflect genuine variation in how opinionated reviews tend to be across different contexts, which is analytically relevant when interpreting polarity trends. A high average polarity score drawn from low-subjectivity observations carries different analytical weight than the same score drawn from strongly opinionated ones, and preserving subjectivity as a continuous measure is what makes that distinction available to analysts.

TextBlob, the tool selected for subjectivity derivation, produces a score normalised within the range of 0.0 to 1.0, where values approaching zero indicate objective expression and values approaching one indicate strongly subjective content. This output is stored directly as the subjectivity_score measure without further representational transformation.

In the proposed data warehouse, subjectivity is therefore modelled as a single continuous numerical measure per fact record, normalised within a bounded range consistent with the polarity representation and subject to average-based aggregation across all relevant analytical dimensions.

4.7.4 Measuring Emotion

The theoretical basis for emotion representation in sentiment analysis, including the distinction between categorical and numerical approaches and the frameworks proposed by Ekman and Plutchik, is covered in Section 2.2.2. The modelling question here is how emotion should be represented within the fact table given the analytical objectives of this work.

The central challenge with emotion is that multiple emotional signals can coexist within the same observation. A review of a hotel stay might simultaneously express joy at the quality of the room and frustration at the waiting time at check-in. A single dominant emotion label would suppress one of those signals entirely, and even a multi-label categorical approach would limit the aggregation operations available at query time. Neither representation is well suited to a warehouse environment where the expectation is that measures can be averaged, compared and grouped across dimensions. Figure 9 illustrates this contrast: where Ekman's model defines six discrete and mutually exclusive categories, Plutchik's structure organises eight primary emotions along axes of similarity and intensity, a configuration that maps more naturally onto a set of concurrent numerical scores than onto a classification scheme.

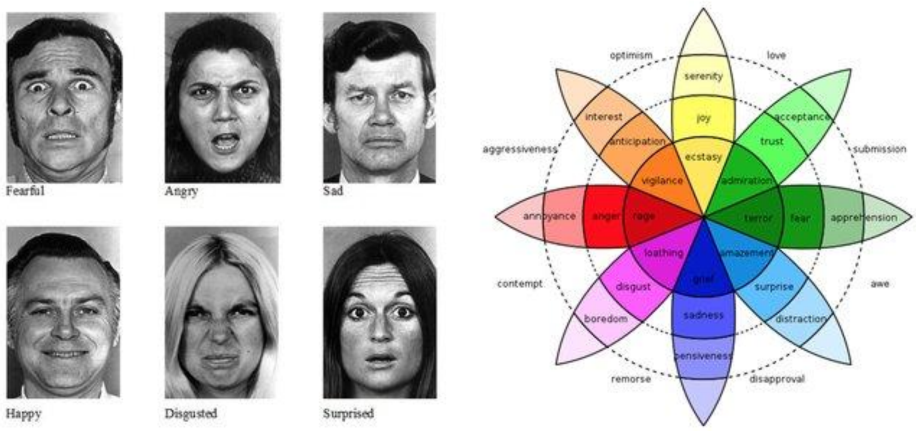

Fig. 9. Comparison of Ekman's six basic emotions and Plutchik's wheel of emotions, illustrating the structural differences between the two frameworks in terms of category discreteness and emotional relationships

Representing emotion as a set of continuous numerical scores resolves this. Each score reflects the intensity of one emotional category within a given observation, and a value of zero simply indicates that the corresponding emotion was not expressed. Multiple emotions can be represented simultaneously within the same fact record, and each score is independently aggregable across any combination of dimensions. This makes it possible to examine, for instance, how average joy and average frustration vary across service categories and seasons without requiring any structural changes to the schema.

The eight emotion categories adopted here correspond to the primary emotions defined in Plutchik's model: joy, trust, fear, surprise, sadness, disgust, anger and anticipation. These are derived using the NRC Emotion Lexicon, which maps lexical items to one or more of these categories, producing intensity scores that are normalised before storage. Each category is stored as a separate measure in F_Aspect_Sentiment_Observation, collectively forming the emotional profile of each aspect-level sentiment observation.

4.7.5 Supporting Measures

In addition to the core affective measures, F_Aspect_Sentiment_Observation includes two supporting measures that enhance the interpretability and analytical usability of sentiment data without introducing new affective dimensions.

The first is **model_confidence**, a numerical score reflecting the degree of certainty associated with the sentiment extraction process for a given observation. Automated sentiment derivation is subject to uncertainty arising from linguistic ambiguity, informal language and the inherent limitations of lexicon-based tools. By storing a confidence score alongside the affective measures, the warehouse allows analysts to account for extraction reliability when interpreting results, for instance by filtering out low-confidence observations in sensitive analytical contexts or by weighting aggregations accordingly. model_confidence is nullable, reflecting the fact that not all derivation tools produce a confidence estimate.

The second is **sentiment_strength**, a derived measure defined as the product of the absolute polarity score and the subjectivity score. The use of absolute polarity is deliberate: raw polarity carries a sign reflecting evaluative direction, and multiplying a signed value by subjectivity would cause strongly negative, highly subjective observations to produce low or negative strength values, which contradicts the intended meaning of the measure. Absolute polarity removes the directional component, so that sentiment_strength captures how forcefully an opinion is expressed regardless of whether it is positive or negative. A highly subjective review expressing strong negative sentiment and a highly subjective review expressing strong positive sentiment should both register as high-strength observations, and the formula as defined reflects that. This measure is proposed as an operational construct specific to this work rather than as an established metric in the sentiment analysis literature, and should be interpreted accordingly. Although it can be computed dynamically at query time, storing it as a pre-calculated measure simplifies common analytical queries involving sentiment intensity.

4.7.6 Fact Table Characterisation and Summary

The structure of F_Aspect_Sentiment_Observation is consolidated in Table 17, which documents the fact table's grain, size characteristics, foreign keys and the full set of measures alongside their types, nullability and default aggregation behaviour.

Table 17. Specification of F_Aspect_Sentiment_Observation

<table>
<tr><th colspan="6">Fact Table</th></tr>
<tr><td>Name</td><td colspan="5">F_Aspect_Sentiment_Observation</td></tr>
<tr><td>Description</td><td colspan="5">Stores sentiment observations associated with specific aspects mentioned in user-generated reviews of tourism and hospitality services.</td></tr>
<tr><td>Grain</td><td colspan="5">One record per sentiment observation regarding one specific aspect in one review</td></tr>
<tr><td>Size Estimation</td><td colspan="5">Expected to be High. Depends on dataset volume and average number of aspects per review.</td></tr>
<tr><td>Growth</td><td colspan="5">Proportional to review ingestion rate</td></tr>
<tr><td colspan="6">Keys</td></tr>
<tr><td>Nr</td><td>Key</td><td>Type</td><td colspan="3">Description</td></tr>
<tr><td>1</td><td>sentiment_fact_key</td><td>Surrogate Key</td><td colspan="3">Unique identifier of the fact record</td></tr>
<tr><td>2</td><td>date_key</td><td>Foreign Key</td><td colspan="3">References the Date dimension</td></tr>
<tr><td>3</td><td>service_key</td><td>Foreign Key</td><td colspan="3">References the Service dimension</td></tr>
<tr><td>4</td><td>review_key</td><td>Foreign Key</td><td colspan="3">References the Review dimension</td></tr>
<tr><td>5</td><td>aspect_key</td><td>Foreign Key</td><td colspan="3">References the Aspect dimension</td></tr>
<tr><td>6</td><td>platform_key</td><td>Foreign Key</td><td colspan="3">References the Platform dimension</td></tr>
<tr><td colspan="6">Measures</td></tr>
<tr><td>Nr</td><td>Measure</td><td>Type</td><td>Nullable</td><td>Description</td><td>Aggregation</td></tr>
<tr><td>1</td><td>polarity_score</td><td>Decimal</td><td>No</td><td>Numerical representation of evaluative orientation associated with the aspect-level sentiment observation</td><td>AVG</td></tr>
</table>

2	subjectivity_score	Decimal	No	Degree to which the sentiment observation reflects personal opinion rather than objective content	AVG
3	joy_score	Decimal	No	Intensity of joy expressed in the sentiment observation	AVG
4	trust_score	Decimal	No	Intensity of trust expressed in the sentiment observation	AVG
5	fear_score	Decimal	No	Intensity of fear expressed in the sentiment observation	AVG
6	surprise_score	Decimal	No	Intensity of surprise expressed in the sentiment observation	AVG
7	sadness_score	Decimal	No	Intensity of sadness expressed in the sentiment observation	AVG
8	disgust_score	Decimal	No	Intensity of disgust expressed in the sentiment observation	AVG
9	anger_score	Decimal	No	Intensity of anger expressed in the sentiment observation	AVG
10	anticipation_score	Decimal	No	Intensity of anticipation expressed in the sentiment observation	AVG
11	sentiment_strength	Decimal	No	Derived measure combining polarity and subjectivity to indicate how strongly an opinion is expressed	AVG
12	model_confidence	Decimal	Yes	Confidence score associated with the sentiment extraction process	AVG

All measures are represented as decimal values and are primarily analysed using average-based aggregation, reflecting their nature as normalised scores. One exception worth noting is model_confidence, which is the only nullable measure in the table. Its absence from a given record does not indicate a data quality issue but rather that the derivation tool used for that observation does not produce a confidence estimate. Queries that incorporate model_confidence as a filtering or weighting criterion should account for this when handling null values to avoid unintended exclusion of otherwise valid observations.

4.8 Data Warehouse Schema Representation

Following the definition of the dimensions and the fact table, the proposed warehouse is represented through three complementary views. Each captures a different level of detail, from the overall analytical organisation of the model down to a concrete illustration of how the declared grain manifests in practice.

4.8.1 Conceptual Schema

The conceptual schema provides a high-level representation of the warehouse, focusing on the main analytical structures and their relationships without detailing individual attributes or keys.

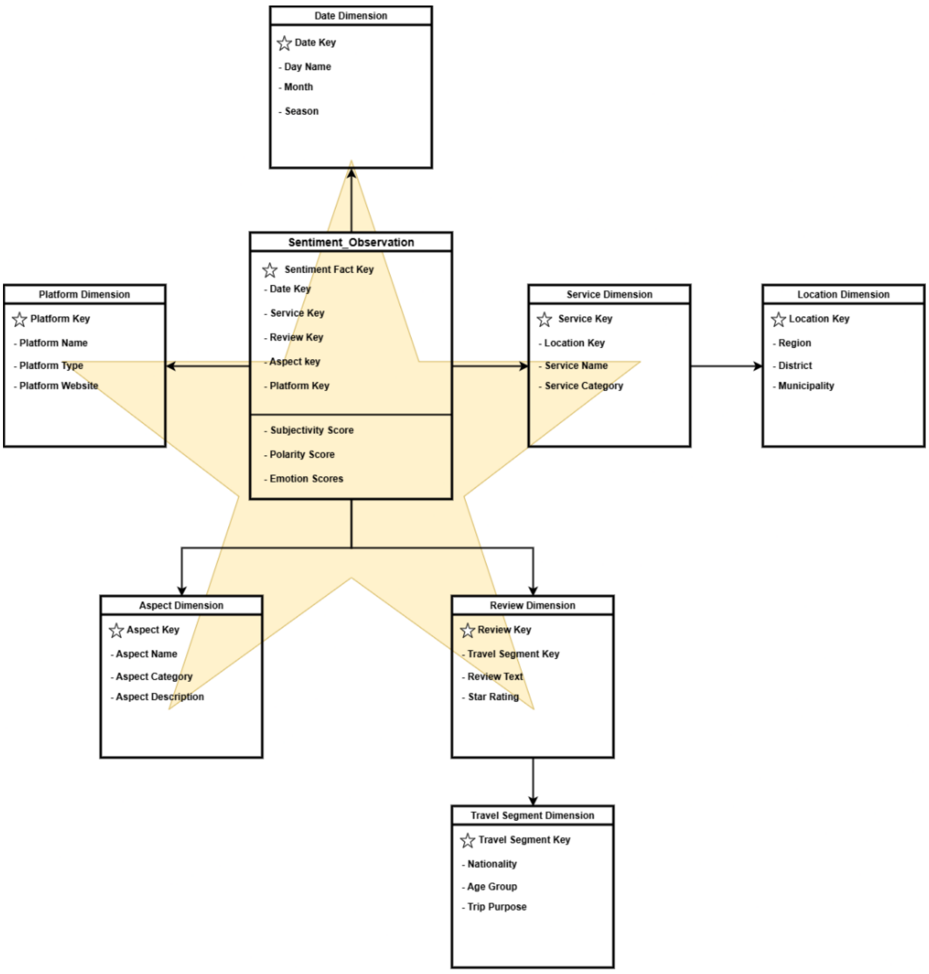

Fig. 10. Conceptual schema of the proposed data warehouse

As shown in Figure 10, the model is organised around F_Aspect_Sentiment_Observation as the central structure, directly connected to the Date, Service, Review, Aspect and Platform dimensions. The two outrigger relationships are also visible: Location depends on Service and Travel Segment depends on Review. This representation makes clear that geographic and reviewer-segment context is not directly addressable from the fact table but remains accessible through their respective parent dimensions, a structural choice whose rationale is discussed in Section 4.4.

4.8.2 Logical Schema

The logical schema translates the conceptual organisation into a more detailed view that makes explicit the tables composing the warehouse, their primary and foreign keys and the main attributes associated with each.

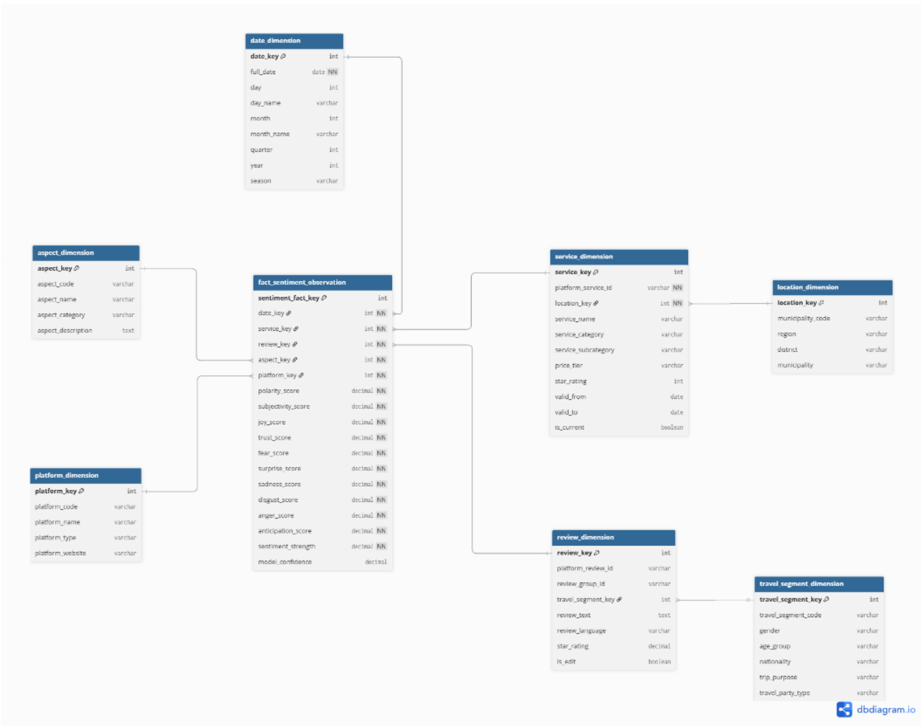

Fig. 11. Logical schema of the proposed data warehouse

This view clarifies the composition of F_Aspect_Sentiment_Observation, showing how the five foreign keys anchor each fact record within the multidimensional space and how the affective and supporting measures are stored alongside them. It also makes the SCD Type 2 structure of the Service dimension visible through its validity period attributes, reinforcing that historical versions of a service record are distinguishable at the schema level. The logical schema acts as the primary reference for physical implementation and for the source-to-target mappings defined in Chapter 5.

4.8.3 Illustration of the Declared Grain

While the conceptual and logical schemas communicate the structure of the model, neither makes the practical meaning of the declared grain immediately intuitive. Figure 12 addresses this through a concrete example.

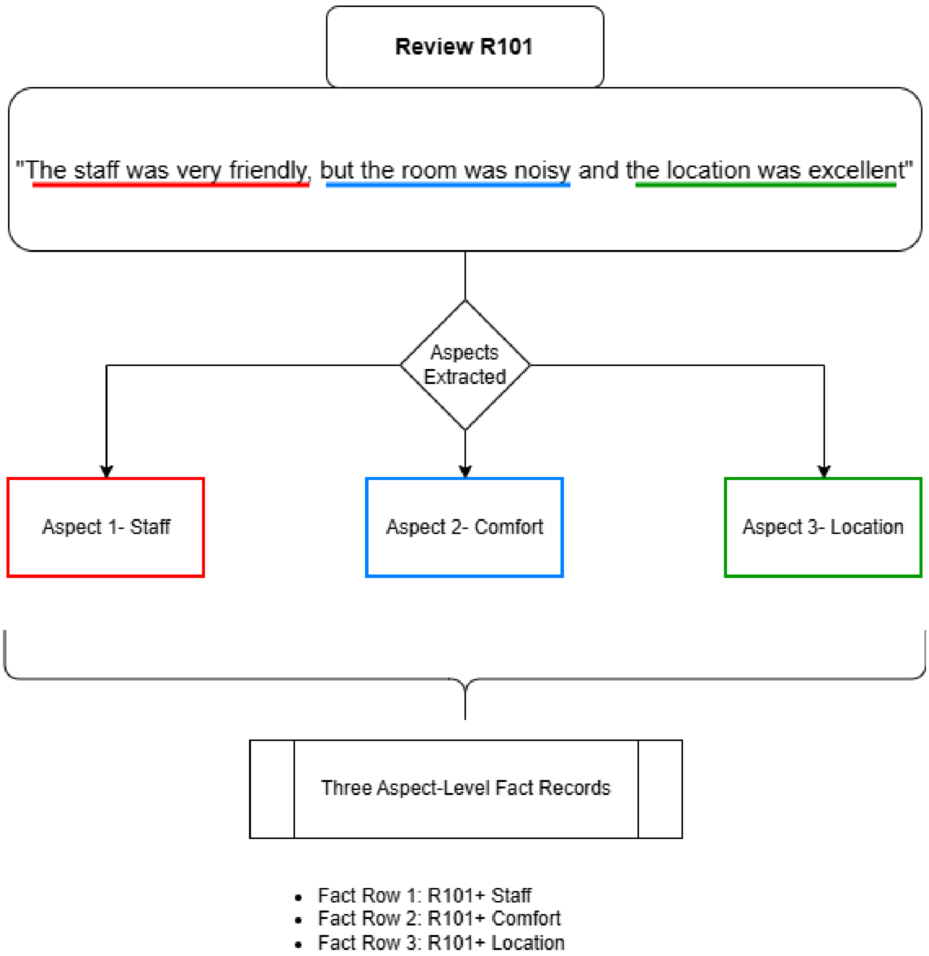

Fig. 12. Illustration of the declared fact table grain

The figure presents a single review containing references to multiple aspects of a tourism or hospitality service. Through the aspect extraction process described in Chapter 5, the review is decomposed into distinct sentiment-bearing observations, each associated with a specific aspect and each generating a separate record in F_Aspect_Sentiment_Observation. This illustration reinforces why the aspect-level grain produces a one-to-many relationship between reviews and fact records, and why aggregations intended to reflect overall service sentiment must account for that multiplicity in their design.

4.9 Summary

Chapter 4 translated the analytical contract established in Chapter 3 into a concrete dimensional design. Following Kimball's four-step methodology, the business process was selected and its analytical character examined, the grain was declared at the aspect level, and seven dimensions were identified and characterised in detail, including the outrigger structures for geographic and reviewer-segment context. The facts were then identified and their representation justified, with the modelling of polarity, subjectivity and emotion as concurrent numerical measures forming the analytical core of the fact table. The chapter concluded with conceptual, logical and grain-level schema representations that consolidate the design into visual form. The resulting model provides the structural foundation for the ETL processes and data integration work described in the following chapter.

5 ETL Process and Data Integration

Chapter 4 produced a complete dimensional design: a declared grain, seven characterised dimensions and a fact table whose measures represent polarity, subjectivity and emotion as concurrent quantitative data. That design, however, describes what the warehouse should contain and how it should be structured, not how data reaches it. Translating the model into a populated analytical environment requires a systematic process through which source data is acquired, reshaped and loaded in a form consistent with the dimensional schema. This chapter presents that process.

The ETL process described here operates over a synthetically generated dataset constructed to emulate realistic tourism and hospitality review data. The extraction phase ingests and stages the source data, the transformation phase decomposes review-level content into aspect-level observations and derives the affective measures defined in Chapter 4, and the loading phase populates the dimensional schema through a set of source-to-target mappings that account for the specific modelling decisions made throughout the design. Together these stages bridge the gap between the raw source data and the structured analytical environment the warehouse is designed to provide.

5.1 From Conceptual Design to Data Integration

In a conventional data warehouse, ETL processes operate primarily over structured operational data. Source records arrive with defined fields, known types and predictable formats, and the transformation work consists mainly of standardising representations, resolving keys and loading records in the correct order. The ETL process described in this chapter operates under different conditions. The primary source of analytical content is unstructured text, and the measures the warehouse is designed to store, namely polarity, subjectivity and emotion, are not present in the source data at all. They must be derived during the transformation phase through the application of sentiment analysis tools to aspect-level textual fragments extracted from each review.

This derivation requirement, combined with the aspect-level grain of the warehouse, means that the transformation phase carries more analytical responsibility than it would in a standard ETL context. A single source review may generate multiple fact records, each requiring its own independently derived affective measures. The ETL process must therefore manage not only the structural transformation from review-level to aspect-level data, but also the normalisation and conformance of derived affective attributes into warehouse-compatible representations. The sections that follow address each stage of this process in detail.

5.2 Role of ETL in Sentiment-Aware Data Warehousing

In most warehousing contexts, ETL processes operate over data that is already structured and whose analytical attributes are present in the source records, even if they require standardisation or transformation. The role of ETL in a sentiment-aware warehouse differs in a fundamental way: the attributes that define the analytical value of each fact record do not exist in the source data. Polarity, subjectivity and emotion must be derived computationally from unstructured text during the transformation phase. This means ETL is not only responsible for integrating data but for generating the measures the warehouse is built around.

A further consequence follows from the aspect-level grain. The source data is organised at the review level, where one record corresponds to one user-generated review. The warehouse requires data at the aspect level, where one record corresponds to one sentiment observation associated with a specific service characteristic. This structural gap cannot be resolved through standard format normalisation or key resolution. It requires the decomposition of review content into aspect-level fragments, the derivation of independent affective measures for each fragment and the preservation of the relationships between those fragments and their source review, service and temporal context throughout the transformation process.

These two requirements, derived measures and grain transformation, shape the design of the ETL process described in the sections that follow and distinguish it from a conventional data integration pipeline.

5.3 Source Data Characterisation

The ETL process described in this chapter operates over a synthetically generated dataset designed to replicate the structural and semantic characteristics of real-world user-generated reviews within the Portuguese tourism and hospitality domain. The dataset is synthetic in origin to avoid the copyright and licensing constraints associated with real platform data, but its structure reflects the properties of review-based platforms, including the presence of unstructured textual content alongside structured contextual metadata. This combination of structured and unstructured components defines the principal challenges the ETL process must address.

The source dataset can be characterised according to the following main elements:

- **Nature of the Data:** The dataset consists of synthetically generated observations emulating real-world review records. Each observation includes both unstructured and structured components, reflecting the dual character of user-generated content in analytical environments.

- **Textual Component:** The primary source of analytical content is the review text, which represents an individual user's evaluation of a tourism or hospitality service. This content is unstructured and carries the sentiment-related information, including polarity, subjectivity and emotion, that the warehouse is designed to store.

None of this information is explicitly labelled in the source record; it must be derived during transformation.

- **Structured Contextual Attributes**: Each review is associated with a set of structured attributes providing analytical context, including service identification, service category, geographic location and review date. These attributes support the dimensional alignment required for warehouse loading but do not require affective derivation.

- **Initial Data Organisation:** At the source level, data is organised at the review level: each record corresponds to one user-generated review linked to one service. This structure reflects the natural organisation of review platforms but differs from the aspect-level grain of the proposed warehouse.

- **Granularity Mismatch:** The review-level structure of the source data does not align with the aspect-level granularity of the fact table. A single review may reference multiple service characteristics, each requiring an independent sentiment observation. Resolving this mismatch is the central transformation challenge addressed in Section 5.6.

These characteristics define the gap between source and target that the ETL process must bridge. The transformation from review-level to aspect-level data, the derivation of affective measures and the alignment of contextual attributes with the defined dimensional structures are the three core operations required to close it.

The nature of this gap is illustrated in Figure 13, which shows an annotated review record drawn from a public review platform. Although this example originates outside the synthetic dataset, it represents the structural properties that the synthetic data is designed to emulate: structured contextual attributes coexist with a block of unstructured textual content in which multiple service aspects, such as queue times, pricing and staff behaviour, are implicitly embedded rather than explicitly labelled.

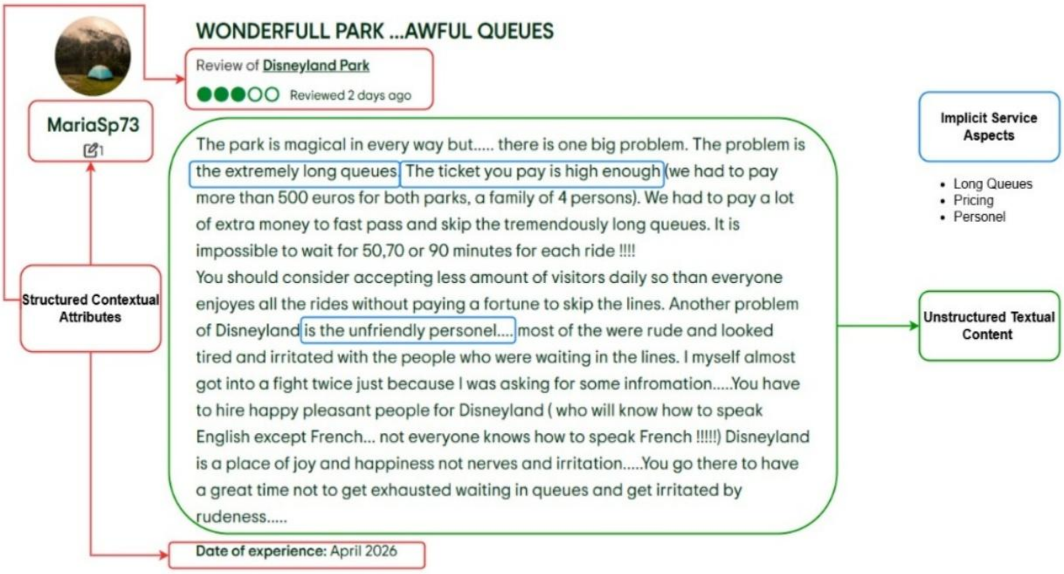

Fig. 13. Example of a user-generated review illustrating the coexistence of structured contextual attributes and unstructured textual content, and the implicit presence of multiple service aspects within the review text

5.4 ETL Requirements and Design Principles

The characteristics of the source data described in Section 5.3, together with the dimensional model defined in Chapter 4, establish a set of requirements that govern the ETL process. The following principles guide its definition and apply across all stages of extraction, transformation and loading.

Alignment with the Defined Data Grain.

The ETL process must preserve consistency with the aspect-level granularity of the dimensional model. Each output record must correspond to a single sentiment observation associated with a specific service aspect within a review, requiring the decomposition of review-level source data into multiple independent analytical observations.

Transformation of Unstructured Textual Data.

The primary source of analytical content consists of unstructured review text. The ETL process must convert this content into structured representations compatible with the multidimensional schema, enabling its storage, aggregation and comparison within the warehouse.

Extraction and Structuring of Service Aspects

Reviews typically reference multiple service characteristics within the same text. The ETL process must identify and isolate these aspects, mapping each to a standardised entry in the Aspect dimension so that each can be represented as an independent analytical element.

Derivation of Sentiment-related Attributes

Polarity, subjectivity and emotion are not present in the source data and must be generated during transformation. The ETL process is responsible for deriving these attributes and representing them in a consistent normalised format suitable for warehouse storage.

Conformance with the Dimensional Model

Transformed data must align with the dimensional structures defined in Chapter 4, including correct population of dimension tables, resolution of surrogate keys and enforcement of the relationships between fact records and their associated dimensions.

Preservation of Contextual Relationships

The relationships between reviews and their associated service, temporal and platform context must be maintained throughout the transformation process, ensuring that each analytical observation remains correctly linked to its source context.

Data Consistency and Integrity

The ETL process must avoid duplicate records, validate attribute values and enforce referential integrity between fact and dimension tables.

Reproducibility

The transformation workflow must be defined in a way that allows consistent re-execution, producing equivalent analytical outputs from the same input data. This is particularly relevant given the synthetic nature of the source dataset.

5.5 Extraction Phase

The extraction phase is the initial stage of the ETL process, responsible for ingesting the source dataset and preparing it for subsequent transformation. In this work, the source is a synthetically generated dataset rather than a live operational system, so the phase does not involve real-time retrieval or connection to external platforms. Its function is to ingest the review-level data into a staging environment where it can be accessed consistently by the transformation operations that follow. The staging layer preserves the original structure of the source data and acts as the controlled entry point for all subsequent processing.

During this phase, basic validation steps are applied to ensure the data is suitable for transformation. These include verification of data completeness, identification of missing or malformed records and standardisation of basic formats such as date representations. No structural modifications are made at this stage. The decomposition of review content into aspect-level observations and the derivation of affective attributes are deferred entirely to the transformation phase described in the following section.

5.6 Transformation Phase

The transformation phase is where the core analytical work of the ETL process occurs. While the extraction phase delivers staged, review-level source data, the transformation phase is responsible for converting that data into the aspect-level, sentiment-enriched records the warehouse is designed to store. This involves two broad lines of processing. The first concerns the unstructured textual content of the reviews: decomposing it into aspect-level fragments, mapping those fragments to the controlled aspect taxonomy and deriving polarity, subjectivity and emotion values for each. The second concerns the structured metadata associated with services, platforms, reviewers and locations: standardising, classifying and conforming it to the dimensional structures defined in Chapter 4 so that surrogate keys can be resolved and fact records can be properly anchored.

These two lines of processing operate in parallel but converge at the conformation stage, which assembles the fact-ready observations that the loading phase will insert into the warehouse. The subsections that follow address each transformation step in the order it occurs within this pipeline.

5.6.1 Data Cleaning and Preprocessing

Before the core transformation operations can proceed, the source data undergoes a set of preliminary cleaning and preprocessing steps. Given that the input is a synthetically generated dataset with a controlled structure, these operations are relatively limited in scope, focusing on standardisation and validation rather than extensive data correction.

The preprocessing steps are applied to both structured attributes and textual content:

- **Validation of structured attributes**, ensuring that required fields such as service identifiers, review identifiers and timestamps are complete and correctly formatted
- **Standardisation of formats**, including consistent date representations, categorical value normalisation and uniform encoding of textual fields
- **Handling of missing or inconsistent values**, through the identification of incomplete records and the application of predefined default values or exclusion rules where appropriate
- **Removal of extraneous characters**, such as formatting artifacts, duplicated whitespace or non-informative symbols that may interfere with subsequent processing
- **Text normalisation**, including case standardisation and basic formatting alignment to reduce variability in textual expressions

These operations ensure that the dataset is consistent and well-formed before it enters the more complex transformation steps that follow. The decomposition of reviews into aspect-level observations and the derivation of affective attributes are not performed here; both are handled in the subsequent stages described in Sections 5.6.2 and 5.6.3.

5.6.2 Aspect Identification and Structuring

Following preprocessing, the next transformation step identifies and structures the service aspects mentioned in each review. Since the warehouse operates at aspect-level grain, review-level textual data must be decomposed into a set of aspect-level observations before affective attributes can be derived and loaded. Each output observation corresponds to a specific service characteristic addressed within a review, and a single review may produce several such observations.

Aspect identification in this work is implemented as a controlled mapping process. Rather than attempting to discover arbitrary aspects from text, the ETL process maps review fragments to a predefined set of service aspects represented in the Aspect dimension. This ensures consistency between the textual transformation process and the dimensional model and maintains analytical comparability across reviews, services and service categories.

The predefined aspect taxonomy is derived from the tourism and hospitality domain and reflects the service characteristics most commonly addressed in user-generated reviews. For accommodation services, relevant aspects include room quality, cleanliness, staff quality, location and perceived value. For restaurants and bars, aspects include food quality, service efficiency, atmosphere and price. For tourist sites, aspects include accessibility, crowding, preservation state, informational value and overall experience. These standardised categories provide the analytical vocabulary through which review content is structured and stored.

The sequence of operations involved in this step is illustrated in Figure 14, which shows the transformation of review-level text into intermediate review-aspect observations.

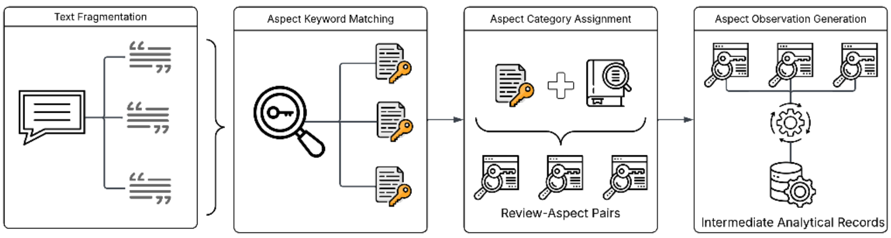

Fig. 14. Aspect identification and structuring process, illustrating the transformation of review-level textual content into intermediate review-aspect observations prior to sentiment derivation

The process follows four main steps:

1. **Text Fragmentation**. Each review is divided into smaller textual fragments, such as sentences or clauses, to isolate distinct statements within the review. This step is necessary because a single review may contain multiple opinions referring to different service aspects.
2. **Aspect Keyword Matching**. Each fragment is compared against a predefined dictionary of keywords and expressions associated with possible service aspects. Terms such as "room", "bed" or "bathroom" may indicate room quality, while terms such as "staff", "reception" or "waiter" may indicate staff quality.

3. **Aspect Category Assignment**. Detected keyword evidence is mapped to the standardised aspect categories in the Aspect dimension. If a fragment references more than one aspect, the process generates multiple observations from the same fragment, one for each assigned category.

4. **Aspect-level Observation Generation**. For each detected review-aspect pair, an intermediate analytical record is created preserving the relationship between the original review, the identified aspect and the corresponding textual fragment. These records are the input for sentiment derivation in the following section.

For example, the fragment "The room was clean and comfortable, but the staff was unfriendly" may be decomposed into separate observations associated with room quality, cleanliness and staff quality, each of which will receive independently derived sentiment measures. Without this decomposition, distinctions between aspects that receive different evaluations within the same review would be lost, collapsing analytically distinct signals into a single aggregate value. At the end of this step the output is not yet a fully populated fact record; polarity, subjectivity and emotion have not yet been derived. The result is a set of aspect-level textual observations ready for the sentiment derivation process described in the following section.

5.6.3 Sentiment Derivation and Representation

After aspect identification, the transformation process proceeds to derive the affective attributes associated with each observation. The input at this stage is no longer the complete review text but a set of intermediate review-aspect observations, each containing a textual fragment associated with a specific service aspect. Applying sentiment derivation to these fragments rather than to the full review text ensures that each affective measure reflects the sentiment expressed toward that particular aspect, preserving the analytical grain the warehouse is built around.

The three affective components considered in this work are discussed in detail in Chapter 2, and their representation as fact table measures is justified in Sections 4.7.2 through 4.7.4. The purpose of this section is narrower: to describe how each is derived operationally within the ETL process. The objective is not to propose new sentiment extraction algorithms but to apply established tools within a structured, reproducible transformation pipeline that produces warehouse-compatible affective attributes.

Polarity Derivation

Polarity derivation is the first affective transformation applied to each aspect-level observation. The input is the textual fragment associated with a specific service aspect generated during the previous step, ensuring that the resulting score reflects the evaluative orientation expressed toward that aspect rather than the review as a whole.

Polarity is represented as a normalised numerical score in the interval [-1.0, 1.0], where negative values indicate negative sentiment, positive values indicate positive sentiment and values close to zero indicate neutral or weakly expressed sentiment. This representation supports aggregation, ranking and trend analysis and is consistent with established sentiment analysis practice, where polarity is commonly expressed as a continuous score rather than a discrete label (Liu, 2012); (Pang & Lee, 2008).

Polarity derivation is implemented through a lexicon-based and rule-supported scoring procedure. Sentiment-bearing terms are associated with predefined orientation values, commonly referred to as valence scores, and the polarity score of the fragment is obtained by aggregating the contribution of detected terms after applying linguistic rules that adjust their strength or orientation. This approach is transparent and reproducible, allowing the transformation logic to be explicitly described and inspected rather than treated as a black-box classification. Taboada et al. (2011) describe this type of procedure as relying on dictionaries of words annotated with semantic orientation and strength, which are combined to compute the sentiment orientation of a text.

VADER, the tool selected for polarity derivation in this work, operationalises this approach for user-generated and informal text. It combines a sentiment lexicon with rules for handling intensifiers, negation, punctuation and capitalisation, producing a compound score normalised between -1 and +1 that is directly compatible with the polarity representation adopted here (Hutto & Gilbert, 2014). TextBlob provides an alternative implementation with a simpler rule set, returning a polarity float in the same interval alongside a subjectivity score, and is referenced here as a supplementary illustration of a warehouse-compatible polarity output.

The polarity derivation process can be summarised as follows:

1. The aspect-level fragment is received as input
2. Sentiment-bearing terms are identified within the fragment
3. Each detected term is associated with a valence score from the sentiment lexicon
4. Modifier rules covering intensification, negation and punctuation emphasis are applied where relevant
5. The adjusted values are aggregated into a single polarity score
6. The resulting score is normalised to the interval [-1.0, 1.0]

These modifier adjustments are analytically important because raw word-level polarity alone may not accurately represent the evaluative meaning of a complete fragment. Intensifiers increase the strength of a sentiment expression, as in "very clean." Negation reverses or reduces the expected orientation of a term, as in "not friendly." Weak modifiers such as "slightly" reduce the intensity of an expression without reversing its direction. Table 18 illustrates these adjustments across a set of aspect-level fragment examples.

Table 18. Example of polarity derivation for aspect-level fragments

Review Fragment	Aspect	Sentiment Evidence	Polarity Adjustment	Final Polarity Score
“The room was very clean”	Cleanliness	clean	intensified by “very”	0.72
“The staff was not friendly”	Staff Quality	friendly	reversed by “not”	-0.45
“The location was excellent”	Location	excellent	strong positive term	0.85
“The queue was slightly long”	Waiting Time	long	weakened by “slightly”	-0.25

The resulting polarity score is stored as a measure in F_Aspect_Sentiment_Observation. Although categorical labels may be derived from the score for reporting purposes, the numerical value remains the primary warehouse representation because it preserves the magnitude distinctions that aggregation and trend analysis depend on.

Subjectivity Derivation

Subjectivity derivation is applied to the same aspect-level fragments used for polarity derivation. Where polarity captures the evaluative direction of a fragment, subjectivity estimates the extent to which it reflects personal opinion or judgment rather than factual description. The distinction matters analytically because two fragments may reference the same aspect with the same polarity score while carrying very different degrees of opinionated expression. A factual statement about location distance and a strongly personal evaluation of the same location are not analytically equivalent, even when both are classified as positive.

Subjectivity is represented as a normalised numerical score in the interval [0.0, 1.0], where values close to zero indicate objective or largely factual statements and values close to one indicate highly subjective expressions. This representation treats subjectivity as a continuous property rather than a binary distinction, which is consistent with the analytical treatment of the measure established in Section 4.7.3 and with prior work on distinguishing private states from objective statements in natural language (Wiebe, Wilson, & Cardie, 2005); (Liu, 2012).

Subjectivity scoring is implemented by identifying linguistic indicators associated with opinionated language, including evaluative adjectives, affective expressions, personal markers and intensifiers. Terms such as "excellent", "terrible" or "disappointing" are characteristic of subjective evaluation, while expressions such as "500 meters from the station" or "open until 10 p.m." carry low subjectivity because they primarily convey factual information. Modifiers affect the final score: "the location was absolutely perfect" expresses stronger subjectivity than "the location was perfect" due to the intensifier, while "the hotel is located near the train station" receives a low score regardless of any positive connotation because its content is descriptive rather than evaluative. TextBlob operationalises this scoring by returning a subjectivity value in the interval [0.0, 1.0] alongside its polarity output, providing a practical mechanism for producing warehouse-compatible subjectivity attributes from aspect-level text.

The subjectivity derivation process proceeds as follows:

1. The aspect-level fragment is received as input
2. Linguistic indicators of opinionated expression are identified, including evaluative adjectives, affective terms and personal markers
3. Each identified term is weighted according to its degree of subjectivity
4. Intensifiers and other modifiers are applied where relevant
5. The weighted evidence is aggregated and normalised to the interval [0.0, 1.0]

Table 19 illustrates subjectivity derivation across a set of contrasting fragment examples, showing the distinction between factual and evaluative expression across the same aspect categories.

Table 19. Example of subjectivity derivation for aspect-level fragments

Review Fragment	Aspect	Subjectivity Evidence	Interpretation	Subjectivity Score
“The hotel is 500 meters from the station”	Location	Factual distance information	mostly objective	0.10
“The location was absolutely perfect”	Location	Perfect, intensified by “absolutely”	highly subjective	0.92
“The room had two beds and a balcony”	Room Quality	Descriptive room attributes	mostly objective	0.18
“The room felt uncomfortable and outdated”	Room Quality	Felt, uncomfortable, outdated	strongly evaluative	0.84
“The staff were available at reception”	Staff Quality	Factual availability statement	mostly objective	0.22
“The staff were incredibly kind and helpful”	Staff Quality	Kind, helpful, intensified by “incredibly”	highly subjective	0.88

The subjectivity score complements polarity by indicating how strongly opinionated a given sentiment expression is, supporting analytical queries that filter or weight observations by their degree of evaluative intensity.

Emotion Derivation

Emotion derivation is the third and final affective transformation applied to each aspect-level observation. Where polarity captures evaluative direction and subjectivity measures the degree of opinionated expression, emotion derivation identifies the qualitative affective state conveyed in the fragment. Two fragments with identical polarity scores may carry meaningfully different emotional content: a negative evaluation driven by anger signals something different analytically from one driven by sadness or fear, and collapsing these into a single polarity value discards information that the warehouse is designed to preserve.

Emotion derivation is applied at the aspect-fragment level, using the same textual input as polarity and subjectivity derivation. This ensures that emotional signals are associated with the specific aspect being evaluated. In a review stating that "the location was beautiful, but the staff were rude and unhelpful," the positive emotional content should be associated with the location observation and the negative with the staff quality observation. Applying derivation to the fragment level rather than the full review preserves that distinction.

The derivation is implemented through a lexicon-based affective scoring approach. Each term or expression in the fragment is matched against a predefined emotion lexicon where entries are annotated with one or more emotion categories. The NRC Emotion Lexicon (Mohammad & Turney, 2013) is the resource used in this work, associating English words with the eight basic emotions defined in Plutchik's model: joy, trust, fear, surprise, sadness, disgust, anger and anticipation. The use of a predefined lexicon ensures that emotion labels are drawn from a controlled set of categories rather than generated arbitrarily during processing, which is necessary for consistent aggregation and comparison within the warehouse.

The emotion derivation process proceeds as follows:

1. The aspect-level textual fragment is received as input
2. Terms and expressions within the fragment are identified and matched against the NRC Emotion Lexicon
3. Each matched term contributes to one or more of the eight predefined emotion categories
4. Contributions across all matched terms are aggregated into per-category emotion scores for the fragment
5. The resulting values are normalised into warehouse-compatible fields corresponding to the eight emotion measures in F_Aspect_Sentiment_Observation

The result is not a single emotion label but a set of independent numerical scores, one per category. A fragment may carry nonzero values across several categories simultaneously, reflecting the coexistence of multiple emotional signals in natural language. The fragment "the staff was rude and frustrating" may produce a high anger score and a lower disgust score, while "the room was cozy and pleasant" may produce stronger joy and trust scores. Table 20 illustrates this across several aspect-level fragment examples.

Table 20. Example of emotion derivation for aspect-level fragments

Review Fragment	Aspect	Emotion Evidence	Emotion Scores
“The staff was rude and frustrating”	Staff Quality	rude, frustrating	anger: 0.80; disgust: 0.40
“The room was cozy and pleasant”	Room Quality	cozy, pleasant	joy: 0.70; trust: 0.45
“The queue was stressful and endless”	Waiting Time	stressful, endless	anger: 0.65; fear: 0.35
“The view was a wonderful surprise”	Location	wonderful, surprise	joy: 0.75; surprise: 0.60

By deriving emotion at the aspect-fragment level and representing it as a set of independent numerical measures, the ETL process preserves affective detail that a single dominant label would suppress. Each emotion score is independently aggregable across any combination of dimensions, allowing analysts to examine, for instance, how average anger scores vary across service categories, how joy scores evolve over time or which aspects are most frequently associated with fear or disappointment. This expressiveness is only possible because emotion is stored as structured quantitative data rather than as a categorical output of the derivation process.

5.6.4 Conformation to the Dimensional Model

After polarity, subjectivity and emotion values have been derived, each observation is sentiment-enriched but not yet warehouse-ready. The intermediate records produced by the transformation pipeline contain source-level identifiers and derived affective measures, but the fact table does not store source-level identifiers directly. Before a record can be inserted into F_Aspect_Sentiment_Observation, each source reference must be resolved to its corresponding warehouse surrogate key. Conformation is the process of performing that resolution while ensuring that the dimensional records the keys reference actually exist.

Since fact records depend on dimension records through foreign keys, the ETL process must confirm that all required dimension entries are present before any fact row is written. Static dimensions such as Date and Location are preloaded and available throughout. Others, such as Service and Review, are populated or updated during the ETL run as review records are processed. Only once the necessary dimension records exist can their surrogate keys be resolved and the fact-ready observation assembled.

Dimension preparation and key resolution

The preparation logic and key resolution method differs by dimension. Table 21 summarises these differences.

Table 21. Dimension preparation and key resolution before fact loading

Dimension / Structure	ETL Preparation	Key Resolution for Fact Loading
Date Dimension	Preloaded with calendar records for the defined temporal window	The standardised review date is matched to date_key
Location Dimension	Preloaded or maintained as a static list of Portuguese municipalities	Location is resolved through the Service dimension, not directly in the fact table
Service Dimension	Inserted or updated before fact loading, applying SCD Type 2 rules where necessary	The source service identifier and review date are used to select the correct service_key
Aspect Dimension	Preloaded or maintained using the controlled aspect taxonomy	The detected aspect is matched to the corresponding aspect_key
Travel Segment Dimension	Populated with the relevant visitor segment combinations	The travel segment is linked through the Review dimension
Review Dimension	Inserted before fact loading, preserving review-level metadata and edited-review grouping	The source review identifier is matched to review_key
Platform Dimension	Preloaded or populated with the source platforms considered in the dataset	The standardised platform identifier is matched to platform_key

Handling dimension-specific rules

Several dimensions require additional logic beyond a straightforward lookup, reflecting the modelling decisions made in Chapter 4.

The **Service dimension** is modelled as SCD Type 2, which means the ETL process cannot retrieve a single current record for each service. The version of the service that was valid at the time the review was published must be identified. This is done by selecting the service record whose valid_from and valid_to interval encompasses the review date, ensuring that the resulting foreign key reflects the state of the service when the reviewer's experience actually occurred.

The **Aspect dimension** uses Type 1 logic, so aspect conformation consists of a direct match between the detected aspect and its corresponding aspect_code in the controlled taxonomy. If taxonomy labels have been refined since the previous ETL run, the updated value replaces the previous one, since such changes are treated as corrections rather than historical events.

The **Review dimension** requires care around traceability. Because one review generates multiple fact records, several rows in F_Aspect_Sentiment_Observation will reference the same review_key. Where a review has been edited, the edited version is stored as a separate review record while its logical connection to the original is preserved through the review_group_id mechanism established in Chapter 4.

Location and Travel Segment are resolved indirectly. Location is accessed through the Service dimension and Travel Segment through the Review dimension, so neither requires a direct foreign key lookup against the fact record during conformation.

At the end of this step, each observation contains valid surrogate keys for all five dimensions alongside its full set of derived affective measures. Table 22 illustrates the transition from intermediate to fact-ready form.

Table 22. Example of conformation from intermediate observation to fact-ready record

Intermediate Element	Conformation Rule	Fact-Ready Output
Review date: 2024-04-15	Match to preloaded Date dimension	date_key = 20240415
Source service ID: TRIP_458921	Match Service dimension version valid on review date	service_key = 10245
Detected aspect: Staff Quality	Match controlled Aspect taxonomy	aspect_key = 12
Source review ID: REV_90031	Match inserted Review dimension record	review_key = 8001
Source platform: TripAdvisor	Match Platform dimension record	platform_key = 3
Derived polarity: -0.68	Preserve as fact measure	polarity_score = -0.68
Derived subjectivity: 0.82	Preserve as fact measure	subjectivity_score = 0.82
Derived emotion scores	Map to fact-level emotion fields	anger_score = 0.76, joy_score = 0.00

These fact-ready records constitute the input to the loading phase described in Section 5.8. Conformation prepares them; loading performs the insertions that make them available for analytical processing.

5.7 Source to Target Mapping

Source-to-target mapping defines the correspondence between pre-ETL source attributes, intermediate transformation outputs and the target structures of the data warehouse. For each dimension and the fact table, it specifies which values are loaded directly, which require transformation and which are generated during the ETL process, ensuring full traceability from synthetic source data through to the populated warehouse schema. The following subsections present these mappings in order, each supported by a source-to-target diagram illustrating the flow from source fields to target attributes.

5.7.1 Date Dimension Mapping

The Date dimension is populated from the review date associated with each source record. Unlike other dimensions whose records are derived from review content or service metadata, the Date dimension follows a predefined calendar structure and can be prepared independently of the remaining source data. The review date is used to resolve the corresponding date_key and to derive the full set of calendar attributes that populate the dimension.

As illustrated in Figure 15, the mapping process applies three transformation rules to the source review date. The first performs date parsing and format standardisation, producing a consistent date representation and generating the date_key in YYYYMMDD integer format alongside the full_date attribute. The second extracts direct calendar components from the standardised date, populating day_of_month, month and year. The third derives higher-level descriptive attributes, including day_name, month_name, quarter and season, to support the analytical perspectives the dimension is designed to enable.

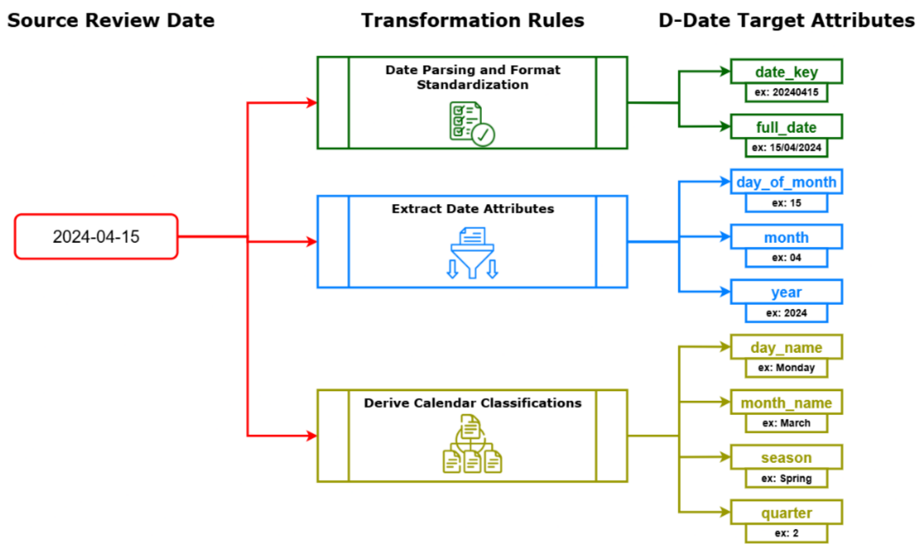

Fig. 15. Source-to-target mapping for the Date dimension, showing the transformation of the source review date into standardised calendar attributes used in the data warehouse.

The season attribute warrants specific mention in the domain context of this work. Tourism and hospitality review patterns vary significantly across seasons, and making season directly available as a dimension attribute avoids requiring analysts to derive it at query time from month or quarter values.

5.7.2 Location Dimension Mapping

The Location dimension is populated from official Portuguese administrative reference data, specifically the Carta Administrativa Oficial de Portugal (CAOP). Unlike the dimensions derived from review or service records, this dimension is constructed entirely from a controlled geographic reference source and can be preloaded independently of any review processing. Its records represent Portuguese municipalities, each assigned to a district and region through the administrative hierarchy that CAOP defines.

As illustrated in Figure 16, the mapping applies three transformation rules to the source reference data. The first extracts and standardises municipality-level records, ensuring that municipality names and codes follow a consistent representation before being loaded as municipality_code and municipality. The second resolves the administrative hierarchy for each municipality, assigning the corresponding district and region values. The third generates the surrogate key for each record, producing the location_key used internally by the warehouse to associate services with their geographic context.

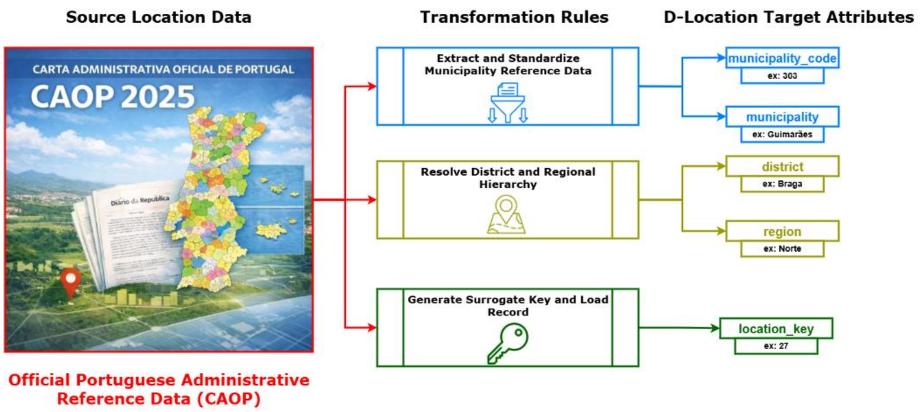

Fig. 16. Source-to-target mapping for the Location dimension, showing how official Portuguese administrative reference data from CAOP is transformed into the attributes of D_LOCATION.

5.7.3 Service Dimension Mapping

The Service dimension is populated from service profile metadata extracted from the source dataset. Each source profile represents a tourism or hospitality entity and contains a broader set of attributes than those ultimately stored in the warehouse, including ranking information, amenities, photo counts and extraction timestamps. Only the attributes required by the dimensional model are selected, standardised and loaded into D_SERVICE.

As illustrated in Figure 17, the mapping applies four transformation rules to the source profile. The first standardises the main descriptive and classification attributes. The service name is cleaned and mapped to service_name, the original platform identifier is preserved as platform_service_id to maintain traceability to the source entity, and raw service type information is mapped to the controlled service_category and service_subcategory values. The star_rating and price_tier attributes are validated and standardised to ensure consistency across services from different source platforms.

The second transformation rule resolves the geographic reference. Since location is modelled separately in D_LOCATION, the municipality information extracted from the source profile is matched against the already-populated Location dimension to produce the corresponding location_key. This avoids repeating geographic attributes in the Service dimension and ensures that all services reference a standardised, hierarchy-consistent location record.

The third rule applies SCD Type 2 versioning logic. When the ETL process detects a change in a tracked service attribute, such as category, subcategory, star rating or price tier, the existing service record is closed by setting its valid_to date and marking is_current as false. A new versioned record is then inserted with the updated attribute values and a new valid_from date. This ensures that each fact record in F_Aspect_Sentiment_Observation references the version of the service that accurately described it at the time the review was written, preserving historical consistency across the warehouse.

The fourth rule generates the internal surrogate key, service_key, and loads the completed record into D_SERVICE.

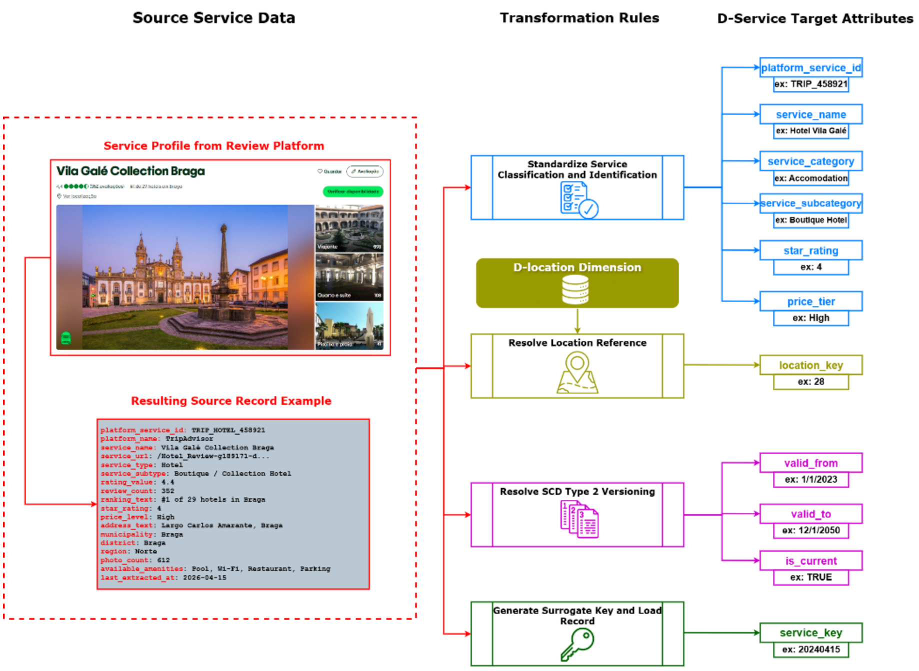

Fig. 17. Source-to-target mapping for the Service dimension, showing how service profile metadata is extracted, standardised, linked to the Location dimension and versioned according to SCD Type 2 rules before being loaded into D_SERVICE. The service profile shown is drawn from a public review platform and is used here as a structural illustration consistent with the synthetic source data the warehouse operates over

5.7.4 Aspect Dimension Mapping

Unlike the dimensions populated from review or service records, the Aspect dimension is constructed from a controlled taxonomy defined specifically for this work. Its records are not extracted from source data but designed in advance to provide the standardised analytical vocabulary against which review content is structured during the aspect identification step in Section 5.6.2.

As illustrated in Figure 18, the construction process takes a set of taxonomy design inputs (service evaluation criteria, aspect identification rules, review analysis requirements and domain knowledge) and applies four construction rules to produce the dimension records and the auxiliary detection structure.

The first rule defines the controlled aspect categories. These provide the higher-level grouping structure used for analytical aggregation, allowing individual aspects to be analysed both at their own level and rolled up to a broader category. The second rule defines the individual aspect entries, assigning each a stable aspect_code, a human-readable aspect_name and an aspect_description that clarifies its analytical meaning. For example, the aspect code STAFF_QUALITY corresponds to the name

"Staff Quality" with a description covering the evaluation of staff behaviour, friendliness and professionalism.

The third rule associates matching terms with each aspect for use during the identification process in Section 5.6.2. Terms such as "staff", "receptionist", "waiter", "rude" and "helpful" are assigned to STAFF_QUALITY as detection evidence. These terms are not loaded into D_ASPECT; they form a separate aspect detection dictionary used only during transformation. Keeping this dictionary outside the dimension preserves the analytical cleanliness of the taxonomy while still supporting flexible keyword-based aspect identification in review text.

The fourth rule generates the surrogate key for each aspect record, producing the aspect_key used by the fact table to reference the corresponding dimension entry.

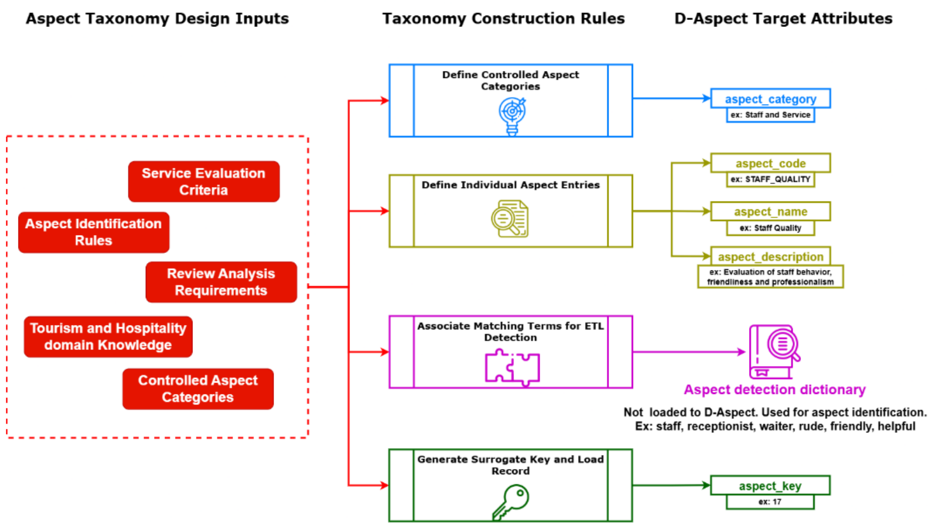

Fig. 18. Source-to-target mapping for the Aspect dimension, showing how taxonomy design inputs are transformed into controlled aspect records and the supporting aspect detection dictionary.

5.7.5 Travel Segment Dimension Mapping

The Travel Segment dimension is populated from reviewer and trip metadata associated with each source review. Unlike dimensions built from controlled reference data, this dimension is populated dynamically during review processing, since segment combinations are only known once the reviewer and trip attributes associated with each review have been extracted and standardised.

As illustrated in Figure 19, the mapping applies three transformation rules to the source reviewer and trip data. The first extracts and standardises the individual segment attributes. Raw trip labels such as "Travelled with family" are normalised into controlled values for travel_party_type, while reviewer profile metadata is standardised into nationality, age_group, gender and trip_purpose. Where segment attributes are unavailable in the source record, null values are preserved rather than imputed.

The second rule generates a stable travel_segment_code from the standardised attribute combination. This code acts as a natural identifier for the segment profile, allowing identical combinations to be detected and reused across multiple reviews. A French reviewer aged 25 to 34 travelling with family for leisure purposes would produce a code such as FR_25_34_FAMILY_LEISURE, which remains consistent regardless of which review it is derived from.

The third rule resolves the surrogate key. If the generated segment combination already exists in D_TRAVEL_SEGMENT, the existing travel_segment_key is reused. If it does not, a new record is inserted and a new key is assigned. This approach ensures the dimension contains only segment profiles observed in the actual dataset rather than a preloaded enumeration of all possible combinations.

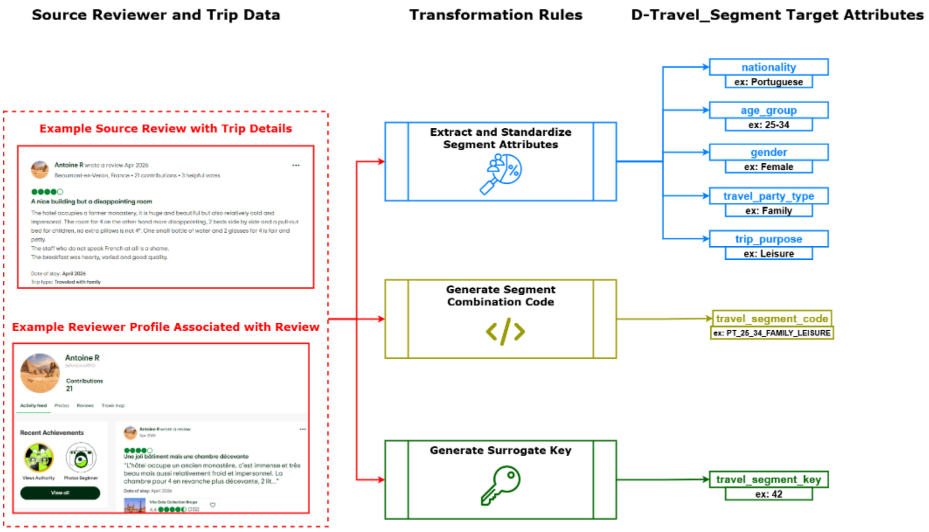

Fig. 19. Source-to-target mapping for the Travel Segment dimension, showing how reviewer and trip metadata are standardised into reusable travel segment combinations. The source review and reviewer profile shown are drawn from a public review platform and are used here as a structural illustration consistent with the synthetic source data the warehouse operates over.

5.7.6 Review Dimension Mapping

The Review dimension is populated from source review records extracted during the ETL process. Each source record represents a review as published on a platform and contains both the review content and its associated metadata. Not all source attributes are loaded into D_REVIEW; fields such as review_title and extraction-level metadata are excluded, and only the attributes required for traceability and contextualisation of aspect-level observations are retained.

As illustrated in Figure 20, the mapping applies four transformation rules. The first standardises review identification and metadata, preserving the source identifier as platform_review_id, cleaning the review text, validating the star_rating and standardising the detected language into review_language.

The second rule resolves the travel segment reference. Reviewer and trip attributes present in the source record are passed to D_TRAVEL_SEGMENT, where the corresponding segment combination is matched or created, and the resulting travel_segment_key is assigned to the review record before loading.

The third rule handles review versioning and edit traceability. Where the source record indicates that a review is an edited version of a previously submitted one, the ETL process assigns a shared review_group_id to both the original and the edited record, and sets is_edit to true for the revised version. This preserves the logical relationship between versions while treating each as a distinct analytical object in the warehouse, consistent with the versioned dimension design established in Section 4.5.6.

The fourth rule generates the surrogate key, review_key, and loads the completed record into D_REVIEW.

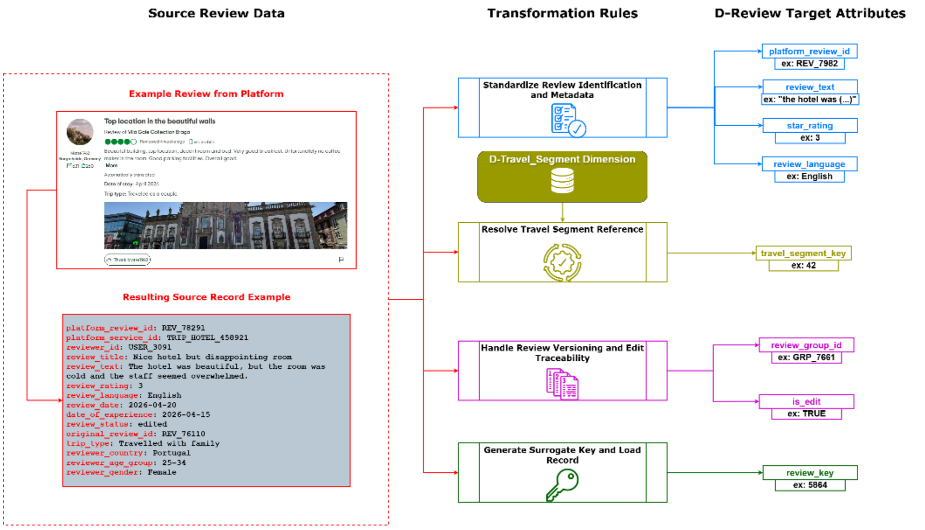

Fig. 20. Source-to-target mapping for the Review dimension, showing how source review data is standardised, linked to the Travel Segment dimension and prepared for edit traceability before being loaded into D_REVIEW. The source review shown is drawn from a public review platform and is used here as a structural illustration consistent with the synthetic source data the warehouse operates over.

By storing review-level metadata separately from the aspect-level fact records, D_REVIEW ensures that multiple sentiment observations derived from the same review remain linked to their shared source context, supporting both traceability and review-level filtering in analytical queries.

5.7.7 Platform Dimension Mapping

The Platform dimension is populated from metadata describing the digital review platform associated with each source record. As with the Service dimension, the source record contains additional fields, including country scope, data access method and extraction timestamp, that are not loaded into D_PLATFORM. Only the attributes required to identify and classify the platform within the warehouse are selected.

As illustrated in Figure 21, the mapping applies three transformation rules. The first standardises platform identification, transforming the source platform identifier into a stable platform_code and cleaning the platform name into a consistent platform_name representation. The second classifies the platform metadata, assigning platform_type to distinguish between platform categories such as review platforms and booking platforms, and preserving platform_url as a stable reference attribute. The third generates the surrogate key, platform_key, and loads the completed record into D_PLATFORM.

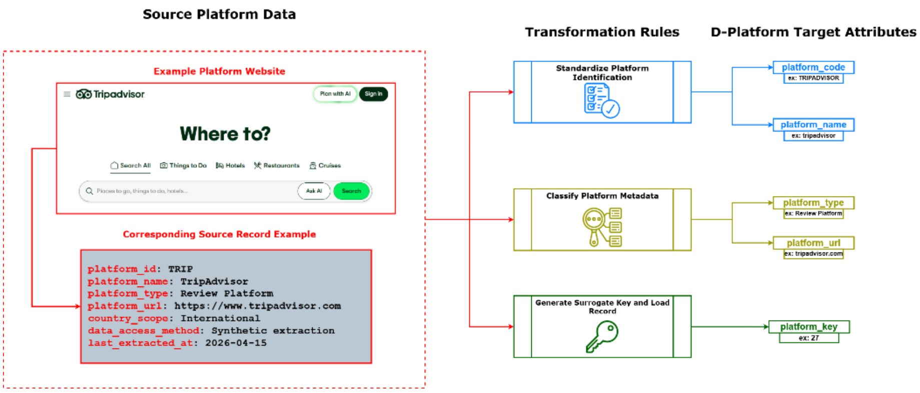

Fig. 21. Source-to-target mapping for the Platform dimension, showing how platform metadata is standardised, classified and loaded into D_PLATFORM. The platform website shown is used here as a structural illustration consistent with the synthetic source data the warehouse operates over.

5.7.8 Fact Table Mapping

The sentiment fact table is populated from the fact-ready observations produced at the end of the conformation stage described in Section 5.6.4. Unlike the dimension mappings in the preceding subsections, the source for this mapping is not raw platform data but the output of the full transformation pipeline: a sentiment-enriched, aspect-level observation in which affective measures have been derived and source-level identifiers are ready to be resolved against the populated warehouse dimensions. The fact table can only be loaded after all dimension records it references are confirmed to exist.

As illustrated in Figure 22, the mapping applies four transformation rules to each fact-ready observation.

The first resolves the dimension foreign keys. The review date is matched against D_DATE to produce date_key. The source service identifier is resolved against D_SERVICE using a temporally scoped lookup that selects the version whose valid_from and valid_to interval encompasses the review date, preserving the historical consistency required by the SCD Type 2 design. The source review identifier is matched against D_REVIEW to retrieve review_key. The detected aspect is matched against the controlled taxonomy in D_ASPECT to produce aspect_key. The source platform identifier is matched against D_PLATFORM to retrieve platform_key. Location and Travel Segment are not resolved as direct foreign keys, since both are accessed indirectly through the Service and Review dimensions respectively.

The second rule standardises and preserves the affective measures. Polarity and subjectivity scores are passed through without further modification, having already been normalised during the transformation phase. The eight Plutchik-aligned emotion scores, covering joy, trust, fear, surprise, sadness, disgust, anger and anticipation, are preserved and mapped directly to their corresponding fact-level fields.

The third rule addresses the supporting measures. The sentiment_strength value is computed as the product of the absolute polarity score and the subjectivity score if not already present in the intermediate record. The model_confidence value is passed through from the derivation pipeline and stored as is, with a null value recorded where the extraction tool does not produce a confidence estimate.

The fourth rule generates the surrogate key, sentiment_fact_key, and performs the insertion into F_Aspect_Sentiment_Observation. Before insertion, a deduplication check verifies that no record with the same combination of date_key, service_key, review_key, aspect_key and platform_key already exists, enforcing the declared grain and preventing duplicate observations from being introduced through reprocessing or incremental loading. The fact table is strictly insert-only: existing records are never updated, as any reprocessing of sentiment derivation produces a new observation rather than overwriting an existing one.

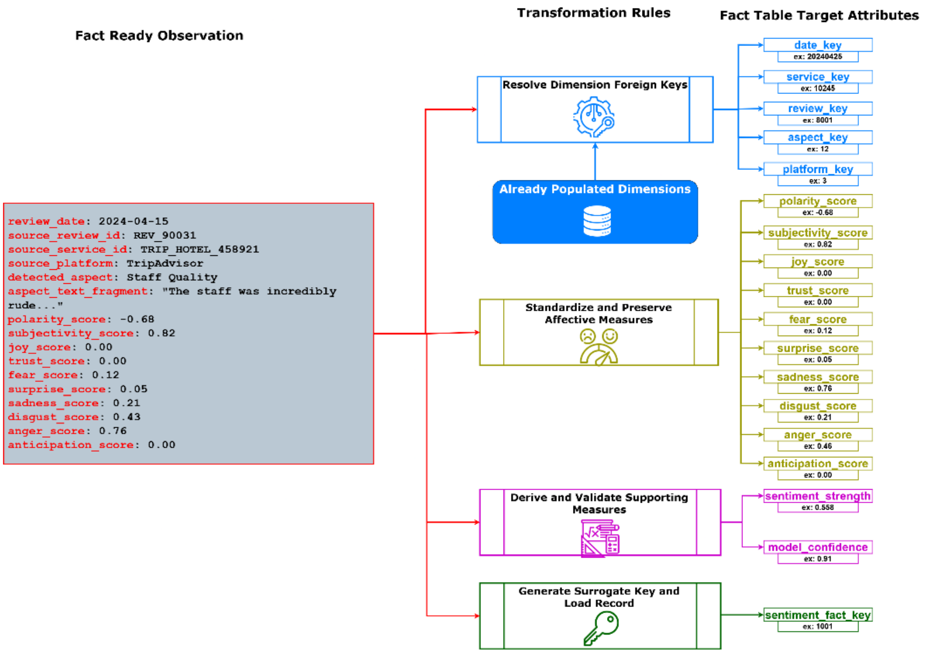

Fig. 22. Source-to-target mapping for the sentiment fact table, showing how fact-ready observations are resolved against populated dimensions and loaded into F_Aspect_Sentiment_Observation as structured affective measures

Each completed record in F_Aspect_Sentiment_Observation integrates the affective output of the transformation pipeline with the contextual structure of the dimensional model, constituting a self-contained analytical unit that can be aggregated, filtered and compared across all the analytical perspectives the warehouse is designed to support.

5.8 Loading Phase

The loading phase is the final stage of the ETL process, responsible for persisting transformed and conformed data into the warehouse structures defined in Chapter 4. While the transformation phase produces sentiment-enriched aspect-level observations and the conformation stage assembles the corresponding fact-ready records, data is only written into the warehouse during this phase. The distinction between conformation and loading is deliberate: conformation prepares records by resolving dimensional keys and validating their completeness, whereas loading performs the actual insertions that make those records available for analytical processing.

The loading phase enforces a strict loading order. Because the fact table references dimension tables through foreign keys, all required dimension records must exist before any fact record can be inserted. Loading fact records before their referenced dimension entries are present would violate referential integrity and produce an analytically inconsistent warehouse. The phase therefore proceeds in two sequential stages: dimension loading followed by fact table loading.

Dimension Loading

Not all dimensions are loaded in the same way or at the same point in ETL execution. The dimensions of the proposed warehouse fall into two groups based on their loading behaviour.

The first group consists of static or reference dimensions, populated independently of review processing and therefore loadable before any review data is ingested. The Date dimension falls into this category, constructed from a predefined calendar structure that does not depend on any source review record. The Location dimension is similarly static, derived from official Portuguese administrative reference data and stable across ETL executions. The Aspect dimension is treated as a reference structure since it is based on a controlled taxonomy defined prior to ETL execution. The Platform dimension, while potentially extensible as new sources are incorporated, is effectively preloaded for the scope of this work. These dimensions are loaded once and remain available for key resolution throughout the remainder of the process.

The second group consists of dynamic dimensions, whose records are created or updated during review processing. The Service dimension belongs to this group because service metadata is derived from source platform profiles and must be loaded before review processing begins. Since it is modelled as SCD Type 2, the loading process must evaluate whether each incoming service record corresponds to a new service or a changed version of an existing one. When a change is detected, the previous record is closed by updating its valid_to date and setting is_current to false, and a new versioned record is inserted. This ensures that sentiment observations can always be associated with the historically accurate state of the service at the time the review was published.

The Travel Segment dimension is populated during review processing, as segment combinations are derived from reviewer and trip metadata associated with each review. Since it functions as an outrigger connected to the Review dimension, travel segment records must be resolved before the corresponding review record is inserted. If the standardised segment combination already exists in D_TRAVEL_SEGMENT, the existing surrogate key is reused. If it does not, a new record is inserted and the generated key is assigned to the review record.

The Review dimension is loaded last among the dimensions, immediately before fact table insertion. Each review record is inserted with its resolved travel_segment_key and its edit-traceability attributes, including review_group_id and is_edit, ensuring that both original and edited review instances are correctly represented before any of their associated aspect-level observations are inserted into the fact table.

Fact Table Loading

Once all required dimension records have been confirmed to exist and their surrogate keys resolved, the fact-ready observations produced by the conformation stage are inserted into F_Aspect_Sentiment_Observation. Each insertion corresponds to one aspect-level sentiment observation and carries the five foreign keys, date_key, service_key, review_key, aspect_key and platform_key, alongside the full set of affective and supporting measures.

The fact table is strictly insert-only. Existing records are never updated or overwritten, since each represents an immutable analytical observation tied to a specific review at the time it was processed. In the event that sentiment derivation is rerun on the same source data, the deduplication check described in Section 5.7.8, based on the unique combination of dimensional foreign keys, prevents duplicate records from being inserted and preserves the integrity of the declared grain.

For incremental loading, where new reviews are ingested in successive ETL runs, the phase applies the same sequence on each execution. Static dimensions are verified but not reloaded. Dynamic dimensions are updated as new services, travel segment combinations or reviews are encountered. Fact records are appended for each new aspect-level observation derived from the incoming reviews. This ensures the warehouse grows consistently with each ETL cycle while maintaining the referential and analytical consistency established during the initial load.

5.9 ETL Pipeline Overview

The preceding sections have addressed each stage of the ETL process individually. This section consolidates them into a unified view of the end-to-end pipeline, as illustrated in Figure 23.

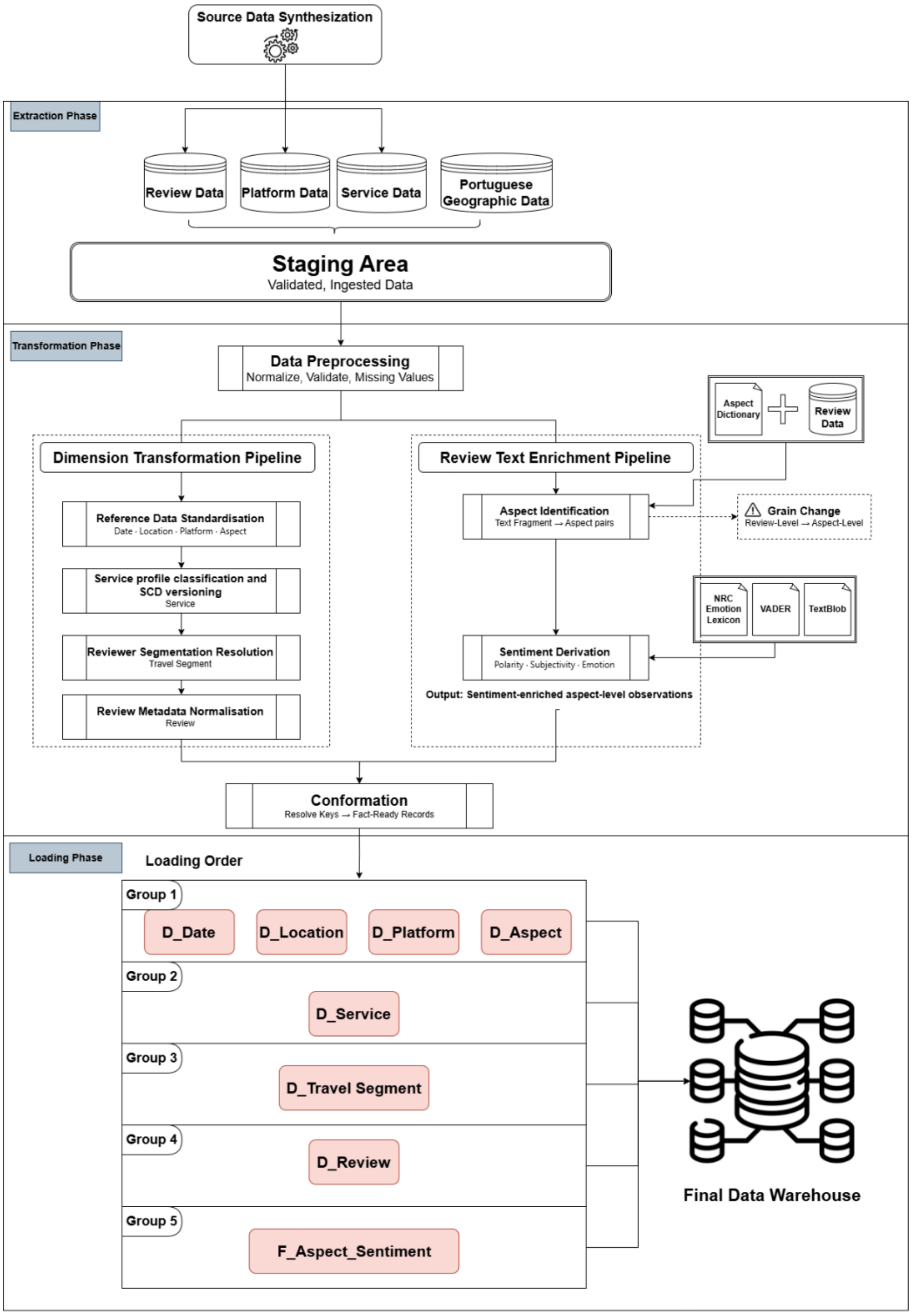

Fig. 23. End-to-end ETL pipeline overview, showing the full sequence from source data synthesis through extraction, parallel transformation pipelines, conformation and dependency-ordered loading into the data warehouse

The pipeline originates from the source data synthesis process described in Section 5.3, which produces three datasets, namely review data, platform data and service data, alongside which Portuguese geographic data is sourced independently from the Carta Administrativa Oficial de Portugal. These datasets are ingested during the extraction phase into a common staging area where basic validation and format standardisation are applied without structural modification of the source content. The staging area acts as the controlled entry point through which all subsequent transformation operations access their inputs in a consistent and reproducible manner.

Upon entering the transformation phase, the staged data is first subjected to a shared preprocessing step that normalises attribute representations, validates completeness and handles missing values across all data types. Following preprocessing, the transformation phase diverges into two parallel pipelines serving distinct purposes.

The dimension transformation pipeline processes the structured metadata associated with services, platforms, locations, reviewers and travel segments. It begins with reference data standardisation covering the static dimensions, Date, Location, Platform and Aspect, whose records are derived from controlled reference sources rather than review data. This is followed by service profile classification and SCD versioning, which applies the Type 2 logic required to preserve the historical evolution of service attributes. Reviewer segmentation resolution then derives the standardised travel segment combinations associated with each reviewer, and review metadata normalisation prepares the review-level attributes including the versioning information required by the versioned dimension design of D_REVIEW.

The review text enrichment pipeline processes the unstructured textual content of the staged review records. Assisted by the Aspect Dictionary, the aspect identification step decomposes each review into aspect-level observations and assigns each fragment to a standardised aspect category. This step introduces the grain change that defines the analytical unit of the warehouse: from this point forward, each observation corresponds to a single aspect within a single review rather than to a review as a whole. Sentiment derivation then operates on these aspect-level observations, drawing on VADER, TextBlob and the NRC Emotion Lexicon to produce polarity, subjectivity and emotion scores for each observation.

Both pipelines converge at the conformation stage, which constitutes the final operation of the transformation phase. Conformation aligns the sentiment-enriched aspect-level observations with the dimensional structures produced by the dimension pipeline, resolving the appropriate surrogate keys and assembling the fact-ready records that the loading phase will insert into the warehouse. As discussed in Section 5.6.4, this step does not perform any warehouse insertions; it prepares the records that make those insertions possible.

The loading phase receives the fact-ready observations from conformation and proceeds according to the dependency-ordered sequence established in Section 5.8. The four static reference dimensions, D_DATE, D_LOCATION, D_PLATFORM and D_ASPECT, are loaded first since they carry no inter-dimensional dependencies. D_SERVICE follows, depending on D_LOCATION for its geographic reference. D_TRAVEL_SEGMENT follows, resolving the segmentation combinations required before review records can be inserted. D_REVIEW is loaded last among the dimensions, incorporating the travel segment references and edit-traceability attributes that define its versioned structure. Only after all dimension records have been confirmed to exist are the fact-ready observations inserted into F_Aspect_Sentiment_Observation as structured aspect-level sentiment observations, completing the population of the warehouse.

Taken as a whole, the pipeline reflects the central argument of this chapter: that integrating sentiment data into a dimensional warehouse requires not only the technical operations of extraction and loading, but a carefully designed transformation architecture capable of converting unstructured affective content into structured analytical records while preserving the contextual relationships, historical consistency and grain integrity that multidimensional analysis demands.

5.10 Summary

Chapter 5 described the ETL process through which the dimensional design established in Chapter 4 is translated into a populated analytical environment. The specific challenges of this process were identified at the outset: the affective measures the warehouse is built around are not present in the source data and must be derived computationally, and the review-level organisation of the source dataset must be restructured into the aspect-level representation the fact table requires. These two constraints shaped every stage of the process described throughout the chapter.

The extraction phase ingested and staged the synthetic source dataset without structural modification. The transformation phase performed the substantive analytical work, decomposing review content into aspect-level observations through a controlled aspect identification process and enriching each observation with polarity, subjectivity and emotion scores derived using VADER, TextBlob and the NRC Emotion Lexicon. A conformation stage then resolved source-level identifiers into warehouse surrogate keys, assembling the fact-ready records that the loading phase would subsequently insert. Source-to-target mappings for all seven dimensions and the fact table documented the correspondence between source attributes, intermediate transformation outputs and target warehouse structures. The loading phase completed the process by inserting dimension and fact records in a dependency-ordered sequence that preserves referential integrity across the schema. The chapter closed with a pipeline overview consolidating all stages into a unified end-to-end view. Together, these processes operationalise the sentiment-aware warehouse proposed in this dissertation and make its analytical capabilities available for exploration.

6 Prototype Implementation

Chapter 4 produced a complete dimensional design, and Chapter 5 translated that design into a systematic ETL process capable of populating it. Both chapters describe the warehouse as it is intended to function, its structure, its measures and the transformations that fill it, but neither chapter demonstrates the design operating over real data. Before the warehouse can be evaluated against the analytical use cases established in Section 3.4, a concrete instance of the design must exist. This chapter presents that instance: a working prototype built directly from the schema and ETL processes defined in Chapters 4 and 5, populated with a synthetically generated dataset and executed end to end.

The prototype described here is the first implemented instance of the design, built to exercise the warehouse at a representative scale rather than at the production volumes projected in Chapter 4. The chapter begins by identifying the technologies used to implement the schema and ETL processes. It then describes the construction of the synthetic dataset used to populate this instance, including the parameters and generation logic that produced it, and notes that later iterations of the implementation may vary these parameters to test the architecture under different conditions. The chapter concludes by walking through the pipeline architecture that connects data generation, transformation and loading into a single reproducible process, preparing the transition to the analytical evaluation carried out further in this work.

6.1 Implementation Environment and Technologies

The prototype implements the schema defined in Chapter 4 and the ETL processes defined in Chapter 5 using a small, self-contained set of technologies chosen for their suitability to a prototype of this scale rather than for production deployment. Python provides the implementation language throughout, handling data generation, transformation and the orchestration of the loading process. SQLite serves as the target database management system. As a serverless, file-based relational database, SQLite requires no separate installation or configuration, supports standard SQL including foreign key constraints and check constraints, and produces a single portable file that can be inspected, shared or version-controlled independently of any particular machine. These properties make it appropriate for validating a dimensional design at the scale exercised here, without the operational overhead a server-based system such as SQL Server or PostgreSQL would introduce for a prototype of this size. Sentiment derivation relies on the three tools already established in Chapter 5:

1. VADER (Hutto & Gilbert, 2014) produces the polarity measure. Its lexicon and rule-based approach to sentiment intensity was selected in Chapter 5 for its transparency and its suitability to short, review-length text, and the prototype applies it exactly as specified there, at the level of individual aspect-tagged sentences rather than whole reviews.

2. TextBlob produces the subjectivity measure, returning a normalised score that distinguishes opinionated content from factual statement, consistent with its role as defined in Section 5.7.2.
3. The NRC Emotion Lexicon (Mohammad & Turney, 2013) produces the eight Plutchik-aligned emotion measures, mapping lexical items in each sentence to their associated emotional categories.

All three tools operate as Python libraries, allowing them to be called directly within the transformation stage of the pipeline without an intermediate export or import step, further in this work.

6.2 Synthetic Dataset Construction

The prototype is populated with a synthetically generated dataset built specifically to exercise the design described in Chapters 4 and 5. This section describes the dataset's composition and the logic used to generate it.

The dataset comprises 80 services distributed across the three service categories defined in D_Service: 40 accommodation services, 24 restaurants or bars and 16 tourist sites, spread across 22 municipalities selected for their relevance to Portuguese tourism. These municipalities span all seven regions defined in D_Location, including the Autonomous Region of the Azores, ensuring that regional comparisons drawn further in this work are made across the full geographic scope the dimension supports rather than a subset of it. Reviews are drawn from five platforms reflecting the variety of sources a real deployment would draw on, including review websites, a booking platform and a social media source. From these services and platforms, 1,000 review records were generated, organised into 900 distinct review threads, of which 100 include a second, edited version of the original text, spanning four complete calendar years, 2022 to 2025. This scale represents a deliberate departure from the production volumes projected in Chapter 4, which describe the design's intended operating scale rather than the scale required to evaluate it; the dataset here is sized to populate every dimension meaningfully and exercise the analytical use cases defined in Section 3.4, not to approximate a live platform's review volume.

Review text is assembled from a library of aspect-tagged sentence fragments, each written to express a specific sentiment tier for a specific aspect of the taxonomy defined in D_Aspect. Each generated review combines two to four fragments, reflecting how a guest's review typically touches on several distinct aspects of a service within a single piece of text rather than addressing only one.

To ensure the dataset supports meaningful comparison rather than converging toward a uniform average, each service is assigned a sentiment profile that governs which fragment tiers it draws from more heavily:

- Mostly positive services (48 of 80) draw predominantly from favourable fragments, consistent with the tendency of real tourism and hospitality services to receive more positive than negative reviews.

- Mixed services (20 of 80) draw evenly across the full sentiment range.
- Mostly negative services (12 of 80) draw predominantly from unfavourable fragments.

This weighting is what allows the analysis carried out further in this work to compare services and categories against one another with genuine contrast in the results, rather than results that reflect only random noise around a single mean. Figure 24 summarises this composition.

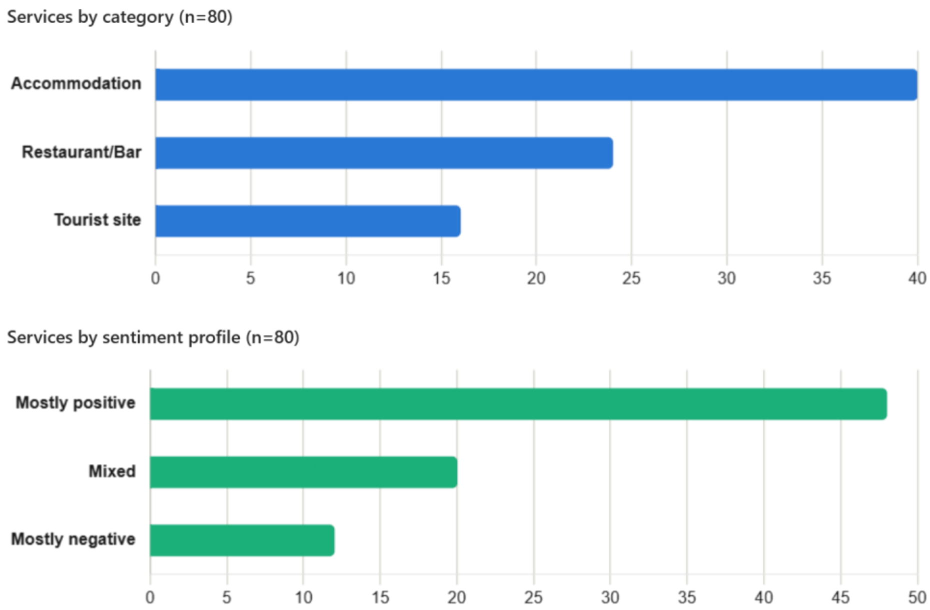

Fig. 24. Composition of the synthetic dataset by service category and sentiment profile

It is important to distinguish what the sentiment profile controls from what it does not. The profile governs only the selection of fragments during text generation; once a review's text is assembled, the sentiment measures stored in the fact table are derived independently by VADER, TextBlob and the NRC Emotion Lexicon from that text, without reference to the profile that produced it. The profile establishes a known, designed contrast in the underlying text so that the analysis carried out further in this work can assess whether the derivation pipeline and dimensional model correctly surface that contrast, not to determine the sentiment values themselves.

Three services in this instance carry a genuine change to a tracked attribute, an official star rating adjustment following a renovation or a change in management, or a price tier reclassification, at a fixed point during the review period. Reviews captured before each service's change date reflect its original attributes; reviews captured after reflect the changed ones. This is what allows D_Service's Type 2 slowly changing dimension logic to be exercised against a real version history during loading, and what allows sentiment before and after a service change to be assessed empirically further in this work, rather than only in principle.

Edited reviews are generated by identifying, for each thread with more than one capture, the earliest submission as the original and any subsequent capture as an edit, consistent with how the mapping described in Section 5.7.6 resolves review_group_id and is_edit. Rather than being generated independently, an edited version is built from the original by either changing the sentiment tier expressed for one previously mentioned aspect or introducing one additional aspect not present in the original text, so that the edited review remains recognisably a revision of the same review rather than an unrelated piece of text.

The generation process is governed by a fixed random seed, so that this dataset can be regenerated identically from the same generation logic. Later iterations of the implementation may vary these parameters further to test the design's behaviour under different conditions, an avenue considered further in this work.

6.3 Pipeline Architecture

The prototype is built and populated by a single orchestrated pipeline, executed in the dependency order established by the schema itself: a dimension must exist before anything can reference it, and the fact table, which references every dimension, is necessarily the final stage. The pipeline is organised into seven stages, implemented across the files summarised in Table 23.

Table 23. Files comprising the prototype pipeline, grouped by stage, distinguishing executable scripts from imported data modules

File	Type	Role
Orchestration		
run_pipeline.py	Python Script	Executes all seven stages below in dependency order from a single command
Stage 1- Schema Creation		
schema.sql	SQL Script	Defines the eight warehouse tables (DW Schema)
raw_schema.sql	SQL Script	Defines the two raw layer tables (Raw Data Schema)
Stage 2- Independent Dimensions		
populate_d_date.py	Python Script	Populates D_Date
populate_d_aspect.py	Python Script	Populates D_Aspect
populate_d_location.py	Python Script	Populates D_Location
municipality_data.py	Python Data Module	Supplies municipality records imported by populate_d_location.py
Stage 3- Raw Review Generation		
generate_raw_reviews.py	Python Script	Generates the synthetic dataset described in Section 6.2
service_data.py	Python Data Module	Supplies the 80 synthetic services
platform_data.py	Python Data Module	Supplies the 5 review platforms
fragment_library.py	Python Data Module	Supplies the aspect-tagged sentence fragments
Stage 4- Keyword Dictionary		
populate_keyword_dict.py	Python Script	Populates the aspect detection dictionary used during transformation
Stage 5- Transformation		
transform.py	Python Script	Splits review text into sentences, identifies aspects and derives sentiment measures
Stage 6- Dependent Dimensions		
load_d_platform.py	Python Script	Populates D_Platform
load_d_service.py	Python Script	Populates D_Service, resolving each service's location against D_Location
load_d_travel_segment.py	Python Script	Populates D_Travel_Segment
load_d_review.py	Python Script	Populates D_Review, requiring D_Travel_Segment to be loaded first
Stage 7- Fact Table Load		
load_fact_table.py	Python Script	Resolves all five dimension keys and populates F_Aspect_Sentiment_Observation

6.3.1 Stages 1 and 2

The first two stages establish the schema and populate the dimensions that carry no dependency on review data. Stage 1 executes schema.sql and raw_schema.sql, creating the DW Schema and the Raw Data Schema respectively. Stage 2 then populates D_Date, D_Aspect and D_Location; because none of the three depends on the others, they may run in any order relative to one another, though D_Location specifically imports its underlying municipality records from municipality_data.py. Figure 25 illustrates both stages.

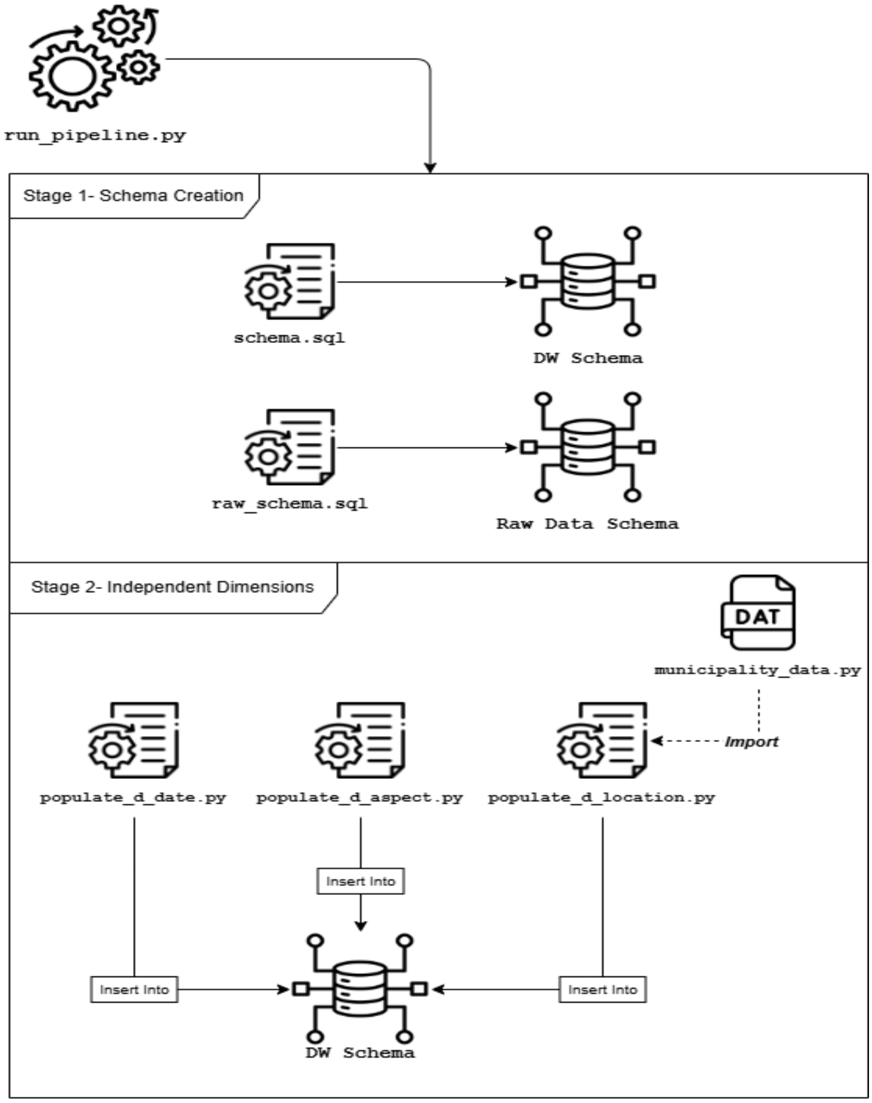

Fig. 25. Pipeline stages 1 and 2: schema creation and independent dimension loading

6.3.2 Stages 3 and 4

Stage 3 generates the synthetic review dataset described in Section 6.2, with generate_raw_reviews.py drawing on the three data modules listed in Table 23 to assemble each review and inserting the result into the Raw Data Schema. Stage 4 then populates the keyword dictionary used for aspect identification, a single script writing directly into the same schema without drawing on any separate data module. Figure 26 illustrates both stages.

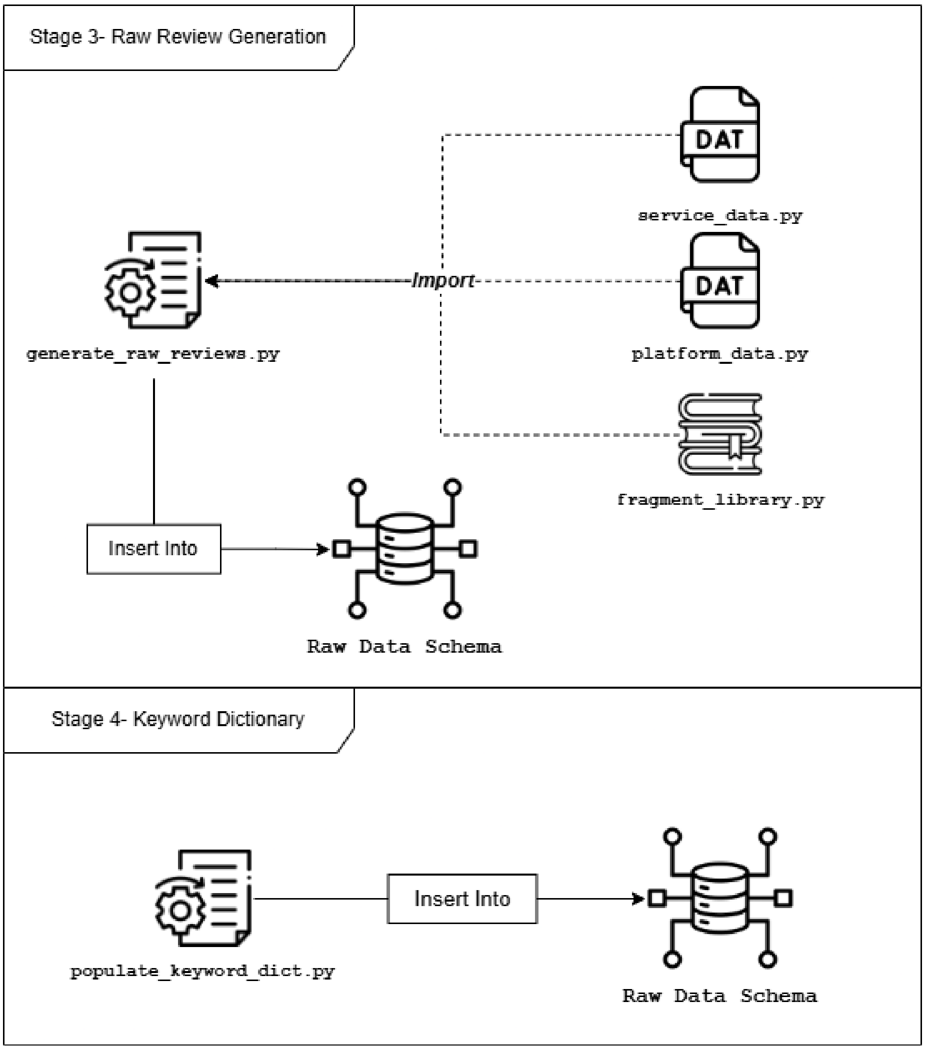

Fig. 26. Pipeline stages 3 and 4: raw review generation and keyword dictionary population

6.3.3 Stages 5 and 6

Stage 5 is the point at which the pipeline moves from raw text to derived sentiment measures. transform.py reads both raw layer tables, splits each review into sentences, identifies the aspect each sentence addresses using the keyword dictionary populated in Stage 4, and derives polarity, subjectivity and the eight emotion measures using the three external libraries introduced in Section 6.1. Rather than writing directly into the warehouse, this stage creates and populates an intermediate staging table, TRANSFORMED_OBSERVATIONS, which exists only for the duration of a pipeline run and is not part of either schema defined in Chapters 4 or 5. Stage 6 then reads from this staging table to populate the four dimensions that depend on it: D_Platform, D_Service, D_Travel_Segment and D_Review. D_Service additionally reads from D_Location during this stage to resolve each service's geographic reference, and D_Review specifically requires D_Travel_Segment to already be loaded, since it resolves a travel segment key during its own insert. Figure 27 illustrates both stages.

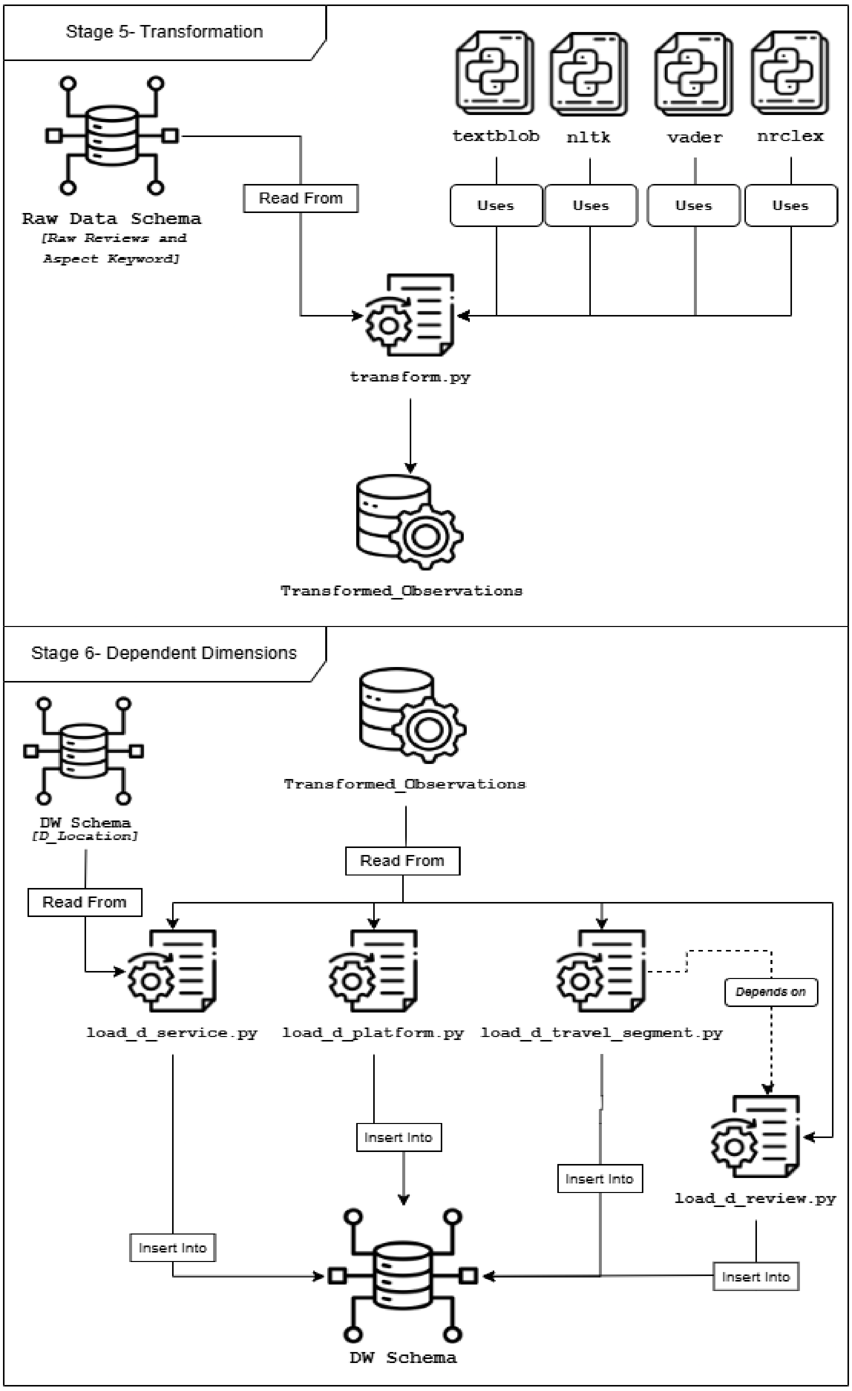

Fig. 27. Pipeline stages 5 and 6: transformation and dependent dimension loading

6.3.4 Stage 7

The seventh and final stage populates the fact table. load_fact_table.py reads from the staging table produced in Stage 5 and from all five dimension tables populated in the stages before it, resolving date_key, service_key, review_key, aspect_key and platform_key for each transformed observation. The service_key resolution is a temporally scoped lookup rather than a simple match, selecting the version of a service whose valid_from and valid_to window contains the date the review was captured, consistent with the SCD Type 2 mapping described in Section 5.7.8. Figure 28 illustrates this stage.

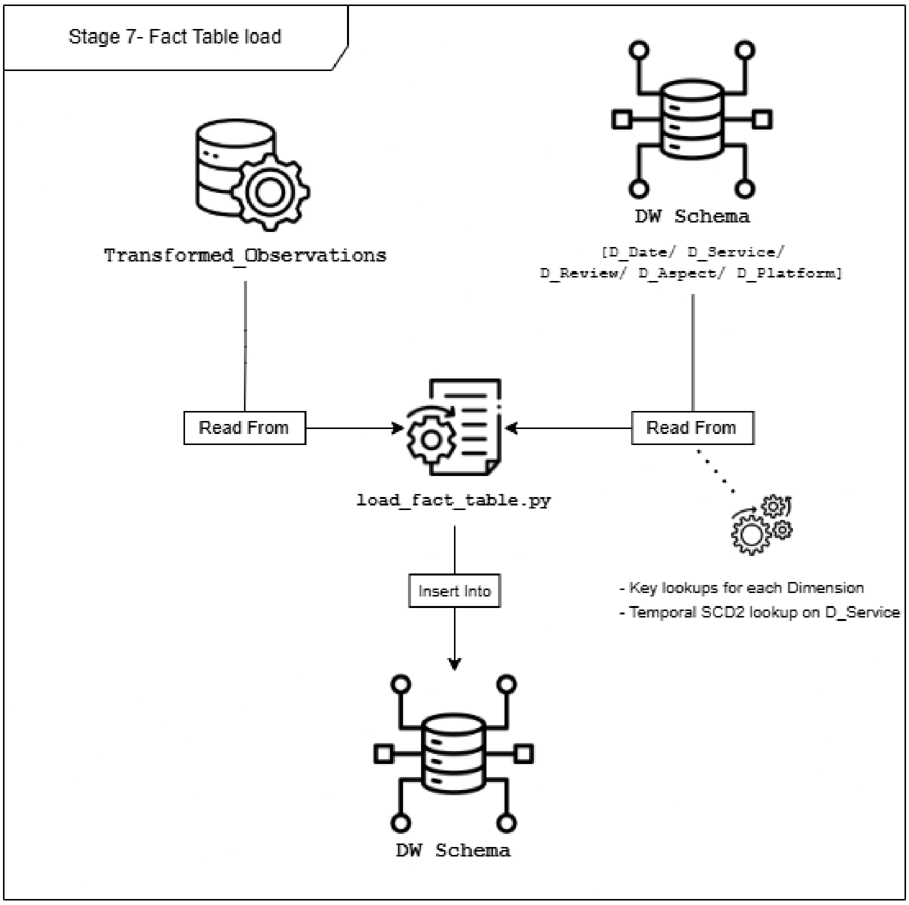

Fig. 28. Pipeline stage 7: fact table load

Two decisions made during this final stage are worth stating explicitly rather than left implicit in the pipeline's behaviour. First, where two sentences within the same review are matched to the same aspect, a case the fact table's grain does not otherwise allow, their sentiment measures are averaged into a single observation rather than one being discarded, so that both sentences contribute to the record rather than one being silently lost. Second, the entire sequence described across Figures 25 to 28 is exposed through run_pipeline.py as a single command, governed by the fixed random seed introduced in Section 6.2, so that the complete prototype, from an empty database to a fully populated warehouse, can be reproduced identically without manual intervention at any stage.

6.4 Summary

Chapter 6 presented an implemented instance of the design established in Chapters 4 and 5, moving the warehouse from specification to a working artefact. The implementation environment was identified at the outset: Python and SQLite as the technologies through which the schema and ETL processes are realised, and VADER, TextBlob and the NRC Emotion Lexicon as the tools through which polarity, subjectivity and emotion are derived. A synthetic dataset was then constructed to populate this instance, comprising 80 services, 5 platforms and 1,000 review records assembled from a library of aspect-tagged sentence fragments, with sentiment profiles introduced deliberately across services, and three services carrying a genuine attribute change over time, to support meaningful comparison in the analysis carried out further in this work. The chapter then traced the pipeline architecture connecting these components, seven dependency-ordered stages beginning with schema creation and concluding with the fact table load, in which raw reviews are transformed into aspect-level sentiment observations and resolved against every dimension of the warehouse. Two implementation decisions, the aggregation of duplicate aspect-level observations within a single review and the pipeline's reproducibility through a fixed random seed, were made explicit rather than left implicit in the pipeline's behaviour. The prototype described in this chapter is now a populated, queryable instance of the warehouse, ready for the analytical evaluation carried out further in this work.

7 Analytical Evaluation

Chapter 6 produced a working, populated instance of the sentiment-aware data warehouse designed in Chapter 4 and built through the ETL processes described in Chapter 5. This chapter evaluates that instance against the analytical use cases established in Section 3.4, assessing whether the completed design supports the multidimensional sentiment analysis it was built to enable.

The chapter begins by stating the methodology through which this evaluation was carried out. It then works through the ten use cases in the same three groups established in Section 3.4, strategic and comparative analysis, aspect-level and operational analysis, and affective and segmentation analysis, presenting each use case's query and result in turn. Beyond these ten use cases, a further analysis tests the extensibility Section 3.4 claims for the warehouse's dimensional structure, particularly through the Travel Segment dimension, before a separate comparison evaluates the warehouse's aspect-level design directly against simpler alternatives, star ratings alone, review-level averages and polarity without emotion, using the populated instance to test empirically what Chapters 3 and 4 argued conceptually. The chapter closes with a discussion drawing these findings together, including the limitations the evaluation surfaced, before a summary transitions into the conclusions drawn further in this work.

7.1 Evaluation Methodology

Validating a data warehouse design against use cases that were only ever described, and never actually run, would leave an obvious gap: a reader has no way to tell whether the schema genuinely supports the analytical questions it was built for, or whether the argument only works on paper. This chapter closes that gap by executing an actual SQL query for each use case established in Section 3.4 against the populated warehouse instance described in Chapter 6, rather than continuing to argue in the abstract. Each query is annotated with the specific OLAP operation it demonstrates, slice and dice for comparative questions across categories, regions or segments, drill-down for aspect-level exploration of service-level aggregates, and roll-up for temporal analysis along the date dimension, so that the evaluation stays traceable to the operation mapping already established at the close of Section 3.4, rather than reintroducing that classification independently.

Testing the ten stated use cases only answers part of what a thorough evaluation should establish, however. Section 3.4 also claims that the warehouse supports further analyses, particularly through the Travel Segment dimension, without requiring any structural change, a claim this chapter should not simply repeat but actually test. A separate, harder question follows from that: even where the aspect-level design does answer these questions correctly, was building at that grain, rather than at the simpler review level Chapter 4 considered and rejected, actually worth the additional complexity it introduced. Answering that requires comparing the warehouse's output directly against what a simpler design would have produced from the same data, using the concrete evidence Chapter 6's prototype makes possible for the first time. Together, these three tests, validating the stated use cases, extending beyond them to check the extensibility claim, and comparing against simpler alternatives, form the methodology carried through the remainder of this chapter.

The results produced by all three should be read with one caveat in mind. Because the dataset is synthetically generated rather than drawn from a live platform, the specific figures reported throughout this chapter, average polarity scores, regional comparisons, seasonal patterns, demonstrate that the warehouse's design correctly supports the analytical operations each question requires, not that they describe actual tourism sentiment in Portugal. A later iteration of the implementation, populated with a different or larger dataset, would be expected to produce different figures while exercising the same underlying analytical capability.

7.2 Strategic and Comparative Analysis

A warehouse that cannot tell a regional planner whether one part of the country is outperforming another, or whether a season genuinely shifts how guests feel about their stay, offers little beyond what a simple average rating already provides. The four use cases addressed here test exactly that capability, whether the aspect-level design surfaces comparative patterns across service categories, regions and time that a coarser representation of sentiment would leave invisible.

Overall sentiment across service categories

Does user sentiment differ between accommodation services, restaurants and bars, and tourist sites? Answering this requires slicing the fact table by service_category, one of the simplest operations the warehouse supports, yet one no review-level or polarity-only system organises data cleanly enough to answer directly.

```
SELECT s.service_category,
       ROUND(AVG(f.polarity_score), 4)     AS avg_polarity,
       ROUND(AVG(f.subjectivity_score), 4) AS avg_subjectivity,
       ROUND(AVG(f.sentiment_strength), 4) AS avg_strength,
       COUNT(*)                            AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_SERVICE s ON f.service_key = s.service_key
GROUP BY s.service_category
ORDER BY avg_polarity DESC
```

Listing 1- Query used to answer UC1

Accommodation records the highest average polarity (0.241, n=1037), followed by tourist sites (0.223, n=461), with restaurants and bars trailing both by a clearer margin (0.176, n=362). The gap between accommodation and dining experiences is more pronounced than a marginal difference, suggesting a genuine tendency for hospitality experiences to be reviewed more favourably than dining experiences within this dataset.

Emotional response across regions and cities

Do emotional responses expressed in user reviews vary across regions and cities? Where the previous query sliced by service category, this one drills further, joining through D_Service into the D_Location outrigger to compare joy, sadness and anger at the municipality level.

```
SELECT l.region, l.municipality,
       ROUND(AVG(f.joy_score), 3)     AS avg_joy,
       ROUND(AVG(f.sadness_score), 3) AS avg_sadness,
       ROUND(AVG(f.anger_score), 3)   AS avg_anger,
       COUNT(*)                       AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_SERVICE s ON f.service_key = s.service_key
JOIN D_LOCATION l ON s.location_key = l.location_key
GROUP BY l.region, l.municipality
ORDER BY avg_joy DESC
LIMIT 5
```

Listing 2- Query used to answer UC2

Beja records the highest average joy score (0.084, n=25), ahead of Albufeira (0.068), Sintra (0.065), Lagos (0.063) and Tavira (0.061). The differences between municipalities remain small but genuine, consistent with what a real destination management organisation would expect to find meaningful variation across locations rather than uniform emotional response, which is precisely the kind of pattern a single, undifferentiated polarity score would not surface.

Consistency of regional performance across categories

Do particular regions consistently exhibit stronger or weaker sentiment across multiple service categories, or does regional performance vary by category? This use case cross-tabulates region against service_category directly, testing whether a region's strength is a property of the region itself or only of the specific services located within it.

```
SELECT l.region, s.service_category,
       ROUND(AVG(f.polarity_score), 3) AS avg_polarity, COUNT(*) AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_SERVICE s ON f.service_key = s.service_key
JOIN D_LOCATION l ON s.location_key = l.location_key
GROUP BY l.region, s.service_category
HAVING COUNT(*) >= 10
ORDER BY l.region, s.service_category
```

Listing 3- Query used to answer UC3

The Azores score consistently high across all three categories, 0.328 for accommodation, 0.247 for restaurants and bars, 0.304 for tourist sites, a spread of only 0.081 between its strongest and weakest category despite being the region with the fewest municipalities represented in the dataset. Centro shows the opposite pattern, scoring consistently weakly across all three, 0.072, 0.098 and 0.058 respectively, a spread of only 0.040. This result demonstrates a genuinely regional effect distinct from category-level variation: a stakeholder asking whether a region is performing well cannot be answered by looking at any single category in isolation, since the Azores' strength and Centro's weakness both hold regardless of which category is examined.

Sentiment evolution across seasons

How does sentiment evolve over time for a given service category, and does it vary across tourism seasons? This use case rolls up the fact table along the D_Date hierarchy, from individual dates to season within year, for accommodation specifically.

```
SELECT d.year, d.season,
       ROUND(AVG(f.polarity_score), 3) AS avg_polarity, COUNT(*) AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_DATE d ON f.date_key = d.date_key
JOIN D_SERVICE s ON f.service_key = s.service_key
WHERE s.service_category = 'Accommodation'
GROUP BY d.year, d.season
ORDER BY d.year,
    CASE d.season WHEN 'Winter' THEN 1 WHEN 'Spring' THEN 2
                  WHEN 'Summer' THEN 3 WHEN 'Autumn' THEN 4 END
```

Listing 4- Query used to answer UC4

Across the four complete years covered by the dataset, seasonal average polarity for accommodation fluctuates within a fairly narrow band, generally between 0.19 and 0.30, without a single season consistently outperforming the others year on year. Every seasonal cell draws on a reasonable number of observations, ranging from 39 to 106, providing a stable basis for comparison across the full period. This roll-up illustrates the warehouse's capacity for temporal drill-down along the calendar hierarchy, distinguishing genuine seasonal variation from the kind of short-term noise a smaller or less complete dataset might otherwise produce.

7.3 Aspect-Level and Operational Analysis

Knowing that accommodation outperforms restaurants and bars on average, as the previous section established, tells a service operator very little about what to actually change. The three use cases addressed here move the evaluation from that comparative level down to the operational one, testing whether the warehouse can isolate which specific aspects of a service drive negative experiences, which emotions accompany different kinds of service, and whether the design can track sentiment as a service itself changes over time.

Aspects most associated with negative sentiment

Which specific aspects of accommodation services are most frequently associated with negative sentiment or negative emotion? This use case drills down from the service-level aggregates examined in Section 7.2 to individual aspect characteristics within a single category.

```
SELECT a.aspect_name,
       ROUND(AVG(f.polarity_score), 3) AS avg_polarity,
       ROUND(AVG(f.anger_score), 3)    AS avg_anger,
       ROUND(AVG(f.disgust_score), 3)  AS avg_disgust,
       COUNT(*)                        AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_SERVICE s ON f.service_key = s.service_key
JOIN D_ASPECT a ON f.aspect_key = a.aspect_key
WHERE s.service_category = 'Accommodation'
GROUP BY a.aspect_name
HAVING COUNT(*) >= 5
ORDER BY avg_polarity ASC
```

Listing 5- Query used to answer UC5

Wifi Connectivity records the lowest average polarity of any accommodation aspect (-0.074, n=32), followed by Decor and Design (-0.048, n=32) and Check-in Experience (-0.009, n=45). None of these three shows any detected negative emotion, anger and disgust both average to 0.000 across all of them, consistent with a documented coverage limitation in the NRC Emotion Lexicon's vocabulary rather than a genuine absence of negative feeling in the underlying text. Hidden Costs, sitting just below neutral (0.000, n=14), shows the opposite pattern: it carries a clearly detected emotional signature, anger at 0.333 and disgust at 0.150, despite its polarity registering as essentially neutral. Taken together these results illustrate why polarity and emotion need to be examined separately rather than assumed to move in lockstep: an aspect can register as clearly negative with no detectable emotion behind it, or as neutral while still carrying a strong emotional charge.

Emotion prevalence across service categories

Are certain emotions more prevalent in specific types of service? Where UC1 compared categories on polarity alone, this use case compares them across all eight Plutchik-aligned emotion measures simultaneously.

```
SELECT s.service_category,
       ROUND(AVG(f.joy_score), 3)          AS joy,
       ROUND(AVG(f.trust_score), 3)        AS trust,
       ROUND(AVG(f.fear_score), 3)         AS fear,
       ROUND(AVG(f.surprise_score), 3)     AS surprise,
       ROUND(AVG(f.sadness_score), 3)      AS sadness,
       ROUND(AVG(f.disgust_score), 3)      AS disgust,
       ROUND(AVG(f.anger_score), 3)        AS anger,
       ROUND(AVG(f.anticipation_score), 3) AS anticipation
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_SERVICE s ON f.service_key = s.service_key
GROUP BY s.service_category
```

Listing 6- Query used to answer UC6

Accommodation records the lowest average trust of the three categories (0.073), while restaurants and bars (0.099) and tourist sites (0.105) sit close together and considerably higher, the clearest categorical split among the eight emotions measured. Anticipation is the most prevalent emotion across all three categories, though it too varies by category, highest for restaurants and bars (0.192), followed by accommodation (0.174) and tourist sites (0.140). Joy, by contrast, is nearly uniform across the three categories (0.050 to 0.055), a reminder that not every emotional measure differentiates categories to the same degree; a genuinely multidimensional representation lets that be seen rather than assumed from polarity alone.

Sentiment change following service changes

Does sentiment shift following a service improvement or operational change, such as a change in category or price tier? Three services in this dataset carry a genuine, dated attribute change, an official star rating adjustment or a price tier reclassification, allowing this question to be answered directly.

```
SELECT platform_service_id, COUNT(*) AS n_versions
FROM D_SERVICE
GROUP BY platform_service_id
HAVING n_versions > 1
```

Listing 7- Query used to answer UC7

The three services show three distinct patterns rather than a single uniform effect. Following a star rating upgrade from four to five stars, one service's average polarity rose from 0.205 to 0.271, consistent with the direction of the change. Following a downgrade from five to four stars, a second service's average polarity remained essentially unchanged, moving from 0.283 to 0.286, showing no meaningful decline despite the downgrade. A third service, whose price tier was reclassified without any change to star rating, saw average polarity rise sharply from 0.205 to 0.381. Because the sentiment profile governing a service's fragment selection, described in Section 6.2, operates independently of a service's official attributes, only one of these three outcomes tracking its official change is expected rather than anomalous: guest sentiment is derived from review text alone and is not mechanically tied to a service's administrative classification. What matters for this use case is not that sentiment moves in step with an official change, but that D_Service's Type 2 versioning correctly attributes each review's sentiment to the specific version of the service that was in effect when that review was written, which the temporally scoped lookup described in Section 5.7.8 performs correctly for all three services regardless of how their sentiment happened to move.

7.4 Affective and Segmentation Analysis

Polarity and category comparisons answer questions about which services and regions perform well. They say very little about how a review's language actually works, whether a strongly worded review is strongly felt, or whether the same aspect lands differently depending on who is travelling and why. The three use cases addressed here test the warehouse's ability to combine polarity, subjectivity and emotion together, and to bring the Travel Segment dimension into the analysis for the first time in this evaluation.

Subjectivity's influence on aggregated sentiment

How does the degree of subjectivity influence the interpretation of aggregated sentiment trends? This use case bands observations by subjectivity level and compares average polarity and sentiment_strength across those bands.

```
SELECT
    CASE
        WHEN f.subjectivity_score < 0.3 THEN 'Low (<0.3)'
        WHEN f.subjectivity_score < 0.6 THEN 'Medium (0.3-0.6)'
        ELSE 'High (>=0.6)'
    END AS subjectivity_band,
    ROUND(AVG(f.polarity_score), 3) AS avg_polarity,
    ROUND(AVG(f.sentiment_strength), 3) AS avg_strength,
    COUNT(*) AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
GROUP BY subjectivity_band
ORDER BY MIN(f.subjectivity_score)
```

Listing 8- Query used to answer UC8

The relationship is clear and consistent: low-subjectivity observations average close to neutral polarity (0.002) and negligible strength (0.026, n=311), while high-subjectivity observations average considerably higher on both (0.303 polarity, 0.268 strength, n=724), with the medium band sitting between the two on every measure. This is a direct empirical confirmation of why sentiment_strength was defined as it was in Section 4.7.5: opinionated language carries more analytical weight than factual statement, and a measure built from polarity alone would treat a flatly descriptive sentence and a strongly felt one as equally significant.

Interaction between aspect category and travel segment

How do polarity, subjectivity and emotion interact when analysed across service aspects and visitor segments together? This use case drills down across two dimensions simultaneously, D_Aspect and D_Travel_Segment, the latter accessed through D_Review, testing whether the warehouse supports genuinely multidimensional affective analysis rather than one dimension at a time.

```
SELECT a.aspect_category, t.travel_party_type,
       ROUND(AVG(f.polarity_score), 3) AS avg_polarity,
       ROUND(AVG(f.subjectivity_score), 3) AS avg_subjectivity,
       ROUND(AVG(f.joy_score), 3) AS avg_joy,
       COUNT(*) AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_ASPECT a ON f.aspect_key = a.aspect_key
JOIN D_REVIEW r ON f.review_key = r.review_key
JOIN D_TRAVEL_SEGMENT t ON r.travel_segment_key = t.travel_segment_key
WHERE t.travel_party_type IS NOT NULL
GROUP BY a.aspect_category, t.travel_party_type
HAVING COUNT(*) >= 8
ORDER BY a.aspect_category, t.travel_party_type
```

Listing 9- Query used to answer UC9

Physical Environment shows the widest segment-based spread of any aspect category, travelling with friends (0.493) and travelling solo (0.416) both score considerably higher than travelling as a couple (0.275) or with family (0.244), a gap of over 0.2 within a single category depending entirely on who is travelling. Which aspect category sits lowest is similarly unfixed across segments: Core Offering is weakest for families and friends (0.161 and 0.111), Staff and Service is weakest for couples (0.166), and Location is weakest for solo travellers (0.147). No single aspect category functions as a uniform floor any more than one functions as a uniform ceiling, reinforcing that aspect and segment genuinely interact rather than either dimension dominating the other on its own.

Sentiment differences across travel party type and purpose

How does sentiment differ between solo travellers, families and business travellers for the same service category? This use case cross-tabulates service_category against both travel_party_type and trip_purpose, drawn from the same outrigger dimension used in UC9.

```
SELECT s.service_category, t.travel_party_type, t.trip_purpose,
       ROUND(AVG(f.polarity_score), 3) AS avg_polarity, COUNT(*) AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_SERVICE s ON f.service_key = s.service_key
JOIN D_REVIEW r ON f.review_key = r.review_key
JOIN D_TRAVEL_SEGMENT t ON r.travel_segment_key = t.travel_segment_key
WHERE t.travel_party_type IN ('Solo', 'Family') OR t.trip_purpose = 'Business'
GROUP BY s.service_category, t.travel_party_type, t.trip_purpose
HAVING COUNT(*) >= 5
ORDER BY s.service_category
```

Listing 10- Query used to answer UC10

The pattern is not uniform across categories, which is itself informative. Solo leisure travellers record the highest polarity of any segment at tourist sites (0.397), well above families travelling for leisure at the same category (0.298), yet the same solo leisure segment records the lowest polarity of any named segment at restaurants and bars (0.128), below every other combination in that category. A stakeholder relying on a single, undifferentiated travel segment measure would miss this entirely: solo leisure travellers are not simply more or less satisfied in general, their sentiment depends on the specific combination of who they are and what kind of service they are reviewing, exactly the kind of interaction a warehouse built on independently queryable dimensions is positioned to surface.

7.5 Extending the Analysis Beyond the Core Use Cases

The ten use cases addressed so far test whether the warehouse answers the specific questions Section 3.4 poses directly. Section 3.4 also makes a further claim: that the dimensional structure supports analyses beyond those ten without any change to the schema, citing the Travel Segment dimension specifically as an example, since nationality and age group, unlike travel party type and trip purpose, are not drawn on by any of the ten use cases themselves. This section tests that claim directly, using two of the specific examples Section 3.4 gives, rather than treating the claim as self-evidently true.

Which visitor profiles are associated with particular service types

No structural change to the schema is needed to answer this, only a query joining D_Travel_Segment through D_Review and cross-tabulating nationality against service_category, exactly the join pattern already used in Section 7.4.

```
SELECT t.nationality, s.service_category,
       ROUND(AVG(f.polarity_score), 3) AS avg_polarity, COUNT(*) AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_SERVICE s ON f.service_key = s.service_key
JOIN D_REVIEW r ON f.review_key = r.review_key
JOIN D_TRAVEL_SEGMENT t ON r.travel_segment_key = t.travel_segment_key
WHERE t.nationality IS NOT NULL
GROUP BY t.nationality, s.service_category
HAVING COUNT(*) >= 8
ORDER BY t.nationality, s.service_category
```

Listing 11- Query used to test the nationality extensibility claim

Spanish visitors record the highest average polarity of any nationality for both accommodation (0.311) and restaurants and bars (0.304), reviewing consistently more favourably across categories than most other nationalities in the dataset. French visitors show the opposite kind of pattern, not a uniformly low reviewer but a category-dependent one: their average polarity for tourist sites (0.256) sits well above their average for accommodation (0.150), a gap of 0.106 within the same nationality. This is precisely the kind of association Section 3.4 anticipated, specific visitor profiles relating to specific service types in ways that a single, undifferentiated nationality measure would not reveal.

How demographic groups differ in their emotional responses to the same service category

The same join pattern, substituting age_group for nationality and comparing emotion measures rather than polarity, tests the second claim.

```
SELECT t.age_group, s.service_category,
       ROUND(AVG(f.joy_score), 3) AS avg_joy,
       ROUND(AVG(f.fear_score), 3) AS avg_fear,
       COUNT(*) AS n
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_SERVICE s ON f.service_key = s.service_key
JOIN D_REVIEW r ON f.review_key = r.review_key
JOIN D_TRAVEL_SEGMENT t ON r.travel_segment_key = t.travel_segment_key
WHERE t.age_group IS NOT NULL
GROUP BY t.age_group, s.service_category
HAVING COUNT(*) >= 8
ORDER BY t.age_group, s.service_category
```

Listing 12-Query used to test the age group extensibility claim

Visitors aged 65 and over record the lowest average joy of any age group for accommodation (0.031) and tourist sites (0.036), yet the highest average joy of any age group for restaurants and bars (0.080). A single, category-independent emotional profile for this age group would obscure this entirely: the same demographic segment feels differently about different kinds of service, which is only visible because emotion, service category and travel segment are each independently queryable dimensions within the same fact table.

Both results were produced without adding a single column, table or relationship to the schema described in Chapter 4. This is the concrete demonstration of Section 3.4's extensibility claim: the warehouse's analytical reach is not limited to the ten questions it was explicitly designed to answer.

The two analyses shown here are illustrative rather than exhaustive. The same join pattern, D_Travel_Segment through D_Review, cross-tabulated against any other dimension in the warehouse, extends equally to gender, trip purpose in combination with nationality, or platform-level comparisons, none of which required a schema change to become queryable either. What matters is not the specific pair of examples chosen, but that the warehouse's dimensional structure makes this entire category of analysis available without redesign, which is the property Section 3.4 claims and this section set out to test.

7.6 Comparative Analysis Against Simpler Sentiment Representations

The aspect-level design defended in Chapters 3 and 4 was argued conceptually: that a single polarity label, a review-level average, or polarity without emotion would each lose analytically meaningful information a finer-grained representation preserves. The populated warehouse makes it possible to test that argument directly rather than take it on faith, comparing what the aspect-level fact table actually contains against what each simpler alternative would have reported instead.

Star rating versus derived aspect sentiment

A system relying on star ratings alone would treat a highly rated review as uniformly positive. Whether that assumption holds can be tested directly by checking how often a highly rated review nonetheless contains a clearly negative aspect.

```
SELECT r.platform_review_id, r.star_rating, a.aspect_name,
       ROUND(f.polarity_score,3) AS polarity_score, r.review_text
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_REVIEW r ON f.review_key = r.review_key
JOIN D_ASPECT a ON f.aspect_key = a.aspect_key
WHERE r.star_rating >= 4.5 AND f.polarity_score < -0.2
ORDER BY f.polarity_score ASC
LIMIT 5
```

Listing 13- Query used for the star rating comparison

Of the 643 reviews rated 4.0 stars or higher, 75 (11.7%) contain at least one aspect with polarity below -0.15. CAP_00571, rated 4.5 stars overall, illustrates this concretely: its Booking Process aspect scores -0.511, describing a booking process that "failed repeatedly," entirely invisible to a system that reports only the overall star rating.

Review-level average versus aspect-level spread

The grain declared in Section 4.3 was chosen specifically to avoid the mixed sentiment a review-level average would produce. This can be measured directly by comparing each review's most positive and most negative aspect.

```
SELECT r.platform_review_id, ROUND(AVG(f.polarity_score),3) AS review_average,
      ROUND(MIN(f.polarity_score),3) AS min_aspect,
      ROUND(MAX(f.polarity_score),3) AS max_aspect, COUNT(*) AS n_aspects
FROM F_ASPECT_SENTIMENT_OBSERVATION f
JOIN D_REVIEW r ON f.review_key = r.review_key
GROUP BY r.platform_review_id
HAVING n_aspects >= 2
ORDER BY (max_aspect - min_aspect) DESC
LIMIT 5
```

Listing 14- Query used for the review-level average comparison

Across the 642 reviews with two or more aspect observations, the average spread between a review's most positive and most negative aspect is 0.406, and 228 of those reviews (35.5%) show a spread greater than 0.5 on a scale running from -1 to 1. CAP_00255 illustrates the effect starkly: its aspects range from -0.440 to 0.872, yet its review-level average sits at a comparatively unremarkable 0.256, precisely the kind of averaging effect the aspect-level grain was designed to avoid.

Polarity alone versus polarity with subjectivity and emotion

A polarity-only system would treat two observations with similar polarity as equivalent, regardless of what kind of negative feeling either actually expresses.

```
SELECT aspect_code, ROUND(polarity_score,3) AS polarity_score,
       ROUND(sadness_score,2) AS sadness, ROUND(disgust_score,2) AS disgust,
       ROUND(anger_score,2) AS anger, ROUND(fear_score,2) AS fear, sentence
FROM TRANSFORMED_OBSERVATIONS
WHERE polarity_score BETWEEN -0.55 AND -0.45
GROUP BY aspect_code, polarity_score
ORDER BY polarity_score
```

Listing 15-Query used for the polarity-versus-emotion comparison

At an identical polarity score, Food Quality and Room Quality (both -0.477) carry different emotional profiles: Food Quality registers both sadness and disgust (0.2 and 0.4), while Room Quality registers disgust alone (0.4, with no detected sadness), a distinction entirely lost to a measure of polarity alone. This comparison also surfaces a limitation worth stating plainly rather than smoothing over: 55 of the 211 clearly negative observations in the dataset (26.1%) show no detected emotion at all, anger, disgust, sadness and fear all register zero. A supplementary keyword lexicon, consulted only when NRC detects nothing and the sentence is already confirmed negative, narrows what was originally a larger gap without altering NRC's own judgement in any case where it has one, though a residual gap of this size remains a genuine limitation of the tool's vocabulary rather than of the dimensional structure that stores its output.

7.7 Discussion

Across ten use cases, an extensibility test and a comparative analysis, the warehouse consistently answered the questions it was built to answer using operations, slicing, drilling down and rolling up, that a coarser representation of sentiment could not support. The comparative analysis in Section 7.6 makes this concrete rather than assumed: star ratings alone missed 11.7% of highly rated reviews containing a clearly negative aspect, a review-level average would have flattened a 0.406 average spread between a review's best and worst aspect into a single unremarkable number, and polarity alone would have treated observations with genuinely different emotional character as equivalent. Together with the extensibility test in Section 7.5, which answered two of Section 3.4's own examples without altering the schema at all, this evaluation validates the objective set out in Section 1.3: that the design supports multidimensional sentiment-aware analytical processing, not merely in principle but against a populated, queryable instance of the warehouse.

The evaluation also surfaced genuine limitations, and presenting only the results that worked would be a thinner account than the one this thesis can actually give. Two limitations concern VADER specifically. The first is a sensitivity to negation and qualification: a sentence describing heating as unreliable and the stay as too cold to relax registers as strongly positive (0.440) despite being written to express the opposite. The second is a tendency to misread figurative language: a sentence describing a view that "stopped us in our tracks" registers as negative (-0.226) despite describing something the reviewer found striking in a positive sense. Neither was corrected, since a proper fix would mean replacing the polarity engine entirely, a change that would ripple through nearly every figure reported in this chapter, and rewording the specific sentences that exposed the issue would hide the two cases found without addressing the underlying tool limitation for any sentence not yet tested.

A third limitation, in NRC's coverage of negative-affect vocabulary, proved more tractable. Word by word verification confirmed that terms such as "failed," "deceptive" and "undercooked" carry no emotional weight in NRC's general-purpose lexicon at all, leaving a substantial share of clearly negative observations with no detected emotion. A small supplementary keyword lexicon, consulted only when NRC detects nothing and polarity is already confirmed negative, narrowed this gap considerably without overriding NRC's own judgement in any case where it had one.

A fourth limitation was found in the ETL itself rather than in either sentiment tool: keyword-based aspect identification, matching a word such as "options" against the MENU_VARIETY dictionary entry regardless of context, was assigning aspects to reviews whose service category could not produce them, a restaurant-only aspect appearing against an accommodation review. This affected 8.4% of the fact table before correction. The fix required no change to the dimensional schema, only a restriction on which aspects the keyword matcher considers, based on the review's own service category, a distinction the ETL already had available to it. All four limitations identified during this evaluation, taken together, concern the specific tools and algorithms used to derive and identify sentiment, not the dimensional structure that stores and organises the results. This separation is itself a property worth noting rather than assuming: it means each of these limitations could be addressed, now or in a future iteration, by adjusting the derivation or identification layer alone, without redesigning the warehouse those layers feed into.

7.8 Summary

Chapter 7 evaluated the prototype built in Chapter 6 against the ten analytical use cases established in Section 3.4, grouped into the same three categories used there: strategic and comparative analysis, aspect-level and operational analysis, and affective and segmentation analysis. Each use case was answered with a real SQL query executed against the populated warehouse, its result discussed rather than assumed, and its OLAP operation identified explicitly. Beyond these ten, an extensibility test confirmed that the warehouse supports further analyses, through nationality and age group specifically, without any change to the schema, and a comparative analysis tested the aspect-level design directly against the simpler alternatives Chapters 3 and 4 argued against conceptually, star ratings alone, review-level averages and polarity without emotion, confirming empirically that each would have lost analytically meaningful information the aspect-level grain preserves. The evaluation also surfaced four genuine limitations, two in VADER's handling of negation and figurative language, one in NRC's vocabulary coverage, and one in the ETL's own keyword-based aspect identification, the latter two of which were identified and corrected during this same evaluation rather than deferred as future work. All four concern the specific tools and algorithms used, not the dimensional structure itself, a distinction that closes this chapter's account of what the warehouse can do and what its current implementation does not yet do perfectly. Chapter 8 draws the conclusions this evaluation supports.

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